



Boolean algebra. Combinational and sequential circuits. Minimization. Number representations and computer arithmetic (fixed and floating point)

Mark Distribution in Previous GATE

Year	2026 - 1	2026 - 2	2025 - 1	2025 - 2	2024 - 1	2024 - 2	2023	2022	2021 - 1	2021 - 2	Minimum
1 Mark Count	2	3	2	3	2	2	2	1	2	3	1
2 Marks Count	3	2	2	3	2	2	2	2	2	2	2
Total Marks	8	7	6	9	6	6	6	5	6	7	5

Welcome to the "Digital Logic" chapter of your GATE Computer Science exam preparation. This foundational subject is the bedrock of computer hardware, providing the essential understanding of how digital circuits process information, store data, and execute instructions. It delves into the fundamental building blocks of modern computing systems, from basic logic gates to complex sequential circuits and memory elements. A strong grasp of Digital Logic is crucial not only for direct questions in the GATE exam but also for understanding related concepts in Computer Architecture, Operating Systems, and even some aspects of programming. Typically, Digital Logic carries a weightage of 8-12 marks in the GATE CS paper, with questions ranging from conceptual understanding and circuit analysis to design problems involving minimization, number representation, and sequential circuit behavior. Questions often include Multiple Choice Questions (MCQs), Multiple Select Questions (MSQs), and Numerical Answer Type (NAT) questions, testing both theoretical knowledge and problem-solving skills.

Topic-wise Key Concepts

Adder

An adder is a digital circuit that performs the addition of numbers. It is a fundamental component in Arithmetic Logic Units (ALUs) of processors.

- **Half Adder (HA):** Adds two single-bit binary numbers. It produces a sum (S) and a carry-out (Cout).
 - Formulas:

$$S = A \oplus B$$

$$C_{out} = A \cdot B$$

- **Full Adder (FA):** Adds three single-bit binary numbers (two input bits and a carry-in). It produces a sum (S) and a carry-out (Cout).
 - Formulas:

$$S = A \oplus B \oplus C_{in}$$

$$C_{out} = (A \cdot B) + (C_{in} \cdot (A \oplus B))$$

$$C_{out} = (A \cdot B) + (B \cdot C_{in}) + (C_{in} \cdot A)$$

- **Ripple Carry Adder:** An n-bit adder constructed by cascading n full adders, where the carry-out of one stage becomes the carry-in of the next.
 - Propagation Delay: The total delay is proportional to n, as carry ripples through all stages.

$$T_{ripple} = n \times T_{FA_carry}$$
- **Carry Look-Ahead Adder (CLA):** A faster adder that generates carries in parallel, reducing propagation delay. It uses carry generate (G) and carry propagate (P) signals.
 - Formulas for a single stage i:

$$P_i = A_i \oplus B_i$$

$$G_i = A_i \cdot B_i$$

$$S_i = P_i \oplus C_i$$

$$C_{i+1} = G_i + (P_i \cdot C_i)$$

- For a 4-bit CLA:

$$C_1 = G_0 + P_0 C_0$$

$$C_2 = G_1 + P_1 C_1 = G_1 + P_1 G_0 + P_1 P_0 C_0$$

$$C_3 = G_2 + P_2 C_2 = G_2 + P_2 G_1 + P_2 P_1 G_0 + P_2 P_1 P_0 C_0$$

$$C_4 = G_3 + P_3 C_3 = G_3 + P_3 G_2 + P_3 P_2 G_1 + P_3 P_2 P_1 G_0 + P_3 P_2 P_1 P_0 C_0$$

- **Common Pitfalls:** Confusing half adder and full adder functionality. Incorrectly calculating propagation delay for ripple carry vs. CLA.
- **Problem-solving:** Be able to draw and analyze basic adder circuits. Calculate delays for different adder types.

Array Multiplier

An array multiplier is a combinational circuit that performs multiplication of two binary numbers using an array of full adders and AND gates. It's a direct hardware implementation of the traditional "paper-and-pencil" multiplication method.

- **Core Idea:** For two n-bit numbers, it requires n rows of partial products. Each partial product is generated by ANDing the multiplicand with one bit of the multiplier. These partial products are then summed using an array of adders (often full adders) in a staggered fashion.
- **Structure:** An n x m bit array multiplier typically uses n*m AND gates to generate partial products and (n-1)*m full adders (or similar) to sum them.
- **Delay:** The delay is proportional to n+m, as carries propagate diagonally across the array.
- **Problem-solving:** Understand how partial products are formed and summed. Be able to trace the data flow for small examples.

Binary Codes

Binary codes are systems used to represent numbers, characters, and instructions in digital systems using binary digits (bits).

- **Weighted Codes:** Each bit position has a specific weight.
 - **BCD (8421):** Each decimal digit is represented by its 4-bit binary equivalent. E.g., $(9)_{10} = (1001)_{BCD}$. $(25)_{10} = (0010\ 0101)_{BCD}$.
 - **Excess-3:** BCD code with 3 added to each digit. Self-complementing.
 - **2421 Code:** Another weighted code.
- **Non-Weighted Codes:** Bit positions do not have fixed weights.
 - **Gray Code:** Only one bit changes between successive numbers. Used to prevent glitches in ADCs and K-maps.
 - Binary to Gray: $G_i = B_i \oplus B_{i+1}$ (for MSB to LSB, $G_{n-1} = B_{n-1}$, $G_i = B_i \oplus B_{i+1}$ for $i < n-1$). More commonly: $G_i = B_i \oplus B_{i+1}$, with $B_n = 0$ for the MSB. Correct: $G_{MSB} = B_{MSB}$, $G_i = B_i \oplus B_{i+1}$ for i from MSB-1 down to 0. Alternative: $G_i = B_i \oplus B_{i-1}$ for $i > 0$, $G_0 = B_0$. Example: Binary 1011 (11) -> Gray: $G_3 = 1$, $G_2 = 1 \oplus 0 = 1$, $G_1 = 0 \oplus 1 = 1$, $G_0 = 1 \oplus 1 = 0$. So 1110.
 - Gray to Binary: $B_{MSB} = G_{MSB}$, $B_i = G_i \oplus B_{i+1}$ (for MSB to LSB). Correct: $B_i = G_i \oplus B_{i+1}$ for i from MSB-1 down to 0, $B_{MSB} = G_{MSB}$. Example: Gray 1110 -> Binary: $B_3 = 1$, $B_2 = 1 \oplus 1 = 0$, $B_1 = 1 \oplus 0 = 1$, $B_0 = 0 \oplus 1 = 1$. So 1011.
 - **ASCII:** 7-bit or 8-bit code for characters.
- **Error Detection Codes (Parity):** Adds an extra bit to detect single-bit errors.
 - **Even Parity:** Total number of 1s (including parity bit) is even.
 - **Odd Parity:** Total number of 1s (including parity bit) is odd.
- **Error Correction Codes (Hamming Code):** Can detect and correct multiple-bit errors.
- **Common Pitfalls:** Confusing BCD with pure binary. Incorrectly converting between binary and Gray code.
- **Problem-solving:** Perform conversions between different binary codes.

Boolean Algebra

Boolean algebra is a mathematical system for analyzing and simplifying digital circuits. It deals with binary variables and logical operations.

- **Basic Postulates:**
 1. Closure: $A + B$ and $A \cdot B$ are in $\{0,1\}$.
 2. Commutative: $A + B = B + A$, $A \cdot B = B \cdot A$.
 3. Associative: $A + (B + C) = (A + B) + C$, $A \cdot (B \cdot C) = (A \cdot B) \cdot C$.

4. Distributive: $A \cdot (B + C) = (A \cdot B) + (A \cdot C)$, $A + (B \cdot C) = (A + B) \cdot (A + C)$.
 5. Identity: $A + 0 = A$, $A \cdot 1 = A$.
 6. Complement: $A + \bar{A} = 1$, $A \cdot \bar{A} = 0$.
- **Theorems and Identities:**
 - Idempotence: $A + A = A$, $A \cdot A = A$.
 - Absorption: $A + (A \cdot B) = A$, $A \cdot (A + B) = A$.
 - Consensus: $(A \cdot B) + (\bar{A} \cdot C) + (B \cdot C) = (A \cdot B) + (\bar{A} \cdot C)$. (If $B \cdot C$ is redundant).
 - De Morgan's Laws: $\overline{A + B} = \bar{A} \cdot \bar{B}$, $\overline{A \cdot B} = \bar{A} + \bar{B}$.
 - Involution: $\overline{\bar{A}} = A$.
 - Null Elements: $A + 1 = 1$, $A \cdot 0 = 0$.
 - **Common Pitfalls:** Forgetting the dual of De Morgan's Law. Incorrectly applying consensus theorem.
 - **Problem-solving:** Simplify Boolean expressions using theorems. Prove identities.

Booths Algorithm

Booth's algorithm is a multiplication algorithm that multiplies two signed binary numbers in 2's complement representation. It is efficient for numbers with long sequences of 0s or 1s.

- **Core Idea:** Instead of adding partial products for every '1' in the multiplier, it looks at pairs of bits in the multiplier.
 - 00 → No operation (0 * multiplicand).
 - 01 → Add multiplicand.
 - 10 → Subtract multiplicand (add 2's complement).
 - 11 → No operation (0 * multiplicand).
- **Steps:**
 1. Initialize Product (P) to 0, Multiplicand (M), Multiplier (Q), and an extra bit $Q_{-1} = 0$.
 2. Repeat n times (where n is the number of bits in Q):
 - Examine Q_0Q_{-1} :
 - If 01: $P = P + M$.
 - If 10: $P = P - M$ (add 2's complement of M).
 - If 00 or 11: No operation.
 - Arithmetic Right Shift (ASHR) the combined register PQ_{-1} .
 3. The final product is in PQ .
- **Common Pitfalls:** Incorrectly performing 2's complement subtraction. Errors in arithmetic right shift (especially for negative numbers). Forgetting the Q_{-1} bit.
- **Problem-solving:** Trace the algorithm step-by-step for given signed numbers.

Canonical Normal Form

Canonical normal forms are standard forms for Boolean expressions, ensuring uniqueness for a given function. They are useful for comparison and synthesis.

- **Canonical Sum of Products (SOP) / Minterm Canonical Form:** A sum of minterms. Each minterm is a product term where all variables appear exactly once, either in true or complemented form.
 - A minterm is 1 for exactly one combination of inputs.
 - Example: For $F(A, B) = A + B$, minterms are $\bar{A}B$, $A\bar{B}$, AB . So $F(A, B) = \bar{A}B + A\bar{B} + AB$.
 - Notation: $\sum m(i, j, k, \dots)$, where i, j, k are decimal equivalents of minterms.
- **Canonical Product of Sums (POS) / Maxterm Canonical Form:** A product of maxterms. Each maxterm is a sum term where all variables appear exactly once, either in true or complemented form.
 - A maxterm is 0 for exactly one combination of inputs.
 - Example: For $F(A, B) = A \cdot B$, maxterms are $A + B$, $A + \bar{B}$, $\bar{A} + B$. So $F(A, B) = (A + B) \cdot (A + \bar{B}) \cdot (\bar{A} + B)$.
 - Notation: $\prod M(i, j, k, \dots)$, where i, j, k are decimal equivalents of maxterms.
- **Relationship:** If $F = \sum m(i, j, k)$, then $\bar{F} = \sum m(\text{other indices})$. Also, $F = \prod M(\text{other indices})$.
- **Common Pitfalls:** Confusing minterms with maxterms, and their corresponding sum/product forms. Incorrectly converting between SOP and POS canonical forms.
- **Problem-solving:** Convert truth tables to canonical SOP/POS. Convert between canonical SOP and POS.

Carry Generator

A carry generator is a combinational circuit used in adders, particularly in Carry Look-Ahead Adders, to quickly compute carry bits, thereby speeding up addition.

- **Core Idea:** It uses generate (G) and propagate (P) signals to calculate carries in parallel, rather than waiting for them to ripple.
 - $G_i = A_i \cdot B_i$ (carry is generated if both input bits are 1).
 - $P_i = A_i \oplus B_i$ (carry is propagated if one input bit is 1).
 - $C_{i+1} = G_i + (P_i \cdot C_i)$ (carry out is generated or propagated from carry in).
- **Look-ahead Logic:** For a 4-bit block, the carry out C_4 can be expressed directly in terms of C_0 and the G_i, P_i terms, as shown in the Adder section.
- **Problem-solving:** Derive carry equations for multi-bit carry look-ahead blocks.

Circuit Output

The circuit output refers to the final value(s) produced by a digital circuit based on its inputs and internal logic. It can be a single bit or a multi-bit word.

- **Combinational Circuits:** Output depends only on the present inputs.
- **Sequential Circuits:** Output depends on present inputs and past inputs (stored state).
- **Analysis:** For a given circuit diagram, determine the Boolean expression for the output(s) in terms of the inputs.
- **Problem-solving:** Derive truth tables or Boolean expressions from circuit diagrams. Evaluate output for specific input combinations.

Combinational Circuit

A combinational circuit is a type of digital circuit whose output depends solely on the current values of its inputs. It has no memory elements.

- **Characteristics:**
 - No feedback loops.
 - Output changes instantaneously with input changes (after propagation delay).
 - Examples: Adders, Subtractors, Decoders, Encoders, Multiplexers, Demultiplexers.
- **Design Steps:**
 1. Define the problem (inputs, outputs).
 2. Derive the truth table.
 3. Obtain simplified Boolean expressions (K-map, Quine-McCluskey).
 4. Draw the logic diagram.
- **Common Pitfalls:** Confusing combinational with sequential circuits. Overlooking propagation delays in timing analysis.
- **Problem-solving:** Design circuits from specifications. Analyze existing circuits.

Conjunctive Normal Form

Conjunctive Normal Form (CNF) is a standardized way of writing Boolean expressions as a conjunction (AND) of clauses, where each clause is a disjunction (OR) of literals.

- **Core Idea:** A product of sums (POS) form where each sum term (clause) contains one or more literals. It is not necessarily canonical (i.e., not all variables need to be present in each clause).
 - Example: $(A + \bar{B}) \cdot (\bar{A} + C)$.
- **Relationship to Canonical POS:** Canonical POS (minterm form) is a specific type of CNF where each clause is a maxterm (contains all variables).
- **Problem-solving:** Convert Boolean expressions to CNF.

Decoder

A decoder is a combinational circuit that converts n input lines into 2^n output lines. Only one output line is active (high or low) at any given time, corresponding to the binary value of the inputs.

- **Types:**
 - **Binary Decoder (n -to- 2^n):** E.g., 2-to-4 decoder, 3-to-8 decoder.
 - **BCD-to-7 Segment Decoder:** Converts BCD input to control a 7-segment display.
- **Enable Input:** Most decoders have an enable input (E). If E is inactive, all outputs are inactive.

- **Applications:** Memory address decoding, data demultiplexing, implementing Boolean functions.
 - An n -to- 2^n decoder can implement any n -variable Boolean function by ORing the appropriate minterms.
- **Common Pitfalls:** Incorrectly interpreting active-high vs. active-low outputs.
- **Problem-solving:** Design decoders. Use decoders to implement Boolean functions.

Digital Circuits

Digital circuits are electronic circuits that operate on discrete voltage levels, typically representing binary 0 and 1. They form the basis of all modern digital systems.

- **Categories:**
 - **Combinational Circuits:** Output depends only on current inputs (e.g., gates, adders, decoders).
 - **Sequential Circuits:** Output depends on current inputs and past inputs (memory elements) (e.g., flip-flops, counters, registers).
- **Logic Gates:** AND, OR, NOT, NAND, NOR, XOR, XNOR are the basic building blocks.
- **Key Properties:** Speed (propagation delay), Power Consumption, Fan-in/Fan-out, Noise Margin.
- **Problem-solving:** Analyze and design circuits using logic gates.

Digital Counter

A digital counter is a sequential circuit that cycles through a predefined sequence of states upon receiving input clock pulses. It's essentially a register that increments or decrements its stored value.

- **Types:**
 - **Asynchronous (Ripple) Counter:** Flip-flops are cascaded, and the output of one FF clocks the next. Simple to design but suffers from propagation delay.
 - **Synchronous Counter:** All flip-flops are clocked simultaneously by a common clock pulse. More complex design but faster and more reliable.
 - **Up/Down Counter:** Can count in increasing or decreasing order.
 - **Mod-N Counter:** Counts from 0 to $N-1$ and then resets.
- **Design:** Involves state diagrams, state tables, flip-flop excitation tables, and K-maps for logic minimization.
- **Formulas:** For an n -bit counter, it can count up to 2^n states (0 to $2^n - 1$).
- **Common Pitfalls:** Incorrectly determining the modulus of a counter. Errors in state table derivation or K-map minimization for synchronous counters.
- **Problem-solving:** Analyze existing counter circuits. Design synchronous counters for specific sequences.

Dual Function

The dual of a Boolean expression is obtained by interchanging OR and AND operations, and 0s and 1s. Variables and their complements remain unchanged.

- **Procedure:**
 1. Replace all '+' with '.' (OR with AND).
 2. Replace all '.' with '+' (AND with OR).
 3. Replace all '0' with '1'.
 4. Replace all '1' with '0'.
- **Example:** Dual of $F = A \cdot B + 0$ is $F^D = (A + B) \cdot 1$.
- **Principle of Duality:** If a Boolean identity is true, its dual is also true.
- **Common Pitfalls:** Accidentally complementing variables when finding the dual.
- **Problem-solving:** Find the dual of a given Boolean expression.

Finite State Machines (FSM)

Finite State Machines (FSMs) are mathematical models of computation used to design sequential circuits. They consist of a finite number of states, transitions between states, and actions based on inputs.

- **Components:** States, Inputs, Outputs, Transitions.
- **Types:**
 - **Mealy Machine:** Output depends on the present state AND the present input.
 - **Moore Machine:** Output depends only on the present state.
- **Representation:** State diagrams, state tables.
- **Design Steps:**

1. State Diagram.
 2. State Table.
 3. State Assignment (binary encoding).
 4. Flip-flop excitation table.
 5. K-maps for input equations and output equations.
 6. Logic diagram.
- **Common Pitfalls:** Confusing Mealy and Moore outputs. Errors in state assignment or deriving excitation equations.
 - **Problem-solving:** Design FSMs from specifications. Analyze given FSMs.

Fixed Point Representation

Fixed-point representation is a method of representing real numbers where the position of the binary point (radix point) is fixed. It's used for numbers with a limited range and high precision.

- **Format:** $I.F$ where I is the integer part and F is the fractional part.
 - For an n-bit number with k bits for the fractional part, the range is -2^{n-k-1} to $2^{n-k-1} - 2^{-k}$ (for signed 2's complement).
 - Value: $\sum_{i=-(k)}^{n-k-1} b_i 2^i$.
- **Signed Fixed-Point:** Typically uses 2's complement for negative numbers.
- **Common Pitfalls:** Incorrectly determining the range or precision for a given bit allocation.
- **Problem-solving:** Convert decimal numbers to fixed-point binary and vice-versa. Perform arithmetic operations.

Flip Flop

A flip-flop is a basic 1-bit memory element (sequential circuit) that can store a binary value (0 or 1). It has two stable states and is often edge-triggered.

- **Types and Characteristics:**
 - **SR Flip-Flop:** Set-Reset.
 - Characteristic Equation: $Q_{next} = S + RQ$, $SR = 0$ (forbidden state if $SR=1$).
 - Excitation Table:

Q_n	Q_{n+1}	SR
0	0	0 X
0	1	1 0
1	0	0 1
1	1	X 0
 - **JK Flip-Flop:** J-K. Overcomes SR's forbidden state (toggle if $JK=11$).
 - Characteristic Equation: $Q_{next} = J\bar{Q} + \bar{K}Q$.
 - Excitation Table:

Q_n	Q_{n+1}	JK
0	0	0 X
0	1	1 X
1	0	X 1
1	1	X 0
 - **D Flip-Flop:** Data. Stores the input D when clocked.
 - Characteristic Equation: $Q_{next} = D$.
 - Excitation Table:

Q_n	Q_{n+1}	D
0	0	0
0	1	1
1	0	0
1	1	1
 - **T Flip-Flop:** Toggle. Toggles its state if $T=1$, holds if $T=0$.
 - Characteristic Equation: $Q_{next} = T\bar{Q} + \bar{T}Q = T \oplus Q$.
 - Excitation Table:

Q_n	Q_{n+1}	T
0	0	0
0	1	1

1	0	1
1	1	0

- **Edge-Triggered vs. Level-Triggered:** Flip-flops are edge-triggered (positive or negative), latches are level-triggered.
- **Setup Time:** Data must be stable before clock edge.
- **Hold Time:** Data must be stable after clock edge.
- **Propagation Delay:** Time for output to change after clock edge.
- **Common Pitfalls:** Confusing characteristic equations with excitation tables. Misinterpreting edge vs. level triggering.
- **Problem-solving:** Analyze flip-flop behavior. Use flip-flops to design sequential circuits.

Floating Point Representation

Floating-point representation is a method to represent real numbers over a wider dynamic range than fixed-point, at the cost of precision. It uses a sign, exponent, and mantissa (significand).

- **Format:** $\pm M \times B^E$, where M is mantissa, B is base, E is exponent.
- **Normalization:** Mantissa is typically normalized to have a leading '1' (implicit in IEEE standard).
- **Bias:** Exponent is stored in biased form to allow representation of both positive and negative exponents without a separate sign bit. Biased exponent = Actual exponent + Bias.
- **Common Pitfalls:** Errors in converting between decimal and floating-point, especially with bias and normalization.
- **Problem-solving:** Convert numbers to/from floating-point format. Understand the implications of range and precision.

Functional Completeness

A set of logic gates is functionally complete if any Boolean function can be implemented using only gates from that set.

- **Minimum Functionally Complete Sets:**
 - {AND, OR, NOT}
 - {NAND} (NAND is universal)
 - {NOR} (NOR is universal)
 - {AND, NOT}
 - {OR, NOT}
- **Properties:** To be functionally complete, a set of gates must be able to implement NOT, AND, and OR (or NAND/NOR). This requires the ability to implement:
 - Inversion (NOT)
 - AND (or OR)
 - The set must not be monotonic (cannot implement NOT).
 - The set must not be self-dual (cannot implement NOT).
 - The set must not be linear (cannot implement AND/OR).
 - The set must not preserve 0 (cannot implement NOT).
 - The set must not preserve 1 (cannot implement NOT).
- **Common Pitfalls:** Incorrectly identifying functionally complete sets.
- **Problem-solving:** Determine if a given set of gates is functionally complete. Implement basic gates using a universal gate.

IEEE Representation

The IEEE 754 standard defines formats for representing floating-point numbers in computers, ensuring consistency across different platforms.

- **Single Precision (32-bit):**
 - 1 bit for Sign (S)
 - 8 bits for Exponent (E) - biased by 127
 - 23 bits for Mantissa (M) - implicit leading '1'
 - Value: $(-1)^S \times (1.M)_{base2} \times 2^{(E-127)}$
 - Range: Approx. $\pm 1.17 \times 10^{-38}$ to $\pm 3.4 \times 10^{38}$
- **Double Precision (64-bit):**
 - 1 bit for Sign (S)
 - 11 bits for Exponent (E) - biased by 1023

- 52 bits for Mantissa (M) - implicit leading '1'
- Value: $(-1)^S \times (1.M)_{base2} \times 2^{(E-1023)}$
- **Special Values:**
 - Zero: $E = 0, M = 0$
 - Denormalized Numbers: $E = 0, M \neq 0$ (for very small numbers, implicit leading '0')
 - Infinity: $E = \text{all } 1\text{s}, M = 0$
 - NaN (Not a Number): $E = \text{all } 1\text{s}, M \neq 0$
- **Common Pitfalls:** Forgetting the implicit leading '1' in the mantissa. Incorrectly applying the exponent bias. Handling special values.
- **Problem-solving:** Convert decimal numbers to IEEE 754 format and vice-versa. Understand the representation of special values.

K Map (Karnaugh Map)

A K-map is a graphical method for simplifying Boolean expressions. It provides a systematic way to find the minimal sum-of-products (SOP) or product-of-sums (POS) form.

- **Core Idea:** Arranges minterms (or maxterms) in a grid such that adjacent cells differ by only one bit, allowing visual identification of adjacent terms for grouping.
- **Grouping Rules:**
 - Groups must be powers of 2 (1, 2, 4, 8, ...).
 - Groups must be rectangular or square.
 - Groups can wrap around the edges.
 - Groups should be as large as possible.
 - Every '1' (for SOP) or '0' (for POS) must be covered at least once.
- **Prime Implicant (PI):** A product term obtained by combining the maximum possible number of adjacent cells in a K-map.
- **Essential Prime Implicant (EPI):** A prime implicant that covers at least one minterm (or maxterm) that no other prime implicant covers. EPIs must be included in the minimal expression.
- **Don't Cares (X):** Input combinations that never occur or whose output doesn't matter. They can be grouped with 1s (for SOP) or 0s (for POS) to make larger groups.
- **Common Pitfalls:** Incorrect grouping (non-power of 2, non-rectangular). Not identifying all EPIs. Not finding the minimal set of PIs.
- **Problem-solving:** Simplify Boolean expressions using K-maps for up to 4-5 variables.

Little Endian Big Endian

These terms refer to the byte ordering conventions used to store multi-byte data (like integers or floating-point numbers) in computer memory.

- **Big-Endian:** The most significant byte (MSB) of a multi-byte data unit is stored at the lowest memory address.
 - Example: For $(0x12345678)_{16}$, at address 1000:

Address	Value
1000	12
1001	34
1002	56
1003	78
- **Little-Endian:** The least significant byte (LSB) of a multi-byte data unit is stored at the lowest memory address.
 - Example: For $(0x12345678)_{16}$, at address 1000:

Address	Value
1000	78
1001	56
1002	34
1003	12
- **Common Pitfalls:** Confusing the two, especially when dealing with memory dumps or network protocols.
- **Problem-solving:** Determine how a given multi-byte value would be stored in memory under both endianness conventions.

Memory Interfacing

Memory interfacing involves connecting memory chips to a processor or other digital system, ensuring proper address decoding, data transfer, and control signaling.

- **Address Decoding:** Logic circuitry that selects the correct memory chip (or block within a chip) based on the address provided by the processor.
 - **Full Decoding:** Every unique address maps to a unique memory location.
 - **Partial Decoding:** Multiple addresses map to the same memory location (simpler but less efficient).
- **Chip Select (CS):** An input pin on memory chips that, when active, enables the chip for read/write operations. Address decoders generate the CS signal.
- **Output Enable (OE) / Read (RD):** Enables data output from memory.
- **Write Enable (WE) / Write (WR):** Enables data input to memory.
- **Memory Map:** A diagram showing how memory addresses are allocated to different memory devices.
- **Formulas:** For a memory chip with N address lines and M data lines, its capacity is $2^N \times M$ bits.
- **Common Pitfalls:** Incorrectly designing address decoding logic. Miscalculating memory capacity or address range.
- **Problem-solving:** Design address decoding circuits for given memory chips and address ranges. Calculate memory capacity.

Min No Gates

Minimizing the number of gates refers to finding the simplest possible logic circuit to implement a given Boolean function, often after minimizing the Boolean expression.

- **Core Idea:** Directly related to Boolean expression simplification (K-maps, Quine-McCluskey). Fewer terms and fewer literals generally mean fewer gates.
- **Considerations:**
 - **Gate Type:** NAND/NOR implementations often require fewer gates than AND/OR/NOT for universal logic.
 - **Fan-in:** Gates have a maximum number of inputs.
 - **Gate Cost:** Sometimes gates have different costs (e.g., XOR is more complex than AND).
- **Problem-solving:** After K-map simplification, draw the circuit using the minimal SOP/POS. Then convert to universal gates (NAND/NOR) if specified, and count the gates.

Min Products of Sum Form (MPOS)

The minimal Product of Sums (POS) form is the simplest Boolean expression representing a function as a product of sum terms (maxterms or sum literals).

- **Derivation:**
 1. Identify the '0's in the truth table or K-map.
 2. Group the '0's in the K-map (following K-map rules).
 3. Each group corresponds to a sum term.
 4. The product of these sum terms is the minimal POS expression.
- **Properties:** Each sum term is an implicant of the complement of the function.
- **Common Pitfalls:** Incorrectly forming sum terms from K-map groups (e.g., $\bar{A}$ for $A=1$).
- **Problem-solving:** Use K-maps to find the minimal POS expression.

Min Sum of Products Form (MSOP)

The minimal Sum of Products (SOP) form is the simplest Boolean expression representing a function as a sum of product terms (minterms or product literals).

- **Derivation:**
 1. Identify the '1's in the truth table or K-map.
 2. Group the '1's in the K-map (following K-map rules).
 3. Each group corresponds to a product term.
 4. The sum of these product terms is the minimal SOP expression.
- **Properties:** Each product term is an implicant of the function.
- **Common Pitfalls:** Not identifying all essential prime implicants. Not selecting the minimal set of prime implicants to cover all 1s.
- **Problem-solving:** Use K-maps to find the minimal SOP expression.

Multiplexer (MUX)

A multiplexer (MUX) is a combinational circuit that selects one of several input data lines and routes it to a single output line. The selection is controlled by a set of select lines.

- **Core Idea:** An n -to-1 MUX has n data inputs, $\log_2 n$ select lines, and 1 output.
- **Functionality:** If there are 's' select lines, it can select from 2^s data inputs.
- **Applications:** Data selection, parallel-to-serial conversion, implementing Boolean functions.
 - An n -variable Boolean function can be implemented using a $(n - 1)$ -variable MUX and external logic on the data inputs.
- **Formulas:** For a 2-to-1 MUX with select S , inputs I_0, I_1 : $Output = \bar{S}I_0 + SI_1$.
- **Common Pitfalls:** Incorrectly mapping inputs to select lines when implementing functions.
- **Problem-solving:** Design MUX circuits. Use MUXes to implement Boolean functions.

Number Representation

Number representation refers to the various ways in which numerical values (integers, real numbers) are stored and manipulated in digital systems.

- **Unsigned Binary:** All bits represent magnitude. Range for n bits: 0 to $2^n - 1$.
- **Signed Magnitude:** MSB is sign bit (0 for positive, 1 for negative), remaining bits are magnitude.
 - Range for n bits: $-(2^{n-1} - 1)$ to $+(2^{n-1} - 1)$.
 - Two representations for zero ($+0, -0$).
- **1's Complement:** Negative numbers are obtained by inverting all bits of the positive number.
 - Range for n bits: $-(2^{n-1} - 1)$ to $+(2^{n-1} - 1)$.
 - Two representations for zero ($+0, -0$).
 - Addition: Add normally, then add end-around carry to the sum.
- **2's Complement:** Negative numbers are obtained by inverting all bits and adding 1 to the result. Most common representation for signed integers.
 - Range for n bits: -2^{n-1} to $+(2^{n-1} - 1)$.
 - Unique representation for zero.
 - Addition/Subtraction: Perform binary addition directly. Overflow detection: if signs of inputs are same but sign of result is different.
- **Fixed-Point and Floating-Point:** Covered in their respective sections.
- **Common Pitfalls:** Errors in converting between signed representations. Incorrectly detecting overflow.
- **Problem-solving:** Convert numbers between different representations. Perform arithmetic operations and detect overflow.

Number System

A number system is a set of symbols and rules for representing numerical quantities. Digital systems primarily use binary, but conversions to/from decimal, octal, and hexadecimal are common.

- **Decimal (Base 10):** Digits 0-9.
- **Binary (Base 2):** Digits 0, 1.
- **Octal (Base 8):** Digits 0-7. Each octal digit corresponds to 3 binary bits.
- **Hexadecimal (Base 16):** Digits 0-9, A-F. Each hex digit corresponds to 4 binary bits.
- **Conversions:**
 - Decimal to Binary/Octal/Hex: Repeated division by base, collecting remainders.
 - Binary/Octal/Hex to Decimal: Sum of (digit * base^{position}).
 - Binary to Octal/Hex: Group bits (3 for octal, 4 for hex) and convert each group.
 - Octal/Hex to Binary: Convert each digit to its 3/4-bit binary equivalent.
- **Common Pitfalls:** Errors in base conversions, especially with fractional parts.
- **Problem-solving:** Perform conversions between any two number systems.

Prime Implicants

In Boolean algebra simplification (e.g., using K-maps), a prime implicant (PI) is a product term that cannot be combined with any other term to eliminate a literal.

- **Core Idea:** It represents a maximal group of adjacent 1s (or 0s for POS) in a K-map.
- **Essential Prime Implicant (EPI):** A prime implicant that covers at least one minterm (or maxterm) that no other prime implicant covers. All EPIs must be included in the minimal expression.

- **Redundant Prime Implicant:** A prime implicant whose covered minterms are all covered by other prime implicants (including EPIs).
- **Problem-solving:** Identify all PIs and EPIs from a K-map. Use them to derive minimal SOP/POS expressions.

ROM (Read-Only Memory)

ROM is a non-volatile memory device whose contents are programmed during manufacturing or by the user and cannot be easily changed during operation. It's used to store fixed programs or lookup tables.

- **Types:**
 - **Mask ROM:** Programmed during manufacturing.
 - **PROM (Programmable ROM):** User-programmable once.
 - **EPROM (Erasable PROM):** Erasable by UV light, reprogrammable.
 - **EEPROM (Electrically Erasable PROM):** Electrically erasable, reprogrammable.
 - **Flash Memory:** A type of EEPROM, block-erasable.
- **Structure:** Consists of an address decoder and an array of memory cells (fuses/transistors).
 - An n-input, m-output ROM can implement m different n-variable Boolean functions.
- **Applications:** Storing BIOS, lookup tables, implementing combinational logic functions.
- **Problem-solving:** Determine the size of ROM needed for a given function. Implement Boolean functions using ROM.

Reduction

Reduction in digital logic refers to the process of simplifying Boolean expressions or minimizing the number of gates/components in a circuit while maintaining its functionality.

- **Methods:**
 - **Boolean Algebra:** Applying theorems and postulates.
 - **Karnaugh Maps (K-maps):** Graphical method for up to 5-6 variables.
 - **Quine-McCluskey Algorithm:** Tabular method for more variables, suitable for computer implementation.
- **Goals:** Reduce cost (fewer gates), reduce power consumption, increase speed (fewer gate delays).
- **Common Pitfalls:** Not finding the absolute minimal form. Errors in applying simplification rules.
- **Problem-solving:** Apply various reduction techniques to simplify Boolean expressions.

Ripple Counter Operation

A ripple counter (asynchronous counter) is a sequential circuit where the output of one flip-flop serves as the clock input for the next flip-flop in the chain.

- **Operation:**
 - The first flip-flop (LSB) is clocked by the external clock.
 - Its output (Q) then clocks the next flip-flop, and so on.
 - This creates a "ripple" effect as the clock signal propagates through the stages.
- **Characteristics:**
 - **Simple Design:** Easy to construct.
 - **Propagation Delay:** The total delay accumulates through each flip-flop, leading to a maximum operating frequency limitation. $T_{total} = n \times T_{FF_delay}$.
 - **Glitches:** Due to varying propagation delays, intermediate states may appear briefly, causing glitches.
- **Modulus:** An n-bit ripple counter counts 2^n states (Mod- 2^n). To achieve Mod-N, external logic is needed to reset the counter at N.
- **Common Pitfalls:** Overlooking the cumulative propagation delay. Incorrectly determining the count sequence or modulus.
- **Problem-solving:** Analyze the timing and counting sequence of ripple counters. Calculate maximum operating frequency.

Shift Registers

A shift register is a sequential circuit capable of storing binary data and shifting it left or right by one or more bit positions on each clock pulse.

- **Types:**
 - **SISO (Serial-In, Serial-Out):** Data enters and exits serially.

- **SIPO (Serial-In, Parallel-Out):** Data enters serially, but all bits are available simultaneously at parallel outputs.
- **PISO (Parallel-In, Serial-Out):** Data enters in parallel, then shifts out serially.
- **PIPO (Parallel-In, Parallel-Out):** Data enters and exits in parallel.
- **Bidirectional Shift Register:** Can shift data both left and right.
- **Universal Shift Register:** Can perform all the above operations (SISO, SIPO, PISO, PIPO, load, hold).
- **Applications:** Data conversion (serial-to-parallel, parallel-to-serial), data manipulation, sequence generation, arithmetic operations (multiplication/division by powers of 2).
- **Structure:** Typically built using D flip-flops.
- **Common Pitfalls:** Incorrectly tracing data movement for different shift register types.
- **Problem-solving:** Analyze the operation of various shift registers. Design circuits using shift registers.

Static Hazard

A static hazard is a momentary undesirable glitch (a brief 0 or 1 pulse) in the output of a combinational circuit when the input changes, even though the output should remain constant.

- **Types:**
 - **Static-1 Hazard:** Output should remain 1 but momentarily goes to 0. Occurs when a K-map has two adjacent 1s that are covered by different prime implicants, and there's no overlapping prime implicant to cover the transition path.
 - **Static-0 Hazard:** Output should remain 0 but momentarily goes to 1. Occurs in POS implementations.
- **Cause:** Unequal propagation delays through different paths in the circuit.
- **Detection:** In a K-map for SOP, a static-1 hazard exists between two adjacent 1s if they are not covered by a common prime implicant.
- **Elimination:** Add redundant prime implicants to cover all adjacent pairs of 1s (for static-1 hazards) or 0s (for static-0 hazards). This means adding an extra AND gate (for SOP) or OR gate (for POS).
- **Common Pitfalls:** Forgetting to check for hazards in minimal circuits. Incorrectly adding redundant terms.
- **Problem-solving:** Identify static hazards in a given circuit or K-map. Eliminate hazards by adding redundant terms.

Synchronous Asynchronous Circuits

This distinction refers to how sequential circuits handle timing and state changes.

- **Synchronous Circuits:**
 - All memory elements (flip-flops) are controlled by a single, common clock signal.
 - State changes occur only at discrete clock edges.
 - Easier to design and debug due to predictable timing.
 - Less prone to race conditions and hazards.
 - Examples: Synchronous counters, shift registers, FSMs.
- **Asynchronous Circuits:**
 - Memory elements are not controlled by a common clock.
 - State changes occur in response to changes in input signals, without a clock.
 - Can be faster (no clock propagation delay) and consume less power.
 - Much harder to design and verify due to potential for race conditions and hazards.
 - Examples: Asynchronous (ripple) counters, latches, arbiters.
- **Common Pitfalls:** Confusing the characteristics and trade-offs of each type. Incorrectly identifying a circuit as synchronous or asynchronous.
- **Problem-solving:** Understand the timing behavior of both types of circuits.

Quick Formula Reference

- **Half Adder:**
 - $S = A \oplus B$
 - $C_{out} = A \cdot B$
- **Full Adder:**
 - $S = A \oplus B \oplus C_{in}$
 - $C_{out} = (A \cdot B) + (C_{in} \cdot (A \oplus B))$
 - $C_{out} = (A \cdot B) + (B \cdot C_{in}) + (C_{in} \cdot A)$
- **Carry Look-Ahead Adder (stage i):**
 - $P_i = A_i \oplus B_i$
 - $G_i = A_i \cdot B_i$

- $S_i = P_i \oplus C_i$
- $C_{i+1} = G_i + (P_i \cdot C_i)$
- **Binary to Gray Code:** $G_i = B_i \oplus B_{i+1}$ (with $B_n = 0$ for MSB, or $G_{MSB} = B_{MSB}$, $G_i = B_i \oplus B_{i+1}$ for $i < MSB$).
- **Gray to Binary Code:** $B_i = G_i \oplus B_{i+1}$ (with $B_{MSB} = G_{MSB}$).
- **De Morgan's Laws:**
 - $\overline{A + B} = \bar{A} \cdot \bar{B}$
 - $\overline{A \cdot B} = \bar{A} + \bar{B}$
- **Absorption Law:**
 - $A + (A \cdot B) = A$
 - $A \cdot (A + B) = A$
- **Consensus Theorem:** $(A \cdot B) + (\bar{A} \cdot C) + (B \cdot C) = (A \cdot B) + (\bar{A} \cdot C)$
- **SR Flip-Flop Characteristic Equation:** $Q_{next} = S + \bar{R}Q$ (with $SR = 0$)
- **JK Flip-Flop Characteristic Equation:** $Q_{next} = J\bar{Q} + \bar{K}Q$
- **D Flip-Flop Characteristic Equation:** $Q_{next} = D$
- **T Flip-Flop Characteristic Equation:** $Q_{next} = T \oplus Q$
- **IEEE 754 Single Precision (32-bit):** $(-1)^S \times (1.M)_{base2} \times 2^{(E-127)}$
- **IEEE 754 Double Precision (64-bit):** $(-1)^S \times (1.M)_{base2} \times 2^{(E-1023)}$
- **Multiplexer (2-to-1):** $Output = \bar{S}I_0 + SI_1$
- **Ripple Counter Delay (n-bit):** $T_{total} = n \times T_{FF_delay}$
- **Memory Capacity (N address, M data):** $2^N \times M$ bits

Important Tips for GATE

1. **Master Boolean Algebra and K-Maps:** These are fundamental. Practice simplification extensively for both SOP and POS forms. Be quick and accurate in identifying prime implicants and essential prime implicants.
2. **Understand Number Representations Thoroughly:** Pay close attention to signed magnitude, 1's complement, and especially 2's complement. Practice conversions, arithmetic operations (addition, subtraction), and overflow detection for different bit lengths. IEEE 754 floating-point is a recurring topic; know the format, bias, and special values.
3. **Differentiate Combinational vs. Sequential Circuits:** Clearly understand their definitions, characteristics, and design methodologies. Be able to identify memory elements (flip-flops, latches) that make a circuit sequential.
4. **Memorize Flip-Flop Characteristic and Excitation Tables:** For SR, JK, D, and T flip-flops, these tables are crucial for analyzing and designing sequential circuits (counters, shift registers, FSMs). Don't confuse them.
5. **Practice FSM Design:** Be comfortable with state diagrams, state tables, state assignment, and deriving next-state and output equations using K-maps. Pay attention to Mealy vs. Moore machine characteristics.
6. **Beware of "Minimum Number of Gates" Questions:** These often require not just Boolean simplification but also considering universal gate implementations (NAND-only or NOR-only logic) and gate fan-in/fan-out constraints.
7. **Focus on Timing and Delays:** Understand propagation delays in combinational circuits (e.g., adders) and sequential circuits (e.g., ripple counters). Be able to calculate critical path delays and maximum operating frequencies.
8. **Don't Overlook Hazards:** Static hazards are a common trap. Learn how to detect them using K-maps and eliminate them by adding redundant terms.

4.1

Adder (9)

 **Practice Test:** Test 1 (12Q)

4.1.1 Adder: GATE CSE 1988 | Question: 4ii



Using binary full adders and other logic gates (if necessary), design an adder for adding 4-bit number (including sign) in 2's complement notation.

gate1988 digital-logic descriptive adder

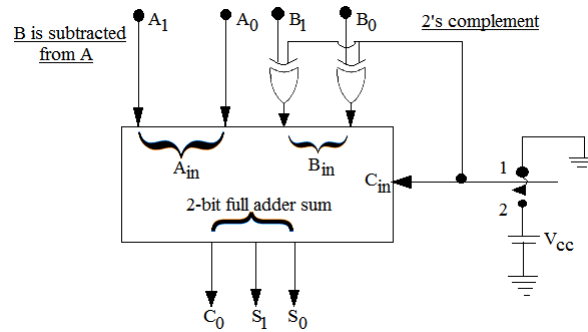
Answer key 

4.1.2 Adder: GATE CSE 1990 | Question: 1-i



Fill in the blanks:

In the two bit full-adder/subtractor unit shown in below figure, when the switch is in position 2 _____ using _____ arithmetic.



gate1990 digital-logic adder fill-in-the-blanks

Answer key

4.1.3 Adder: GATE CSE 1997 | Question: 2.5

An N -bit carry lookahead adder, where N is a multiple of 4, employs ICs 74181 (4 bit ALU) and 74182 (4 bit carry lookahead generator).

The minimum addition time using the best architecture for this adder is

- A. proportional to N
- B. proportional to $\log N$
- C. a constant
- D. None of the above

gate1997 digital-logic normal adder

Answer key

4.1.4 Adder: GATE CSE 1999 | Question: 2.16

The number of full and half-adders required to add 16-bit numbers is

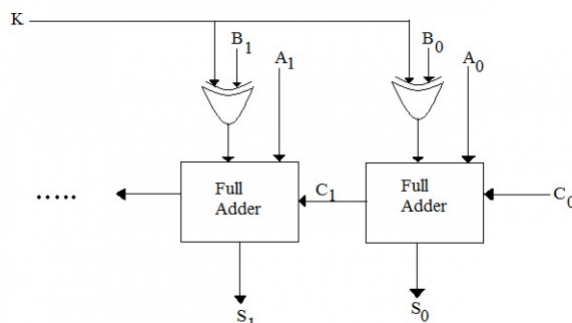
- A. 8 half-adders, 8 full-adders
- B. 1 half-adder, 15 full-adders
- C. 16 half-adders, 0 full-adders
- D. 4 half-adders, 12 full-adders

gate1999 digital-logic normal adder

Answer key

4.1.5 Adder: GATE CSE 2003 | Question: 46

Consider the ALU shown below.



If the operands are in 2 's complement representation, which of the following operations can be performed by suitably setting the control lines K and C_0 only (+ and – denote addition and subtraction respectively)?

- A. $A + B$, and $A - B$, but not $A + 1$
- B. $A + B$, and $A + 1$, but not $A - B$
- C. $A + B$, but not $A - B$ or $A + 1$
- D. $A + B$, and $A - B$, and $A + 1$

gatecse-2003 digital-logic normal adder

Answer key

4.1.6 Adder: GATE CSE 2004 | Question: 62



A 4-bit carry look ahead adder, which adds two 4-bit numbers, is designed using AND, OR, NOT, NAND, NOR gates only. Assuming that all the inputs are available in both complemented and uncomplemented forms and the delay of each gate is one time unit, what is the overall propagation delay of the adder? Assume that the carry network has been implemented using two-level AND-OR logic.

- A. 4 time units
- B. 6 time units
- C. 10 time units
- D. 12 time units

gatecse-2004 digital-logic normal adder

Answer key

4.1.7 Adder: GATE CSE 2015 | Set 2 | Question: 48



A half adder is implemented with XOR and AND gates. A full adder is implemented with two half adders and one OR gate. The propagation delay of an XOR gate is twice that of an AND/OR gate. The propagation delay of an AND/OR gate is 1.2 microseconds. A 4-bit-ripple-carry binary adder is implemented by using four full adders. The total propagation time of this 4-bit binary adder in microseconds is _____.

gatecse-2015-set2 digital-logic adder normal numerical-answers

Answer key

4.1.8 Adder: GATE CSE 2016 | Set 1 | Question: 33



Consider a carry look ahead adder for adding two n -bit integers, built using gates of fan-in at most two. The time to perform addition using this adder is

- A. $\Theta(1)$
- B. $\Theta(\log(n))$
- C. $\Theta(\sqrt{n})$
- D. $\Theta(n)$

gatecse-2016-set1 digital-logic adder normal

Answer key

4.1.9 Adder: GATE CSE 2016 | Set 2 | Question: 07



Consider an eight-bit ripple-carry adder for computing the sum of A and B , where A and B are integers represented in 2's complement form. If the decimal value of A is one, the decimal value of B that leads to the longest latency for the sum to stabilize is _____

gatecse-2016-set2 digital-logic adder normal numerical-answers

Answer key

4.2 Array Multiplier (2)

4.2.1 Array Multiplier: GATE CSE 1999 | Question: 1.21



The maximum gate delay for any output to appear in an array multiplier for multiplying two n bit numbers is

- A. $O(n^2)$
- B. $O(n)$
- C. $O(\log n)$
- D. $O(1)$

gate1999 digital-logic normal array-multiplier

Answer key

4.2.2 Array Multiplier: GATE CSE 2003 | Question: 11



Consider an array multiplier for multiplying two n bit numbers. If each gate in the circuit has a unit delay, the total delay of the multiplier is

- A. $\Theta(1)$
- B. $\Theta(\log n)$
- C. $\Theta(n)$
- D. $\Theta(n^2)$

gatecse-2003 digital-logic normal array-multiplier

Answer key

4.3.1 Binary Codes: GATE CSE 2006 | Question: 40



Consider numbers represented in 4-bit Gray code. Let $h_3h_2h_1h_0$ be the Gray code representation of a number n and let $g_3g_2g_1g_0$ be the Gray code of $(n+1)(\text{modulo } 16)$ value of the number. Which one of the following functions is correct?

- A. $g_0(h_3h_2h_1h_0) = \sum(1, 2, 3, 6, 10, 13, 14, 15)$
 B. $g_1(h_3h_2h_1h_0) = \sum(4, 9, 10, 11, 12, 13, 14, 15)$
 C. $g_2(h_3h_2h_1h_0) = \sum(2, 4, 5, 6, 7, 12, 13, 15)$
 D. $g_3(h_3h_2h_1h_0) = \sum(0, 1, 6, 7, 10, 11, 12, 13)$

gatecse-2006 digital-logic number-representation binary-codes normal

Answer key

Practice Tests: [Test 1 \(15Q\)](#) [Test 2 \(15Q\)](#) [Test 3 \(15Q\)](#) [Test 4 \(1Q\)](#) [Weekly Quiz 1 \(15Q\)](#) [Weekly Quiz 2 \(10Q\)](#)
[Weekly Quiz 3 \(15Q\)](#) [Weekly Quiz 4 \(15Q\)](#)

4.4.1 Boolean Algebra: GATE CSE 1987 | Question: 1-II



The total number of Boolean functions which can be realised with four variables is:

- A. 4 B. 17 C. 256 D. 65,536

gate1987 digital-logic boolean-algebra functions combinatory

Answer key

4.4.2 Boolean Algebra: GATE CSE 1987 | Question: 12-a



The Boolean expression $A \oplus B \oplus A$ is equivalent to

- A. $AB + \bar{A} \bar{B}$ B. $\bar{A} B + A \bar{B}$
 C. B D. $\bar{A}$

gate1987 digital-logic boolean-algebra easy

Answer key

4.4.3 Boolean Algebra: GATE CSE 1988 | Question: 2-iii



Let $*$ be defined as a Boolean operation given as $x * y = \bar{x} \bar{y} + xy$ and let $C = A * B$. If $C = 1$ then prove that $A = B$.

gate1988 digital-logic descriptive boolean-algebra

Answer key

4.4.4 Boolean Algebra: GATE CSE 1989 | Question: 4-x



A switching function is said to be neutral if the number of input combinations for which its value is 1 is equal to the number of input combinations for which its value is 0. Compute the number of neutral switching functions of n variables (for a given n).

gate1989 descriptive digital-logic boolean-algebra

Answer key

4.4.5 Boolean Algebra: GATE CSE 1989 | Question: 5-a



Find values of Boolean variables A, B, C which satisfy the following equations:

- $A + B = 1$
- $AC = BC$

- $A + C = 1$
- $AB = 0$

gate1989 descriptive digital-logic boolean-algebra

Answer key

4.4.6 Boolean Algebra: GATE CSE 1992 | Question: 02-i



The operation which is commutative but not associative is:

- A. AND B. OR C. EX-OR D. NAND

gate1992 easy digital-logic boolean-algebra multiple-selects

Answer key

4.4.7 Boolean Algebra: GATE CSE 1994 | Question: 4



A. Let $*$ be a Boolean operation defined as $A * B = AB + \overline{A}\overline{B}$. If $C = A * B$ then evaluate and fill in the blanks:

- $A * A = \underline{\hspace{2cm}}$
- $C * A = \underline{\hspace{2cm}}$

B. Solve the following boolean equations for the values of A, B and C :

$$\begin{aligned} AB + \overline{A}C &= 1 \\ AC + B &= 0 \end{aligned}$$

gate1994 digital-logic normal boolean-algebra descriptive

Answer key

4.4.8 Boolean Algebra: GATE CSE 1995 | Question: 2.5



What values of A, B, C and D satisfy the following simultaneous Boolean equations?

$$\overline{A} + AB = 0, AB = AC, AB + \overline{A}C + CD = \overline{C}D$$

- A. $A = 1, B = 0, C = 0, D = 1$ B. $A = 1, B = 1, C = 0, D = 0$
 C. $A = 1, B = 0, C = 1, D = 1$ D. $A = 1, B = 0, C = 0, D = 0$

gate1995 digital-logic boolean-algebra easy

Answer key

4.4.9 Boolean Algebra: GATE CSE 1997 | Question: 2-1



Let $*$ be defined as $x * y = \overline{x} + y$. Let $z = x * y$. Value of $z * x$ is

- A. $\overline{x} + y$ B. x C. 0 D. 1

gate1997 digital-logic normal boolean-algebra

Answer key

4.4.10 Boolean Algebra: GATE CSE 1998 | Question: 1.13



What happens when a bit-string is XORed with itself n -times as shown:

$$[B \oplus (B \oplus (B \oplus (B \dots n \text{ times})))]$$

- A. complements when n is even B. complements when n is odd
 C. divides by 2^n always D. remains unchanged when n is even

gate1998 digital-logic normal boolean-algebra

Answer key

4.4.11 Boolean Algebra: GATE CSE 1998 | Question: 2.8



Which of the following operations is commutative but not associative?

- A. AND B. OR C. NAND D. EXOR

gate1998 digital-logic easy boolean-algebra

Answer key

4.4.12 Boolean Algebra: GATE CSE 1999 | Question: 1.7



Which of the following expressions is not equivalent to $\bar{x}$?

- A. $x \text{ NAND } x$ B. $x \text{ NOR } x$ C. $x \text{ NAND } 1$ D. $x \text{ NOR } 1$

gate1999 digital-logic easy boolean-algebra

Answer key

4.4.13 Boolean Algebra: GATE CSE 2000 | Question: 2.10



The simultaneous equations on the Boolean variables x, y, z and w ,

- $x + y + z = 1$
- $xy = 0$
- $xz + w = 1$
- $xy + \bar{z}\bar{w} = 0$

have the following solution for x, y, z and w , respectively:

- A. 0 1 0 0 B. 1 1 0 1
C. 1 0 1 1 D. 1 0 0 0

gatecse-2000 digital-logic boolean-algebra easy

Answer key

4.4.14 Boolean Algebra: GATE CSE 2002 | Question: 2-3



Let $f(A, B) = A' + B$. Simplified expression for function $f(f(x + y, y), z)$ is

- A. $x' + z$ B. xyz C. $xy' + z$ D. None of the above

gatecse-2002 digital-logic boolean-algebra normal

Answer key

4.4.15 Boolean Algebra: GATE CSE 2004 | Question: 17



A Boolean function $x'y' + xy + x'y$ is equivalent to

- A. $x' + y'$ B. $x + y$ C. $x + y'$ D. $x' + y$

gatecse-2004 digital-logic easy boolean-algebra

Answer key

4.4.16 Boolean Algebra: GATE CSE 2007 | Question: 32



Let $f(w, x, y, z) = \sum (0, 4, 5, 7, 8, 9, 13, 15)$. Which of the following expressions are NOT equivalent to f ?

P: $x'y'z' + w'xy' + wy'z + xz$

Q: $w'y'z' + wx'y' + xz$

R: $w'y'z' + wx'y' + xyz + xy'z$

S: $x'y'z' + wx'y' + w'y$

- A. P only B. Q and S C. R and S D. S only

gatecse-2007 digital-logic normal boolean-algebra

Answer key

4.4.17 Boolean Algebra: GATE CSE 2007 | Question: 33



Define the connective $*$ for the Boolean variables X and Y as:

$$X * Y = XY + X'Y'$$

Let $Z = X * Y$. Consider the following expressions P , Q and R .

$$\begin{aligned}P &: X = Y * Z, \\Q &: Y = X * Z, \\R &: X * Y * Z = 1\end{aligned}$$

Which of the following is **TRUE**?

- A. Only P and Q are valid. B. Only Q and R are valid.
C. Only P and R are valid. D. All P , Q , R are valid.

gatecse-2007 digital-logic normal boolean-algebra

Answer key

4.4.18 Boolean Algebra: GATE CSE 2008 | Question: 26



If P, Q, R are Boolean variables, then

$(P + \bar{Q})(P \cdot \bar{Q} + P \cdot R)(\bar{P} \cdot \bar{R} + \bar{Q})$ simplifies to

- A. $P \cdot \bar{Q}$ B. $P \cdot \bar{R}$ C. $P \cdot \bar{Q} + R$ D. $P \cdot \bar{R} + Q$

gatecse-2008 easy digital-logic boolean-algebra

Answer key

4.4.19 Boolean Algebra: GATE CSE 2012 | Question: 6



The truth table

X	Y	(X,Y)
0	0	0
0	1	0
1	0	1
1	1	1

represents the Boolean function

- A. X B. $X + Y$ C. $X \oplus Y$ D. Y

gatecse-2012 digital-logic easy boolean-algebra

Answer key

4.4.20 Boolean Algebra: GATE CSE 2013 | Question: 21



Which one of the following expressions does **NOT** represent exclusive NOR of x and y ?

- A. $xy + x'y'$ B. $x \oplus y'$
C. $x' \oplus y$ D. $x' \oplus y'$

gatecse-2013 digital-logic easy boolean-algebra

Answer key

4.4.21 Boolean Algebra: GATE CSE 2014 | Set 3 | Question: 55



Let $\oplus$ denote the exclusive OR (XOR) operation. Let '1' and '0' denote the binary constants. Consider the following Boolean expression for F over two variables P and Q :

$$F(P, Q) = ((1 \oplus P) \oplus (P \oplus Q)) \oplus ((P \oplus Q) \oplus (Q \oplus 0))$$

The equivalent expression for F is

A. $P + Q$

C. $P \oplus Q$

B. $\overline{P + Q}$

D. $\overline{P \oplus Q}$

gatecse-2014-set3 digital-logic normal boolean-algebra

Answer key

4.4.22 Boolean Algebra: GATE CSE 2015 | Set 2 | Question: 37



The number of min-terms after minimizing the following Boolean expression is _____.

$$[D' + AB' + A'C + AC'D + A'C'D']'$$

gatecse-2015-set2 digital-logic boolean-algebra normal numerical-answers

Answer key

4.4.23 Boolean Algebra: GATE CSE 2016 | Set 1 | Question: 06



Consider the Boolean operator $\#$ with the following properties :

$x\#0 = x, x\#1 = \bar{x}, x\#x = 0$ and $x\#\bar{x} = 1$. Then $x\#y$ is equivalent to

A. $x\bar{y} + \bar{x}y$

C. $\bar{x}y + xy$

B. $x\bar{y} + \bar{x}\bar{y}$

D. $xy + \bar{x}\bar{y}$

gatecse-2016-set1 digital-logic boolean-algebra easy

Answer key

4.4.24 Boolean Algebra: GATE CSE 2016 | Set 2 | Question: 08



Let, $x_1 \oplus x_2 \oplus x_3 \oplus x_4 = 0$ where x_1, x_2, x_3, x_4 are Boolean variables, and $\oplus$ is the XOR operator.

Which one of the following must always be **TRUE**?

A. $x_1x_2x_3x_4 = 0$

B. $x_1x_3 + x_2 = 0$

C. $\bar{x}_1 \oplus \bar{x}_3 = \bar{x}_2 \oplus \bar{x}_4$

D. $x_1 + x_2 + x_3 + x_4 = 0$

gatecse-2016-set2 digital-logic boolean-algebra normal

Answer key

4.4.25 Boolean Algebra: GATE CSE 2017 | Set 2 | Question: 27



If w, x, y, z are Boolean variables, then which one of the following is **INCORRECT**?

A. $wx + w(x + y) + x(x + y) = x + wy$

C. $(w\bar{x}(y + x\bar{z}) + \bar{w}\bar{x})y = x\bar{y}$

B. $\overline{w\bar{x}(y + \bar{z})} + \bar{w}x = \bar{w} + x + \bar{y}z$

D. $(w + y)(wxy + wyz) = wxy + wyz$

gatecse-2017-set2 digital-logic boolean-algebra normal

Answer key

4.4.26 Boolean Algebra: GATE CSE 2018 | Question: 4



Let $\oplus$ and $\odot$ denote the Exclusive OR and Exclusive NOR operations, respectively. Which one of the following is **NOT CORRECT**?

A. $\overline{P \oplus Q} = P \odot Q$

B. $\overline{P \oplus Q} = P \odot Q$

C. $\overline{P \oplus Q} = P \oplus Q$

D. $P \oplus \overline{P \oplus Q} = (P \odot \overline{P \odot Q})$

gatecse-2018 digital-logic normal boolean-algebra one-mark

Answer key

4.4.27 Boolean Algebra: GATE CSE 2019 | Question: 6



Which one of the following is NOT a valid identity?

- A. $(x \oplus y) \oplus z = x \oplus (y \oplus z)$
 B. $(x + y) \oplus z = x \oplus (y + z)$
 C. $x \oplus y = x + y$, if $xy = 0$
 D. $x \oplus y = (xy + x'y')'$

gatecse-2019 digital-logic boolean-algebra one-mark

Answer key

4.4.28 Boolean Algebra: GATE CSE 2021 | Set 1 | Question: 42



Consider the following Boolean expression.

$$F = (X + Y + Z)(\bar{X} + Y)(\bar{Y} + Z)$$

Which of the following Boolean expressions is/are equivalent to $\bar{F}$ (complement of F)?

- A. $(\bar{X} + \bar{Y} + \bar{Z})(X + \bar{Y})(Y + \bar{Z})$
 B. $X\bar{Y} + \bar{Z}$
 C. $(X + \bar{Z})(\bar{Y} + \bar{Z})$
 D. $X\bar{Y} + Y\bar{Z} + \bar{X}\bar{Y}\bar{Z}$

gatecse-2021-set1 multiple-selects digital-logic boolean-algebra two-marks

Answer key

4.4.29 Boolean Algebra: GATE CSE 2024 | Set 2 | Question: 20



For a Boolean variable x , which of the following statements is/are FALSE?

- A. $x.1 = x$ B. $x + 1 = x$ C. $x \cdot x = 0$ D. $x + \bar{x} = 1$

gatecse-2024-set2 digital-logic boolean-algebra easy multiple-selects one-mark

Answer key

4.4.30 Boolean Algebra: GATE CSE 2025 | Set 1 | Question: 14



Let X be a 3-variable Boolean function that produces output as '1' when at least two of the input variables are '1'. Which of the following statement(s) is/are CORRECT, where a, b, c, d, e are Boolean variables?

- A. $X(a, b, X(c, d, e)) = X(X(a, b, c), d, e)$
 B. $X(a, b, X(a, b, c)) = X(a, b, c)$
 C. $X(a, b, X(a, c, d)) = (X(a, b, a) \text{ AND } X(c, d, c))$
 D. $X(a, b, c) = X(a, X(a, b, c), X(a, c, c))$

gatecse2025-set1 digital-logic boolean-algebra multiple-selects one-mark

Answer key

4.4.31 Boolean Algebra: GATE CSE 2025 | Set 2 | Question: 40



Which of the following Boolean algebraic equation(s) is/are CORRECT?

- A. $\bar{A}BC + A\bar{B}\bar{C} + \bar{A}\bar{B}C + A\bar{B}C + ABC = BC + \bar{B}\bar{C} + \bar{A}\bar{B}$
 B. $AB + \bar{A}C + BC = AB + \bar{A}C$
 C. $(A + C)(\bar{A} + B) = AB + \bar{A}C$
 D. $(A + \bar{B} + \bar{D})(C + D)(\bar{A} + C + D)(A + B + \bar{D}) = \bar{A}D + \bar{C}\bar{D}$

gatecse2025-set2 digital-logic boolean-algebra multiple-selects easy two-marks

Answer key

4.4.32 Boolean Algebra: GATE CSE 2026 | Set 2 | Question: 6



Which one of the following options is **not** a property of Boolean Algebra?

Note: $+$ is OR operation, $\cdot$ is AND operation, and $'$ is NOT operation

- A. $a + b = b + a$ B. $a \cdot a' = 1$
C. $a + a' = 1$ D. $a \cdot b = b \cdot a$

gatecse-2026-set2 digital-logic boolean-algebra easy one-mark

Answer key

4.4.33 Boolean Algebra: GATE IT 2004 | Question: 44



The function $\bar{A}\bar{B}C + \bar{A}BC + AB\bar{C} + \bar{A}\bar{B}C + \bar{A}\bar{B}\bar{C}$ is equivalent to

- A. $\bar{A}\bar{C} + AB + \bar{A}\bar{C}$ B. $\bar{A}\bar{B} + \bar{A}\bar{C} + \bar{A}\bar{C}$
C. $\bar{A}\bar{B} + \bar{A}\bar{C} + \bar{A}\bar{B}$ D. $\bar{A}\bar{B} + \bar{A}\bar{C} + \bar{A}\bar{B}$

gateit-2004 digital-logic boolean-algebra easy

Answer key

4.4.34 Boolean Algebra: GATE IT 2005 | Question: 7



Which of the following expressions is equivalent to $(A \oplus B) \oplus C$

- A. $(A + B + C)(\bar{A} + \bar{B} + \bar{C})$ B. $(A + B + C)(\bar{A} + \bar{B} + C)$
C. $ABC + \bar{A}(B \oplus C) + \bar{B}(A \oplus C)$ D. None of these

gateit-2005 digital-logic normal boolean-algebra

Answer key

4.5

Booths Algorithm (7)

Practice Test: Test 1 (6Q)

4.5.1 Booths Algorithm: GATE CSE 1990 | Question: 8b



State the Booth's algorithm for multiplication of two numbers. Draw a block diagram for the implementation of the Booth's algorithm for determining the product of two 8-bit signed numbers.

gate1990 descriptive digital-logic booths-algorithm

Answer key

4.5.2 Booths Algorithm: GATE CSE 1996 | Question: 1.23



Booth's algorithm for integer multiplication gives worst performance when the multiplier pattern is

- A. 101010...1010 B. 100000...0001
C. 111111...1111 D. 011111...1110

gate1996 digital-logic booths-algorithm normal

Answer key

4.5.3 Booths Algorithm: GATE CSE 1999 | Question: 1.20



Booth's coding in 8 bits for the decimal number -57 is:

- A. $0 - 100 + 1000$ B. $0 - 100 + 100 - 1$ C. $0 - 1 + 100 - 10 + 1$ D. $00 - 10 + 100 - 1$

gate1999 digital-logic number-representation booths-algorithm normal

Answer key

4.5.4 Booths Algorithm: GATE CSE 2025 | Set 2 | Question: 22



The following two signed 2's complement numbers (multiplicand M and multiplier Q) are being multiplied using Booth's algorithm:

M: 1100110111101101 and Q: 1010010010101010

The total number of addition and subtraction operations to be performed is _____. (Answer in integer)

gatecse2025-set2 digital-logic booths-algorithm numerical-answers one-mark

Answer key

4.5.5 Booths Algorithm: GATE IT 2005 | Question: 8



Using Booth's Algorithm for multiplication, the multiplier -57 will be recoded as

- A. 0-1 00 1 00 -1 B. 1 1 0 0 0 1 1 1
C. 0-1 0 0 1 0 0 0 D. 0 1 0 0 -1 0 0 1

gateit-2005 digital-logic booths-algorithm normal

Answer key

4.5.6 Booths Algorithm: GATE IT 2006 | Question: 38



When multiplicand Y is multiplied by multiplier $X = x_{n-1}x_{n-2} \dots x_0$ using bit-pair recoding in Booth's algorithm, partial products are generated according to the following table.

Row	x_{i+1}	x_i	x_{i-1}	Partial Product
1	0	0	0	0
2	0	0	1	Y
3	0	1	0	Y
4	0	1	1	$2Y$
5	1	0	0	?
6	1	0	1	$-Y$
7	1	1	0	$-Y$
8	1	1	1	?

The partial products for rows 5 and 8 are

- A. $2Y$ and Y B. $-2Y$ and $2Y$ C. $-2Y$ and 0 D. 0 and Y

gateit-2006 digital-logic booths-algorithm difficult

Answer key

4.5.7 Booths Algorithm: GATE IT 2008 | Question: 42



The two numbers given below are multiplied using the Booth's algorithm.

Multiplicand : 0101 1010 1110 1110

Multiplier: 0111 0111 1011 1101

How many additions/Subtractions are required for the multiplication of the above two numbers?

- A. 6 B. 8 C. 10 D. 12

gateit-2008 digital-logic booths-algorithm normal

Answer key

4.6

Canonical Normal Form (10)

Practice Test: Test 1 (11Q)

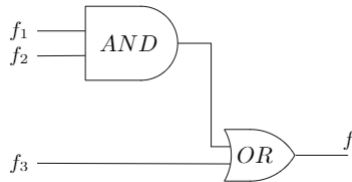
4.6.1 Canonical Normal Form: GATE CSE 1988 | Question: 2-v



Three switching functions f_1 , f_2 and f_3 are expressed below as sum of minterms.

- $f_1(w, x, y, z) = \sum 0, 1, 2, 3, 5, 12$
- $f_2(w, x, y, z) = \sum 0, 1, 2, 10, 13, 14, 15$
- $f_3(w, x, y, z) = \sum 2, 4, 5, 8$

Express the function f realised by the circuit shown in the below figure as the sum of minterms (in decimal notation).



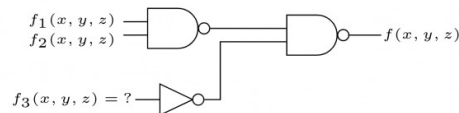
gate1988 descriptive digital-logic easy circuit-output canonical-normal-form digital-circuits

Answer key

4.6.2 Canonical Normal Form: GATE CSE 2002 | Question: 2-1



Consider the following logic circuit whose inputs are functions f_1 , f_2 , f_3 and output is f



Given that

- $f_1(x, y, z) = \sum(0, 1, 3, 5)$
- $f_2(x, y, z) = \sum(6, 7)$, and
- $f(x, y, z) = \sum(1, 4, 5)$.

f_3 is

- A. $\sum(1, 4, 5)$
- B. $\sum(6, 7)$
- C. $\sum(0, 1, 3, 5)$
- D. None of the above

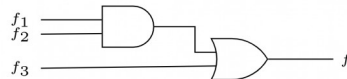
gatecse-2002 digital-logic normal canonical-normal-form circuit-output

Answer key

4.6.3 Canonical Normal Form: GATE CSE 2008 | Question: 8



Given f_1 , f_3 and f in canonical sum of products form (in decimal) for the circuit



$$f_1 = \sum m(4, 5, 6, 7, 8)$$

$$f_3 = \sum m(1, 6, 15)$$

$$f = \sum m(1, 6, 8, 15)$$

then f_2 is

- A. $\sum m(4, 6)$
- B. $\sum m(4, 8)$
- C. $\sum m(6, 8)$
- D. $\sum m(4, 6, 8)$

gatecse-2008 digital-logic canonical-normal-form easy

Answer key

4.6.4 Canonical Normal Form: GATE CSE 2010 | Question: 6



The minterm expansion of $f(P, Q, R) = PQ + QR + PR$ is

- A. $m_2 + m_4 + m_6 + m_7$ B. $m_0 + m_1 + m_3 + m_5$
 C. $m_0 + m_1 + m_6 + m_7$ D. $m_2 + m_3 + m_4 + m_5$

gatecse-2010 digital-logic canonical-normal-form normal

Answer key

4.6.5 Canonical Normal Form: GATE CSE 2015 | Set 3 | Question: 43



The total number of prime implicants of the function $f(w, x, y, z) = \sum(0, 2, 4, 5, 6, 10)$ is _____

gatecse-2015-set3 digital-logic canonical-normal-form normal numerical-answers

Answer key

4.6.6 Canonical Normal Form: GATE CSE 2015 | Set 3 | Question: 44



Given the function $F = P' + QR$, where F is a function in three Boolean variables P, Q and R and $P' = \neg P$, consider the following statements.

- (S1) $F = \sum(4, 5, 6)$
 (S2) $F = \sum(0, 1, 2, 3, 7)$
 (S3) $F = \prod(4, 5, 6)$
 (S4) $F = \prod(0, 1, 2, 3, 7)$

Which of the following is true?

- A. (S1)-False, (S2)-True, (S3)-True, (S4)-False
 B. (S1)-True, (S2)-False, (S3)-False, (S4)-True
 C. (S1)-False, (S2)-False, (S3)-True, (S4)-True
 D. (S1)-True, (S2)-True, (S3)-False, (S4)-False

gatecse-2015-set3 digital-logic canonical-normal-form normal

Answer key

4.6.7 Canonical Normal Form: GATE CSE 2019 | Question: 30



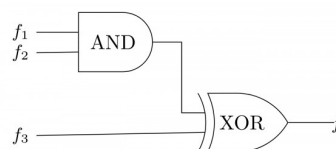
Consider three 4-variable functions f_1, f_2 , and f_3 , which are expressed in sum-of-minterms as

$$f_1 = \sum(0, 2, 5, 8, 14),$$

$$f_2 = \sum(2, 3, 6, 8, 14, 15),$$

$$f_3 = \sum(2, 7, 11, 14)$$

For the following circuit with one AND gate and one XOR gate the output function f can be expressed as:



- A. $\sum(7, 8, 11)$ B. $\sum(2, 7, 8, 11, 14)$
 C. $\sum(2, 14)$ D. $\sum(0, 2, 3, 5, 6, 7, 8, 11, 14, 15)$

gatecse-2019 digital-logic digital-circuits two-marks canonical-normal-form

Answer key

4.6.8 Canonical Normal Form: GATE CSE 2019 | Question: 50



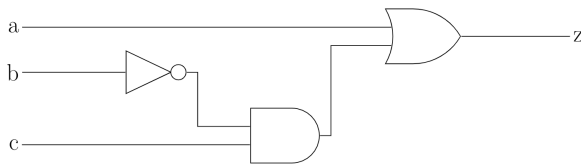
What is the minimum number of 2-input NOR gates required to implement a 4 -variable function expressed in sum-of-minterms form as $f = \sum(0, 2, 5, 7, 8, 10, 13, 15)$? Assume that all the inputs and their complements are available. Answer: _____

Answer key

4.6.9 Canonical Normal Form: GATE CSE 2020 | Question: 28



Consider the Boolean function $z(a, b, c)$.



Which one of the following minterm lists represents the circuit given above?

A. $z = \sum(0, 1, 3, 7)$
 C. $z = \sum(2, 4, 5, 6, 7)$

B. $z = \sum(1, 4, 5, 6, 7)$
 D. $z = \sum(2, 3, 5)$

gatecse-2020 digital-logic canonical-normal-form two-marks

Answer key

4.6.10 Canonical Normal Form: GATE CSE 2024 | Set 2 | Question: 40



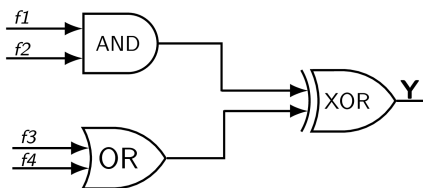
Consider 4-variable functions f_1, f_2, f_3, f_4 expressed in sum-of-minterms form as given below.

$$f_1 = \sum(0, 2, 3, 5, 7, 8, 11, 13)$$

$$f_2 = \sum(1, 3, 5, 7, 11, 13, 15)$$

$$f_3 = \sum(0, 1, 4, 11)$$

$$f_4 = \sum(0, 2, 6, 13)$$



With respect to the circuit given above, which of the following options is/are CORRECT?

A. $Y = \sum(0, 1, 2, 11, 13)$

B. $Y = \Pi(3, 4, 5, 6, 7, 8, 9, 10, 12, 14, 15)$

C. $Y = \sum(0, 1, 2, 3, 4, 5, 6, 7)$

D. $Y = \Pi(8, 9, 10, 11, 12, 13, 14, 15)$

gatecse-2024-set2 digital-logic canonical-normal-form multiple-selects two-marks

Answer key

4.7

Carry Generator (2)

4.7.1 Carry Generator: GATE CSE 2006 | Question: 36



Given two three bit numbers $a_2a_1a_0$ and $b_2b_1b_0$ and c the carry in, the function that represents the carry generate function when these two numbers are added is:

A. $a_2b_2 + a_2a_1b_1 + a_2a_1a_0b_0 + a_2a_0b_1b_0 + a_1b_2b_1 + a_1a_0b_2b_0 + a_0b_2b_1b_0$

B. $a_2b_2 + a_2b_1b_0 + a_2a_1b_1b_0 + a_1a_0b_2b_1 + a_1a_0b_2 + a_1a_0b_2b_0 + a_2a_0b_1b_0$

C. $a_2 + b_2 + (a_2 \oplus b_2)(a_1 + b_1 + (a_1 \oplus b_1) + (a_0 + b_0))$

D. $a_2b_2 + \overline{a_2}a_1b_1 + \overline{a_2}\overline{a_1}a_0b_0 + \overline{a_2}a_0\overline{b_1}b_0 + a_1\overline{b_2}b_1 + \overline{a_1}a_0\overline{b_2}b_0 + a_0\overline{b_2}b_1b_0$

gatecse-2006 digital-logic normal carry-generator adder

Answer key

4.7.2 Carry Generator: GATE CSE 2007 | Question: 35



In a look-ahead carry generator, the carry generate function G_i and the carry propagate function P_i for inputs A_i and B_i are given by:

$$P_i = A_i \oplus B_i \text{ and } G_i = A_i B_i$$

The expressions for the sum bit S_i and the carry bit C_{i+1} of the look ahead carry adder are given by:

$$S_i = P_i \oplus C_i \text{ and } C_{i+1} = G_i + P_i C_i, \text{ where } C_0 \text{ is the input carry.}$$

Consider a two-level logic implementation of the look-ahead carry generator. Assume that all P_i and G_i are available for the carry generator circuit and that the AND and OR gates can have any number of inputs. The number of AND gates and OR gates needed to implement the look-ahead carry generator for a 4-bit adder with S_3, S_2, S_1, S_0 and C_4 as its outputs are respectively:

- A. 6,3 B. 10,4 C. 6,4 D. 10,5

gatecse-2007 digital-logic normal carry-generator adder

Answer key

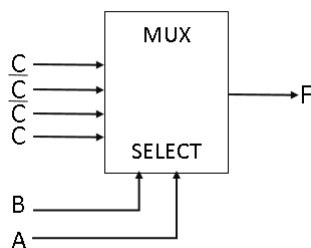
4.8 Circuit Output (40)

Practice Tests: [Test 1 \(15Q\)](#) [Test 2 \(15Q\)](#) [Test 3 \(8Q\)](#)

4.8.1 Circuit Output: GATE CSE 1987 | Question: 1-IV



The output F of the below multiplexer circuit can be represented by



- A. $AB + B\bar{C} + \bar{C}A + \bar{B}\bar{C}$ B. $A \oplus B \oplus C$
C. $A \oplus B$ D. $\bar{A}\bar{B}C + \bar{A}B\bar{C} + A\bar{B}\bar{C}$

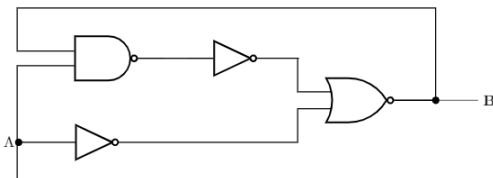
gate1987 digital-logic combinational-circuit multiplexer circuit-output

Answer key

4.8.2 Circuit Output: GATE CSE 1989 | Question: 4-ix



Explain the behaviour of the following logic circuit with level input A and output B .



gate1989 descriptive digital-logic circuit-output

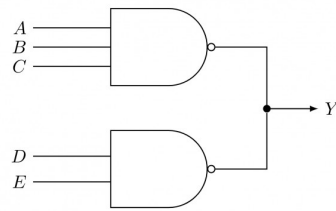
Answer key

4.8.3 Circuit Output: GATE CSE 1990 | Question: 3-i



Choose the correct alternatives (More than one may be correct).

Two NAND gates having open collector outputs are tied together as shown in below figure.



The logic function Y , implemented by the circuit is,

A. $Y = ABC + DE$

B. $Y = \overline{ABC + DE}$

C. $Y = ABC.DE$

D. $Y = \overline{ABC.DE}$

gate1990 normal digital-logic circuit-output

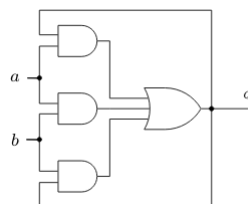
Answer key

4.8.4 Circuit Output: GATE CSE 1991 | Question: 5-a



Analyse the circuit in Fig below and complete the following table

a	b	Q_n
0	0	
0	1	
1	0	
1	1	



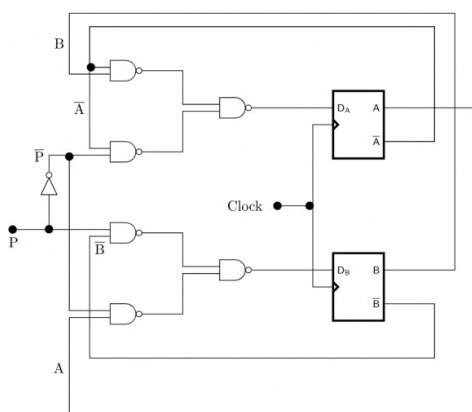
gate1991 digital-logic normal circuit-output sequential-circuit descriptive

Answer key

4.8.5 Circuit Output: GATE CSE 1993 | Question: 19



A control algorithm is implemented by the NAND – gate circuitry given in figure below, where A and B are state variable implemented by D flip-flops, and P is control input. Develop the state transition table for this controller.



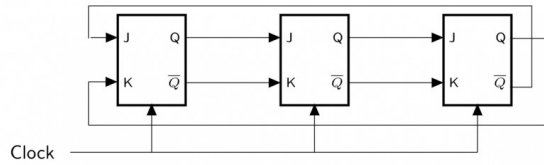
gate1993 digital-logic sequential-circuit flip-flop circuit-output normal descriptive

Answer key

4.8.6 Circuit Output: GATE CSE 1993 | Question: 6-3



For the initial state of 000, the function performed by the arrangement of the J-K flip-flops in figure is:



- A. Shift Register
- B. Mod- 3 Counter
- C. Mod- 6 Counter
- D. Mod- 2 Counter
- E. None of the above

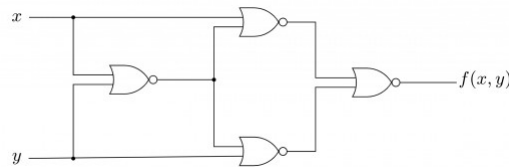
gate1993 digital-logic sequential-circuit flip-flop digital-counter circuit-output multiple-selects

Answer key

4.8.7 Circuit Output: GATE CSE 1993 | Question: 6.1



Identify the logic function performed by the circuit shown in figure.



- A. exclusive OR
- B. exclusive NOR
- C. NAND
- D. NOR
- E. None of the above

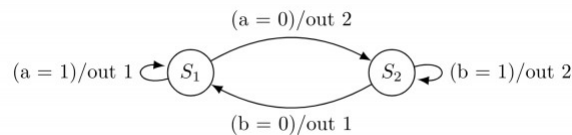
gate1993 digital-logic combinational-circuit circuit-output normal

Answer key

4.8.8 Circuit Output: GATE CSE 1993 | Question: 6.2



If the state machine described in figure should have a stable state, the restriction on the inputs is given by



- A. $a \cdot b = 1$
- B. $a + b = 1$
- C. $\bar{a} + \bar{b} = 0$
- D. $\bar{a} \cdot \bar{b} = 1$
- E. $\overline{a + b} = 1$

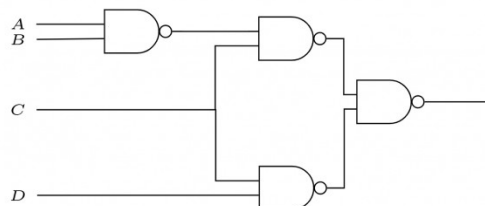
gate1993 digital-logic normal circuit-output sequential-circuit

Answer key

4.8.9 Circuit Output: GATE CSE 1994 | Question: 1.8



The logic expression for the output of the circuit shown in figure below is:



A. $\overline{AC} + \overline{BC} + CD$

B. $\overline{AC} + \overline{BC} + CD$

C. $ABC + \overline{C}\overline{D}$

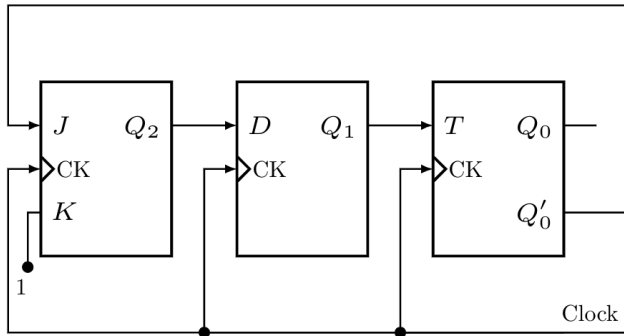
D. $\overline{A}\overline{B} + \overline{B}\overline{C} + CD$

gate1994 digital-logic circuit-output normal

Answer key

4.8.10 Circuit Output: GATE CSE 1994 | Question: 11

Find the contents of the flip-flop Q_2, Q_1 and Q_0 in the circuit of figure, after giving four clock pulses to the clock terminal. Assume $Q_2Q_1Q_0 = 000$ initially.

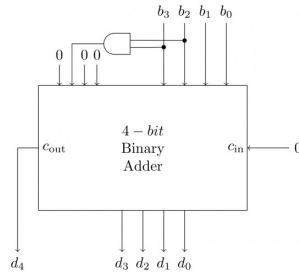


gate1994 digital-logic sequential-circuit digital-counter circuit-output normal descriptive

Answer key

4.8.11 Circuit Output: GATE CSE 1996 | Question: 2.21

Consider the circuit in below figure which has a four bit binary number $b_3b_2b_1b_0$ as input and a five bit binary number, $d_4d_3d_2d_1d_0$ as output.



A. Binary to Hex conversion

B. Binary to BCD conversion

C. Binary to Gray code conversion

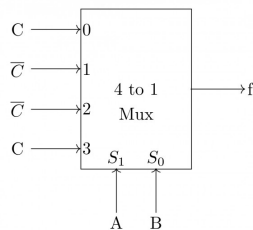
D. Binary to *radix* – 12 conversion

gate1996 digital-logic circuit-output normal

Answer key

4.8.12 Circuit Output: GATE CSE 1996 | Question: 2.22

Consider the circuit in figure. f implements



A. $\overline{ABC} + \overline{ABC} + ABC$

B. $A + B + C$

C. $A \oplus B \oplus C$

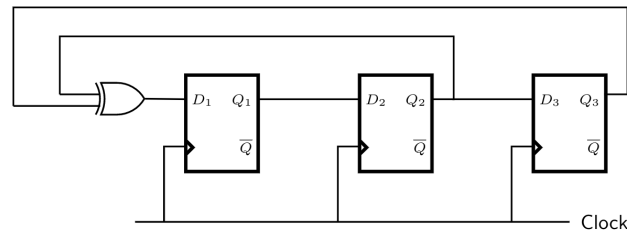
D. $AB + BC + CA$

Answer key

4.8.13 Circuit Output: GATE CSE 1996 | Question: 24-a



Consider the synchronous sequential circuit in the below figure



Draw a state diagram, which is implemented by the circuit. Use the following names for the states corresponding to the values of flip-flops as given below.

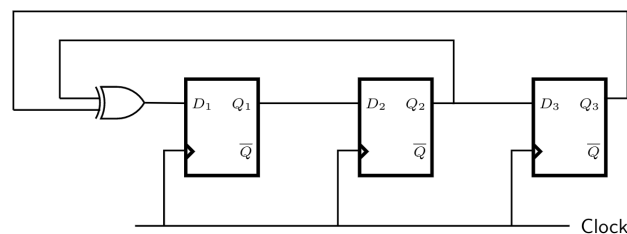
Q1	Q2	Q3	State
0	0	0	S_0
0	0	1	S_1
—	—	—	—
—	—	—	—
—	—	—	—
1	1	1	S_7

Answer key

4.8.14 Circuit Output: GATE CSE 1996 | Question: 24-b



Consider the synchronous sequential circuit in the below figure



Given that the initial state of the circuit is S_4 , identify the set of states, which are not reachable.

Answer key

4.8.15 Circuit Output: GATE CSE 1997 | Question: 5.5

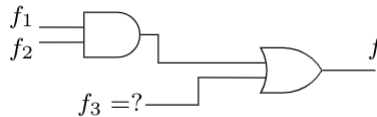


Consider a logic circuit shown in figure below. The functions f_1 , f_2 and f (in canonical sum of products form in decimal notation) are :

$$f_1(w, x, y, z) = \sum 8, 9, 10$$

$$f_2(w, x, y, z) = \sum 7, 8, 12, 13, 14, 15$$

$$f(w, x, y, z) = \sum 8, 9$$



The function f_3 is

- A. $\sum 9, 10$ B. $\sum 9$ C. $\sum 1, 8, 9$ D. $\sum 8, 10, 15$

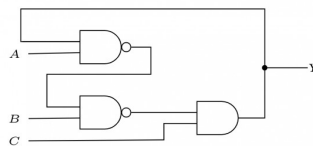
gate1997 digital-logic circuit-output normal

Answer key

4.8.16 Circuit Output: GATE CSE 1999 | Question: 2.8



Consider the circuit shown below. In a certain steady state, the line Y is at '1'. What are the possible values of A, B and C in this state?



- A. $A = 0, B = 0, C = 1$ B. $A = 0, B = 1, C = 1$
C. $A = 1, B = 0, C = 1$ D. $A = 1, B = 1, C = 1$

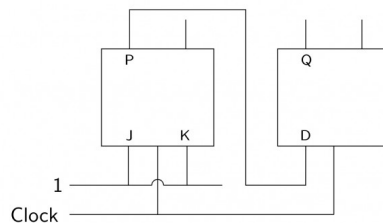
gate1999 digital-logic circuit-output normal

Answer key

4.8.17 Circuit Output: GATE CSE 2000 | Question: 2.12



The following arrangement of master-slave flip flops



has the initial state of P, Q as 0, 1 (respectively). After three clock cycles the output state P, Q is (respectively),

- A. 1, 0 B. 1, 1 C. 0, 0 D. 0, 1

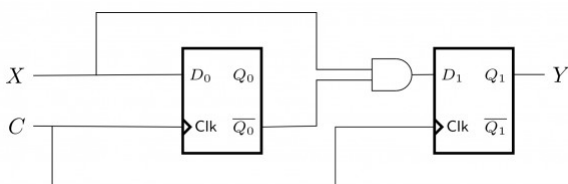
gatecse-2000 digital-logic circuit-output normal flip-flop

Answer key

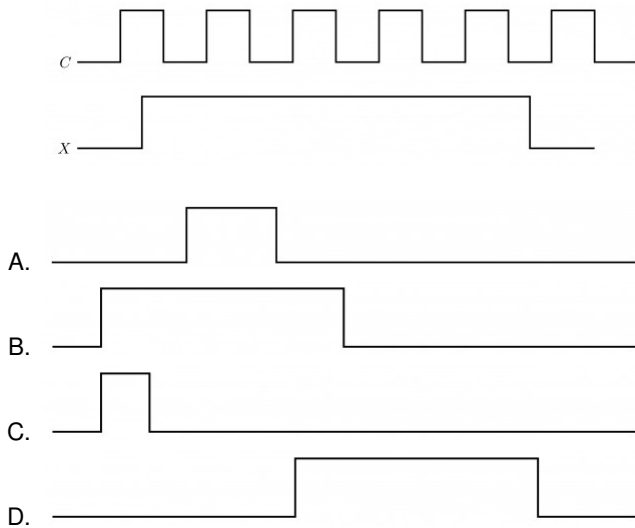
4.8.18 Circuit Output: GATE CSE 2001 | Question: 2.8



Consider the following circuit with initial state $Q_0 = Q_1 = 0$. The D Flip-flops are positive edged triggered and have set up times 20 nanosecond and hold times 0.



Consider the following timing diagrams of X and C . The clock period of $C \geq 40$ nanosecond. Which one is the correct plot of Y ?



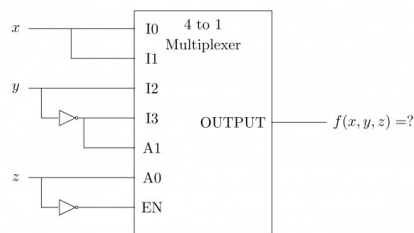
gatecse-2001 digital-logic circuit-output normal

Answer key

4.8.19 Circuit Output: GATE CSE 2002 | Question: 2.2



Consider the following multiplexer where I_0, I_1, I_2, I_3 are four data input lines selected by two address line combinations $A_1A_0 = 00, 01, 10, 11$ respectively and f is the output of the multiplexor. EN is the Enable input.



The function $f(x, y, z)$ implemented by the above circuit is

- A. xyz' B. $xy + z$ C. $x + y$ D. None of the above

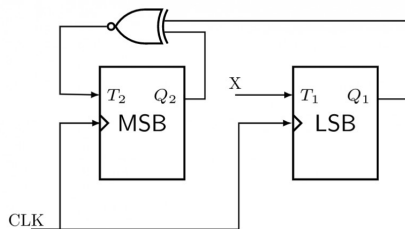
gatecse-2002 digital-logic circuit-output normal

Answer key

4.8.20 Circuit Output: GATE CSE 2004 | Question: 61



Consider the partial implementation of a 2-bit counter using T flip-flops following the sequence $0 - 2 - 3 - 1 - 0$, as shown below.



To complete the circuit, the input X should be

- A. Q_2^c B. $Q_2 + Q_1$ C. $(Q_1 + Q_2)^c$ D. $Q_1 \oplus Q_2$

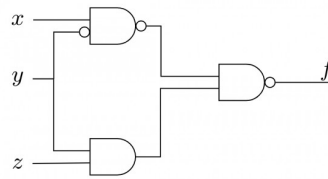
gatecse-2004 digital-logic circuit-output normal

Answer key

4.8.21 Circuit Output: GATE CSE 2005 | Question: 15



Consider the following circuit.



Which one of the following is TRUE?

- A. f is independent of x
- B. f is independent of y
- C. f is independent of z
- D. None of x, y, z is redundant

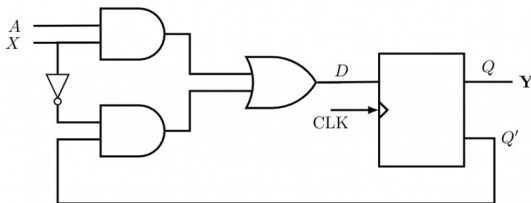
gatecse-2005 digital-logic circuit-output normal

Answer key

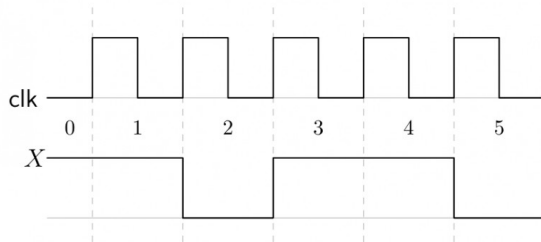
4.8.22 Circuit Output: GATE CSE 2005 | Question: 62



Consider the following circuit involving a positive edge triggered D FF.



Consider the following timing diagram. Let A_i represents the logic level on the line A in the i -th clock period.



Let A' represent the complement of A . The correct output sequence on Y over the clock periods 1 through 5 is:

- A. $A_0 A_1 A'_1 A_3 A_4$
- B. $A_0 A_1 A'_2 A_3 A_4$
- C. $A_1 A_2 A'_2 A_3 A_4$
- D. $A_1 A'_2 A_3 A_4 A'_5$

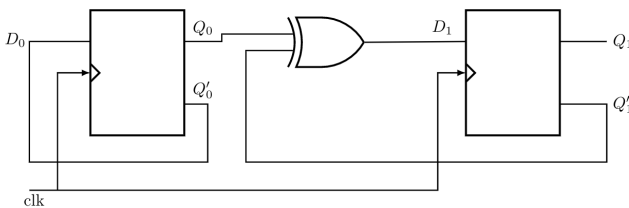
gatecse-2005 digital-logic circuit-output normal

Answer key

4.8.23 Circuit Output: GATE CSE 2005 | Question: 64



Consider the following circuit:

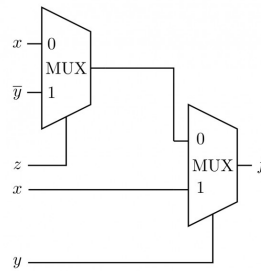


The flip-flops are positive edge triggered D FFs. Each state is designated as a two-bit string $Q_0 Q_1$. Let the initial state be 00. The state transition sequence is

- A. $00 \rightarrow 11 \rightarrow 01$
- B. $00 \rightarrow 11$
- C. $00 \rightarrow 10 \rightarrow 01 \rightarrow 11$
- D. $00 \rightarrow 11 \rightarrow 01 \rightarrow 10$

Answer key

4.8.24 Circuit Output: GATE CSE 2006 | Question: 35



Consider the circuit above. Which one of the following options correctly represents $f(x, y, z)$

- A. $x\bar{z} + xy + \bar{y}z$ B. $x\bar{z} + xy + \bar{y}\bar{z}$
 C. $xz + xy + \bar{y}\bar{z}$ D. $xz + x\bar{y} + \bar{y}z$

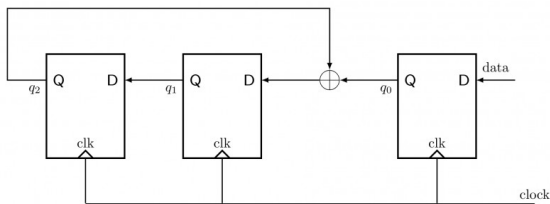
gatecse-2006 digital-logic circuit-output normal

Answer key

4.8.25 Circuit Output: GATE CSE 2006 | Question: 37



Consider the circuit in the diagram. The $\oplus$ operator represents Ex-OR. The D flip-flops are initialized to zeroes (cleared).



The following data: 100110000 is supplied to the "data" terminal in nine clock cycles. After that the values of $q_2q_1q_0$ are:

- A. 000 B. 001 C. 010 D. 101

gatecse-2006 digital-logic circuit-output easy

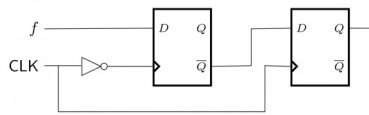
Answer key

4.8.26 Circuit Output: GATE CSE 2006 | Question: 8



You are given a free running clock with a duty cycle of 50% and a digital waveform f which changes only at the negative edge of the clock. Which one of the following circuits (using clocked D flip-flops) will delay the phase of f by 180° ?

- A.
- B.
- C.



D.

gatecse-2006 digital-logic normal circuit-output

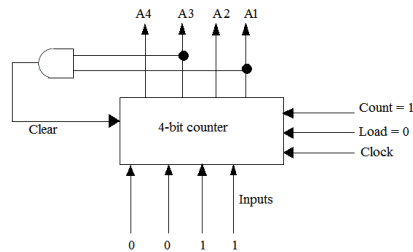
Answer key

4.8.27 Circuit Output: GATE CSE 2007 | Question: 36

The control signal functions of a 4-bit binary counter are given below (where X is “don’t care”):

Clear	Clock	Load	Count	Function
1	X	X	X	Clear to 0
0	X	0	0	No Change
0	↑	1	X	Load Input
0	↑	0	1	Count Next

The counter is connected as follows:



Assume that the counter and gate delays are negligible. If the counter starts at 0, then it cycles through the following sequence:

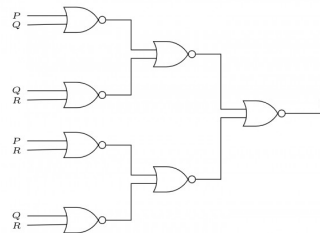
- A. 0,3,4 B. 0,3,4,5 C. 0,1,2,3,4 D. 0,1,2,3,4,5

gatecse-2007 digital-logic circuit-output normal

Answer key

4.8.28 Circuit Output: GATE CSE 2010 | Question: 31

What is the boolean expression for the output f of the combinational logic circuit of NOR gates given below?



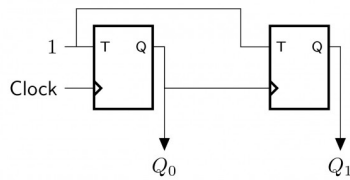
- A. $\overline{Q + R}$ B. $\overline{P + Q}$
 C. $\overline{P + R}$ D. $\overline{P + Q + R}$

gatecse-2010 digital-logic circuit-output normal

Answer key

4.8.29 Circuit Output: GATE CSE 2010 | Question: 32

In the sequential circuit shown below, if the initial value of the output Q_1Q_0 is 00. What are the next four values of Q_1Q_0 ?



- A. 11, 10, 01, 00
C. 10, 00, 01, 11

- B. 10, 11, 01, 00
D. 11, 10, 00, 01

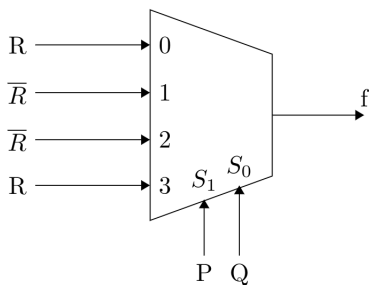
gatecse-2010 digital-logic circuit-output normal

Answer key

4.8.30 Circuit Output: GATE CSE 2010 | Question: 9



The Boolean expression of the output f of the multiplexer shown below is



- A. $\overline{P \oplus Q \oplus R}$
C. $P + Q + R$

- B. $P \oplus Q \oplus R$
D. $\overline{P + Q + R}$

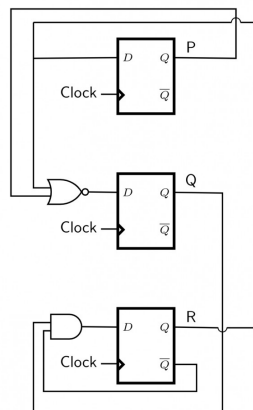
gatecse-2010 digital-logic circuit-output easy multiplexer

Answer key

4.8.31 Circuit Output: GATE CSE 2011 | Question: 50



Consider the following circuit involving three D-type flip-flops used in a certain type of counter configuration.



If at some instance prior to the occurrence of the clock edge, P, Q and R have a value 0, 1 and 0 respectively, what shall be the value of PQR after the clock edge?

- A. 000

- B. 001

- C. 010

- D. 011

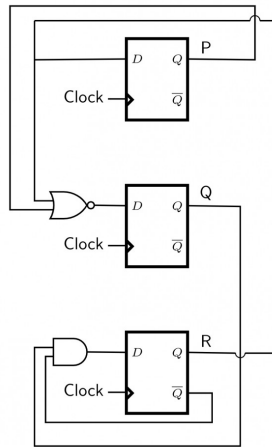
gatecse-2011 digital-logic circuit-output flip-flop normal

Answer key

4.8.32 Circuit Output: GATE CSE 2011 | Question: 51



Consider the following circuit involving three D-type flip-flops used in a certain type of counter configuration.



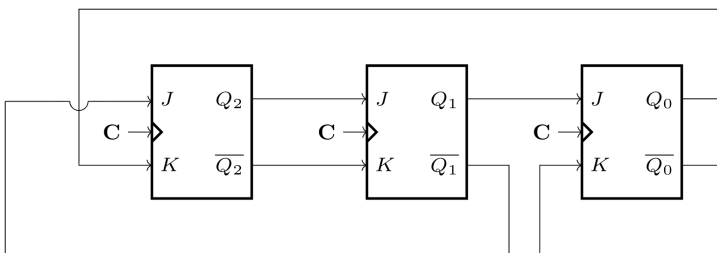
If all the flip-flops were reset to 0 at power on, what is the total number of distinct outputs (states) represented by PQR generated by the counter?

- A. 3 B. 4 C. 5 D. 6

gatecse-2011 digital-logic circuit-output normal

Answer key

4.8.33 Circuit Output: GATE CSE 2014 | Set 3 | Question: 45



The above synchronous sequential circuit built using JK flip-flops is initialized with $Q_2Q_1Q_0 = 000$. The state sequence for this circuit for the next 3 clock cycles is

- A. 001,010,011 B. 111,110,101
C. 100,110,111 D. 100,011,001

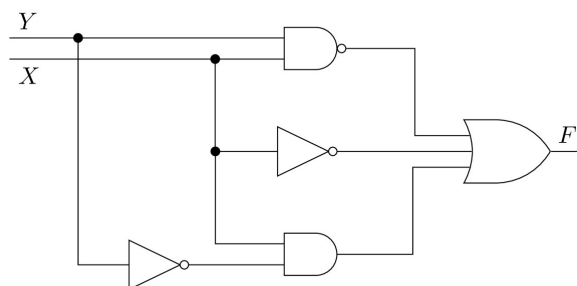
gatecse-2014-set3 digital-logic circuit-output normal

Answer key

4.8.34 Circuit Output: GATE CSE 2025 | Set 2 | Question: 21



Consider the following logic circuit diagram.



Which is/are the CORRECT option(s) for the output function F ?

- A. $\overline{XY}$ B. $\overline{X} + \overline{Y} + XY$
C. $\overline{XY} + \overline{X} + XY$ D. $X + \overline{Y}$

gatecse2025-set2 digital-logic circuit-output multiple-selects easy one-mark



- 



- 

Answer key

4.8.38 Circuit Output: GATE IT 2007 | Question: 38



The following expression was to be realized using 2-input AND and OR gates. However, during the fabrication all 2-input AND gates were mistakenly substituted by 2-input NAND gates.

$$(a.b).c + (a'.c).d + (b.c).d + a.d$$

What is the function finally realized ?

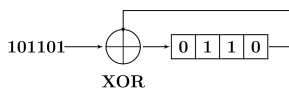
- A. 1
 B. $a' + b' + c' + d'$
 C. $a' + b + c' + d'$
 D. $a' + b' + c + d'$

Answer key

4.8.39 Circuit Output: GATE IT 2007 | Question: 40



What is the final value stored in the linear feedback shift register if the input is 101101?



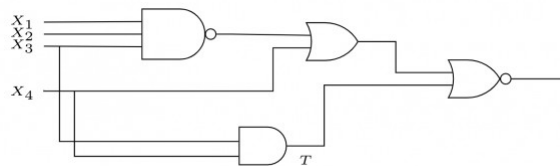
- A. 0110
 B. 1011
 C. 1101
 D. 1111

Answer key

4.8.40 Circuit Output: GATE IT 2007 | Question: 45



The line T in the following figure is permanently connected to the ground.



Which of the following inputs ($X_1X_2X_3X_4$) will detect the fault ?

- A. 0000
 B. 0111
 C. 1111
 D. None of these

Answer key

4.9

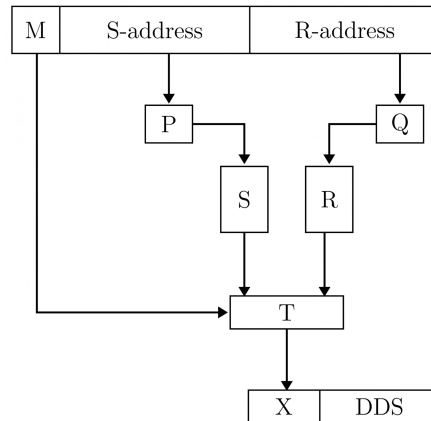
Combinational Circuit (2)

Practice Test: Test 1 (7Q)

4.9.1 Combinational Circuit: GATE CSE 2022 | Question: 30



Consider a digital display system (DDS) shown in the figure that displays the contents of register X . A 16-bit code word is used to load a word in X , either from S or from R . S is a 1024-word memory segment and R is a 32-word register file. Based on the value of mode bit M , T selects an input word to load in X . P and Q interface with the corresponding bits in the code word to choose the addressed word. Which one of the following represents the functionality of P , Q , and T ?



- A. P is 10 : 1 multiplexer; Q is 5 : 1 multiplexer; T is 2 : 1 multiplexer
 B. P is 10 : 2^{10} decoder; Q is 5 : 2^5 decoder; T is 2 : 1 encoder
 C. P is 10 : 2^{10} decoder; Q is 5 : 2^5 decoder; T is 2 : 1 multiplexer
 D. P is 1 : 10 de-multiplexer; Q is 1 : 5 de-multiplexer; T is 2 : 1 multiplexer

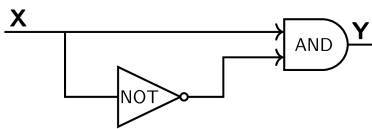
gatecse-2022 digital-logic combinational-circuit two-marks

Answer key

4.9.2 Combinational Circuit: GATE CSE 2024 | Set 1 | Question: 18



Consider the circuit shown below where the gates may have propagation delays. Assume that all signal transitions occur instantaneously and that wires have no delays. Which of the following statements about the circuit is/are CORRECT?



- A. With no propagation delays, the output Y is always logic Zero
 B. With no propagation delays, the output Y is always logic One
 C. With propagation delays, the output Y can have a transient logic One after X transitions from logic Zero to logic One
 D. With propagation delays, the output Y can have a transient logic Zero after X transitions from logic One to logic Zero

gatecse-2024-set1 multiple-selects digital-logic combinational-circuit one-mark

Answer key

4.10

Conjunctive Normal Form (1)

4.10.1 Conjunctive Normal Form: GATE CSE 2007 | Question: 48



Which of the following is TRUE about formulae in Conjunctive Normal Form?

- A. For any formula, there is a truth assignment for which at least half the clauses evaluate to true.
 B. For any formula, there is a truth assignment for which all the clauses evaluate to true.
 C. There is a formula such that for each truth assignment, at most one-fourth of the clauses evaluate to true.
 D. None of the above.

gatecse-2007 digital-logic normal conjunctive-normal-form canonical-normal-form

Answer key

4.11

Decoder (3)

 **Practice Test:** Test 1 (7Q)

4.11.1 Decoder: GATE CSE 2007 | Question: 8, ISRO2011-31



How many 3-to-8 line decoders with an enable input are needed to construct a 6-to-64 line decoder without using any other logic gates?

- A. 7 B. 8 C. 9 D. 10

gatecse-2007 digital-logic normal isro2011 decoder

Answer key 

4.11.2 Decoder: GATE CSE 2020 | Question: 20



If there are m input lines and n output lines for a decoder that is used to uniquely address a byte addressable 1 KB RAM, then the minimum value of $m + n$ is _____.

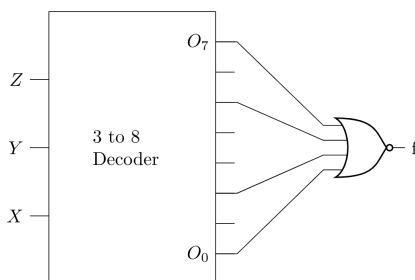
gatecse-2020 numerical-answers digital-logic decoder one-mark

Answer key 

4.11.3 Decoder: GATE IT 2008 | Question: 9



What Boolean function does the circuit below realize?



- A. $xz + \bar{x}\bar{z}$ B. $x\bar{z} + \bar{x}z$
 C. $\bar{x}\bar{y} + yz$ D. $xy + \bar{y}\bar{z}$

gateit-2008 digital-logic circuit-output decoder normal

Answer key 

4.12

Digital Circuits (7)

 **Practice Test:** Test 1 (8Q)

4.12.1 Digital Circuits: GATE CSE 1996 | Question: 5



A logic network has two data inputs A and B , and two control inputs C_0 and C_1 . It implements the function F according to the following table.

C_1	C_0	F
0	0	$\overline{A+B}$
0	1	$A+B$
1	0	$A \oplus B$

Implement the circuit using one 4 to 1 Multiplexer, one 2-input Exclusive OR gate, one 2-input AND gate, one 2-input OR gate and one Inverter.

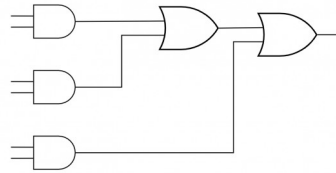
gate1996 digital-logic normal digital-circuits descriptive

Answer key 

4.12.2 Digital Circuits: GATE CSE 2002 | Question: 7



- A. Express the function $f(x, y, z) = xy' + yz'$ with only one complement operation and one or more AND/OR operations. Draw the logic circuit implementing the expression obtained, using a single NOT gate and one or more AND/OR gates.
- B. Transform the following logic circuit (without expressing its switching function) into an equivalent logic circuit that employs only 6 NAND gates each with 2-inputs.



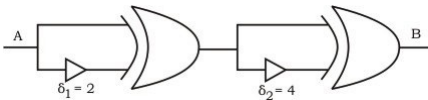
gatecse-2002 digital-logic normal descriptive digital-circuits

Answer key

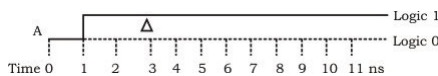
4.12.3 Digital Circuits: GATE CSE 2003 | Question: 47



Consider the following circuit composed of XOR gates and non-inverting buffers.



The non-inverting buffers have delays $\delta_1 = 2ns$ and $\delta_2 = 4ns$ as shown in the figure. Both XOR gates and all wires have zero delays. Assume that all gate inputs, outputs, and wires are stable at logic level 0 at time 0. If the following waveform is applied at input A , how many transition(s) (change of logic levels) occur(s) at B during the interval from 0 to 10 ns?



- A. 1 B. 2 C. 3 D. 4

gatecse-2003 digital-logic digital-circuits

Answer key

4.12.4 Digital Circuits: GATE CSE 2011 | Question: 13



Which one of the following circuits is **NOT** equivalent to a 2-input *XNOR* (exclusive *NOR*) gate?

- A.
- B.
- C.
- D.

Answer key

4.12.5 Digital Circuits: GATE CSE 2013 | Question: 5



In the following truth table, $V = 1$ if and only if the input is valid.

Inputs				Outputs		
D_0	D_1	D_2	D_3	X_0	X_1	V
0	0	0	0	x	x	0
1	0	0	0	0	0	1
x	1	0	0	0	1	1
x	x	1	0	1	0	1
x	x	x	1	1	1	1

What function does the truth table represent?

- A. Priority encoder
 B. Decoder
 C. Multiplexer
 D. Demultiplexer

Answer key

4.12.6 Digital Circuits: GATE CSE 2014 | Set 3 | Question: 8



Consider the following combinational function block involving four Boolean variables x, y, a, b where x, a, b are inputs and y is the output.

```
f(x, a, b, y)
{
  if(x is 1) y = a;
  else y = b;
}
```

Which one of the following digital logic blocks is the most suitable for implementing this function?

- A. Full adder
 B. Priority encoder
 C. Multiplexor
 D. Flip-flop

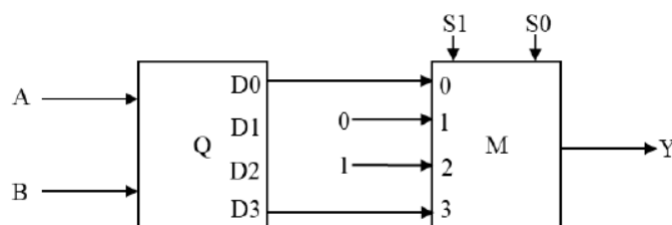
Answer key

4.12.7 Digital Circuits: GATE CSE 2016 | Set 2 | Question: 49



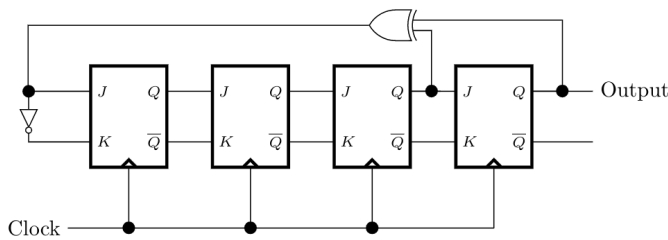
Consider the digital circuit shown below with two input lines A and B , two select lines S_0 and S_1 , and an output line Y . The blocks Q and M represent active high 2 : 4 decoder and 4-to-1 multiplexer, respectively. Out of 16 possible input combinations, the number of combinations that produce $Y = 1$ is _____ (answer in integer)

Note: One input combination is an instance of $[A \ B \ S_1 \ S_0]$.



Answer key

4.13.1 Digital Counter: GATE CSE 1987 | Question: 1-III



The above circuit produces the output sequence:

- A. 1111 1111 0000 0000
C. 1111 0001 0011 0101

- B. 1111 0000 1111 0000
D. 1010 1010 1010 1010

gate1987 digital-logic sequential-circuit flip-flop digital-counter

Answer key

4.13.2 Digital Counter: GATE CSE 1987 | Question: 10c



Give a minimal DFA that performs as a $\text{mod } -3$, 1's counter, i.e. outputs a 1 each time the number of 1's in the input sequence is a multiple of 3.

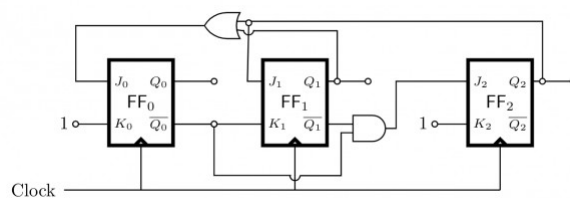
gate1987 digital-logic digital-counter descriptive

Answer key

4.13.3 Digital Counter: GATE CSE 1990 | Question: 5-c



For the synchronous counter shown in Fig.3, write the truth table of Q_0, Q_1 , and Q_2 after each pulse, starting from $Q_0 = Q_1 = Q_2 = 0$ and determine the counting sequence and also the modulus of the counter.



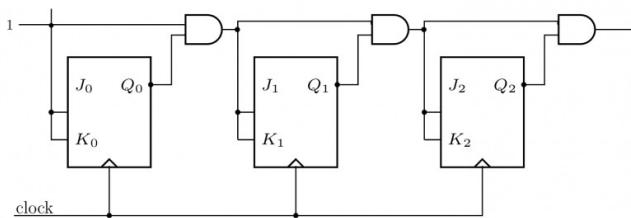
gate1990 descriptive digital-logic sequential-circuit flip-flop digital-counter

Answer key

4.13.4 Digital Counter: GATE CSE 1991 | Question: 5-c



Find the maximum clock frequency at which the counter in the figure below can be operated. Assume that the propagation delay through each flip flop and each AND gate is 10 ns . Also, assume that the setup time for the JK inputs of the flip flops is negligible.



gate1991 digital-logic sequential-circuit flip-flop digital-counter

Answer key

4.13.5 Digital Counter: GATE CSE 1994 | Question: 2-1



The number of flip-flops required to construct a binary modulo N counter is _____

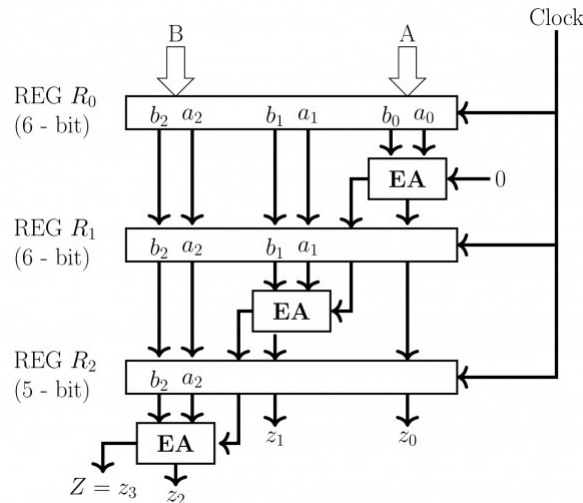
gate1994 digital-logic sequential-circuit flip-flop digital-counter fill-in-the-blanks

Answer key

4.13.6 Digital Counter: GATE CSE 2002 | Question: 8



Consider the following circuit. $A = a_2a_1a_0$ and $B = b_2b_1b_0$ are three bit binary numbers input to the circuit. The output is $Z = z_3z_2z_1z_0$. R_0 , R_1 and R_2 are registers with loading clock shown. The registers are loaded with their input data with the falling edge of a clock pulse (signal CLOCK shown) and appears as shown. The bits of input number A , B and the full adders are as shown in the circuit. Assume Clock period is greater than the settling time of all circuits.



a. For 8 clock pulses on the CLOCK terminal and the inputs A, B as shown, obtain the output Z (sequence of 4-bit values of Z). Assume initial contents of R_0, R_1 and R_2 as all zeros.

A	110	011	111	101	000	000	000	000
B	101	101	011	110	000	000	000	000
Clock No	1	2	3	4	5	6	7	8

b. What does the circuit implement?

gatecse-2002 digital-logic normal descriptive digital-counter

Answer key

4.13.7 Digital Counter: GATE CSE 2011 | Question: 15



The minimum number of D flip-flops needed to design a mod-258 counter is

- A. 9 B. 8 C. 512 D. 258

gatecse-2011 digital-logic easy digital-counter

Answer key

4.13.8 Digital Counter: GATE CSE 2014 | Set 2 | Question: 7



Let $k = 2^n$. A circuit is built by giving the output of an n -bit binary counter as input to an n -to- 2^n bit decoder. This circuit is equivalent to a

- A. k -bit binary up counter. B. k -bit binary down counter.
C. k -bit ring counter. D. k -bit Johnson counter.

gatecse-2014-set2 digital-logic normal digital-counter

Answer key

4.13.9 Digital Counter: GATE CSE 2015 | Set 1 | Question: 20



Consider a 4-bit Johnson counter with an initial value of 0000. The counting sequence of this counter is

- A. 0,1,3,7,15,14,12,8,0
 B. 0,1,3,5,7,9,11,13,15,0
 C. 0,2,4,6,8,10,12,14,0
 D. 0,8,12,14,15,7,3,1,0

gatecse-2015-set1 digital-logic digital-counter easy

Answer key

4.13.10 Digital Counter: GATE CSE 2015 | Set 2 | Question: 7



The minimum number of JK flip-flops required to construct a synchronous counter with the count sequence (0, 0, 1, 1, 2, 2, 3, 3, 0, 0, ...) is _____.

gatecse-2015-set2 digital-logic digital-counter normal numerical-answers

Answer key

4.13.11 Digital Counter: GATE CSE 2016 | Set 1 | Question: 8



We want to design a synchronous counter that counts the sequence 0 – 1 – 0 – 2 – 0 – 3 and then repeats. The minimum number of J-K flip-flops required to implement this counter is _____.

gatecse-2016-set1 digital-logic digital-counter flip-flop normal numerical-answers

Answer key

4.13.12 Digital Counter: GATE CSE 2017 | Set 2 | Question: 42



The next state table of a 2-bit saturating up-counter is given below.

Q_1	Q_0	Q_1^+	Q_0^+
0	0	0	1
0	1	1	0
1	0	1	1
1	1	1	1

The counter is built as a synchronous sequential circuit using T flip-flops. The expressions for T_1 and T_0 are

- A. $T_1 = Q_1 Q_0$, $T_0 = \bar{Q}_1 \bar{Q}_0$
 B. $T_1 = \bar{Q}_1 Q_0$, $T_0 = \bar{Q}_1 + \bar{Q}_0$
 C. $T_1 = Q_1 + Q_0$, $T_0 = \bar{Q}_1 \bar{Q}_0$
 D. $T_1 = \bar{Q}_1 Q_0$, $T_0 = Q_1 + Q_0$

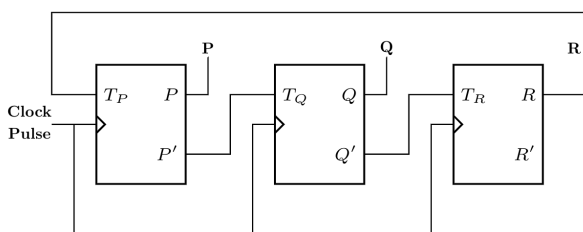
gatecse-2017-set2 digital-logic digital-counter

Answer key

4.13.13 Digital Counter: GATE CSE 2021 | Set 1 | Question: 28



Consider a 3-bit counter, designed using T flip-flops, as shown below:



Assuming the initial state of the counter given by PQR as 000, what are the next three states?

A. 011,101,000

B. 001,010,111

C. 011,101,111

D. 001,010,000

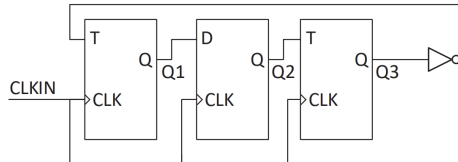
gatecse-2021-set1 digital-logic sequential-circuit digital-counter two-marks

Answer key

4.13.14 Digital Counter: GATE CSE 2023 | Question: 33



Consider a sequential digital circuit consisting of T flip-flops and D flip-flops as shown in the figure. CLKIN is the clock input to the circuit. At the beginning, Q₁, Q₂ and Q₃ have values 0, 1 and 1, respectively.



Which one of the given values of (Q₁, Q₂, Q₃) can NEVER be obtained with this digital circuit?

A. (0,0,1)

B. (1,0,0)

C. (1,0,1)

D. (1,1,1)

gatecse-2023 digital-logic flip-flop two-marks digital-counter

Answer key

4.13.15 Digital Counter: GATE CSE 2026 | Set 1 | Question: 27



Consider a 2-bit saturating up/down counter that performs the saturating up count when the input P is 0, and the saturating down count when P is 1. The Next State table of the counter is as shown. The counter is built as a synchronous sequential circuit using D flip-flops.

Input	Curent State		Next State	
P	Q ₁	Q ₀	Q ₁ ⁺	Q ₀ ⁺
0	0	0	0	1
0	0	1	1	0
0	1	0	1	1
0	1	1	1	1
1	0	0	0	0
1	0	1	0	0
1	1	0	0	1
1	1	1	1	0

Which one of the following options corresponds to the expressions for the inputs of the D flip-flops, D_1 and D_0 ?

A. $D_1 = PQ_1 + \bar{P}Q_0 + Q_1Q_0$ $D_0 = PQ_0 + \bar{P}Q_1 + Q_1\bar{Q}_0$

B. $D_1 = \bar{P}Q_1 + \bar{P}Q_0 + Q_1Q_0$ $D_0 = \bar{P}Q_0 + \bar{P}Q_1 + Q_1\bar{Q}_0$

C. $D_1 = \bar{P}Q_1 + \bar{P}Q_0 + Q_1Q_0$ $D_0 = \bar{P}Q_0 + \bar{P}Q_1 + Q_1Q_0$

D. $D_1 = PQ_1 + \bar{P}Q_0 + Q_1Q_0$ $D_0 = PQ_0 + \bar{P}Q_1 + Q_1Q_0$

gatecse-2026-set1 digital-logic digital-counter sequential-circuit flip-flop two-marks

Answer key

4.13.16 Digital Counter: GATE IT 2005 | Question: 11



How many pulses are needed to change the contents of a 8-bit up counter from 10101100 to 00100111 (rightmost bit is the LSB)?

A. 134

B. 133

C. 124

D. 123

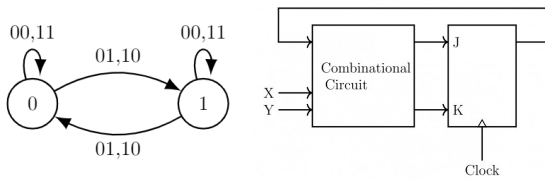
gateit-2005 digital-logic digital-counter normal

Answer key

4.13.17 Digital Counter: GATE IT 2008 | Question: 37



Consider the following state diagram and its realization by a JK flip flop



The combinational circuit generates J and K in terms of x, y and Q.
The Boolean expressions for J and K are :

- A. $\overline{x \oplus y}$ and $\overline{x \oplus y}$
 B. $\overline{x \oplus y}$ and $x \oplus y$
 C. $x \oplus y$ and $\overline{x \oplus y}$
 D. $x \oplus y$ and $x \oplus y$

gateit-2008 digital-logic boolean-algebra normal digital-counter

Answer key

4.13.18 Digital Counter: GATE1992-04-c



Design a 3-bit counter using D-flip flops such that not more than one flip-flop changes state between any two consecutive states.

gate1992 digital-logic sequential-circuit flip-flop digital-counter normal descriptive

Answer key

4.14 Dual Function (1)

4.14.1 Dual Function: GATE CSE 2014 | Set 2 | Question: 6



The dual of a Boolean function $F(x_1, x_2, \dots, x_n, +, \cdot, ')$, written as F^D is the same expression as that of F with $+$ and $\cdot$ swapped. F is said to be self-dual if $F = F^D$. The number of self-dual functions with n Boolean variables is

- A. 2^n
 B. 2^{n-1}
 C. 2^{2^n}
 D. $2^{2^{n-1}}$

gatecse-2014-set2 digital-logic normal dual-function boolean-algebra

Answer key

4.15 Finite State Machines (4)

4.15.1 Finite State Machines: GATE CSE 1994 | Question: 3.3



State True or False with one line explanation

A FSM (Finite State Machine) can be designed to add two integers of any arbitrary length (arbitrary number of digits).

gate1994 digital-logic normal true-false finite-state-machines

Answer key

4.15.2 Finite State Machines: GATE CSE 1995 | Question: 2.23



A finite state machine with the following state table has a single input x and a single out z .

present state	next state, z	
	x=1	x=0
A	D,0	B,0
B	B,1	C,1
C	B,0	D,1
D	B,1	C,0

If the initial state is unknown, then the shortest input sequence to reach the final state C is:

- A. 01 B. 10 C. 101 D. 110

gate1995 digital-logic normal finite-state-machines

Answer key

4.15.3 Finite State Machines: GATE CSE 1996 | Question: 2.23



Consider the following state table for a sequential machine. The number of states in the minimized machine will be

		Input	
		0	1
Present State	A	D,0	B,1
	B	A,0	C,1
	C	A,0	B,1
	D	A,1	C,1
		Next state, Output	

- A. 4 B. 3 C. 2 D. 1

gate1996 normal digital-logic finite-state-machines

Answer key

4.15.4 Finite State Machines: GATE CSE 2009 | Question: 27



Given the following state table of an FSM with two states A and B , one input and one output.

PRESENT STATE A	PRESENT STATE B	Input	Next State A	Next State B	Output
0	0	0	0	0	1
0	1	0	1	0	0
1	0	0	0	1	0
1	1	0	1	0	0
0	0	1	0	1	0
0	1	1	0	0	1
1	0	1	0	1	1
1	1	1	0	0	1

If the initial state is $A = 0, B = 0$ what is the minimum length of an input string which will take the machine to the state $A = 0, B = 1$ with *output* = 1.

- A. 3 B. 4 C. 5 D. 6

gatecse-2009 digital-logic normal finite-state-machines

Answer key

4.16

Fixed Point Representation (2)

4.16.1 Fixed Point Representation: GATE CSE 2017 | Set 1 | Question: 7



The n -bit fixed-point representation of an unsigned real number X uses f bits for the fraction part. Let $i = n - f$. The range of decimal values for X in this representation is

- A. 2^{-f} to 2^i B. 2^{-f} to $(2^i - 2^{-f})$
C. 0 to 2^i D. 0 to $(2^i - 2^{-f})$

gatecse-2017-set1 digital-logic number-representation fixed-point-representation

Answer key

4.16.2 Fixed Point Representation: GATE CSE 2018 | Question: 33



Consider the unsigned 8-bit fixed point binary number representation, below,

$$b_7 \ b_6 \ b_5 \ b_4 \ b_3 \cdot b_2 \ b_1 \ b_0$$

where the position of the primary point is between b_3 and b_2 . Assume b_7 is the most significant bit. Some of the decimal numbers listed below **cannot** be represented **exactly** in the above representation:

- i. 31.500
- ii. 0.875
- iii. 12.100
- iv. 3.001

Which one of the following statements is true?

- A. None of i, ii, iii, iv can be exactly represented
- B. Only ii cannot be exactly represented
- C. Only iii and iv cannot be exactly represented
- D. Only i and ii cannot be exactly represented

gatecse-2018 digital-logic number-representation fixed-point-representation normal two-marks

Answer key

4.17

Flip Flop (7)

Practice Tests: Test 1 (15Q) Test 2 (6Q)

4.17.1 Flip Flop: GATE CSE 2001 | Question: 11



A sequential circuit takes an input stream of 0's and 1's and produces an output stream of 0's and 1's. Initially it replicates the input on its output until two consecutive 0's are encountered on the input. From then onward, it produces an output stream, which is the bit-wise complement of input stream until it encounters two consecutive 1's, whereupon the process repeats. An example input and output stream is shown below.

The input stream: 101100|01001011|011

The desired output: 101100|10110100|011

J-K master-slave flip-flops are to be used to design the circuit.

- a. Give the state transition diagram
- b. Give the minimized sum-of-product expression for J and K inputs of one of its state flip-flops

gatecse-2001 digital-logic normal descriptive flip-flop

Answer key

4.17.2 Flip Flop: GATE CSE 2004 | Question: 18, ISRO2007-31



In an SR latch made by cross-coupling two NAND gates, if both S and R inputs are set to 0, then it will result in

- A. $Q = 0, Q' = 1$
- B. $Q = 1, Q' = 0$
- C. $Q = 1, Q' = 1$
- D. Indeterminate states

gatecse-2004 digital-logic easy isro2007 flip-flop

Answer key

4.17.3 Flip Flop: GATE CSE 2015 | Set 1 | Question: 37



A positive edge-triggered D flip-flop is connected to a positive edge-triggered JK flip-flop as follows. The Q

output of the D flip-flop is connected to both the J and K inputs of the JK flip-flop, while the Q output of the JK flip-flop is connected to the input of the D flip-flop. Initially, the output of the D flip-flop is set to logic one and the output of the JK flip-flop is cleared. Which one of the following is the bit sequence (including the initial state) generated at the Q output of the JK flip-flop when the flip-flops are connected to a free-running common clock? Assume that $J = K = 1$ is the toggle mode and $J = K = 0$ is the state holding mode of the JK flip-flops. Both the flip-flops have non-zero propagation delays.

- A. 0110110...
 B. 0100100...
 C. 011101110...
 D. 011001100...

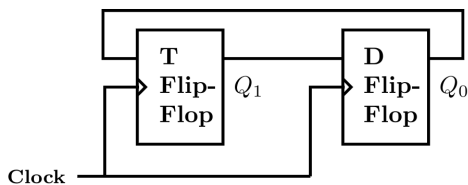
gatecse-2015-set1 digital-logic flip-flop normal

Answer key

4.17.4 Flip Flop: GATE CSE 2017 | Set 1 | Question: 33



Consider a combination of T and D flip-flops connected as shown below. The output of the D flip-flop is connected to the input of the T flip-flop and the output of the T flip-flop is connected to the input of the D flip-flop.



Initially, both Q_0 and Q_1 are set to 1 (before the 1st clock cycle). The outputs

- A. Q_1Q_0 after the 3rd cycle are 11 and after the 4th cycle are 00 respectively.
 B. Q_1Q_0 after the 3rd cycle are 11 and after the 4th cycle are 01 respectively.
 C. Q_1Q_0 after the 3rd cycle are 00 and after the 4th cycle are 11 respectively.
 D. Q_1Q_0 after the 3rd cycle are 01 and after the 4th cycle are 01 respectively.

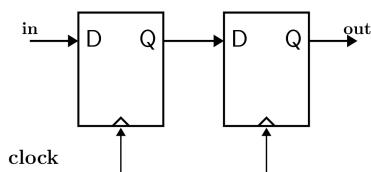
gatecse-2017-set1 digital-logic flip-flop normal

Answer key

4.17.5 Flip Flop: GATE CSE 2018 | Question: 22



Consider the sequential circuit shown in the figure, where both flip-flops used are positive edge-triggered D flip-flops.



The number of states in the state transition diagram of this circuit that have a transition back to the same state on some value of "in" is ____

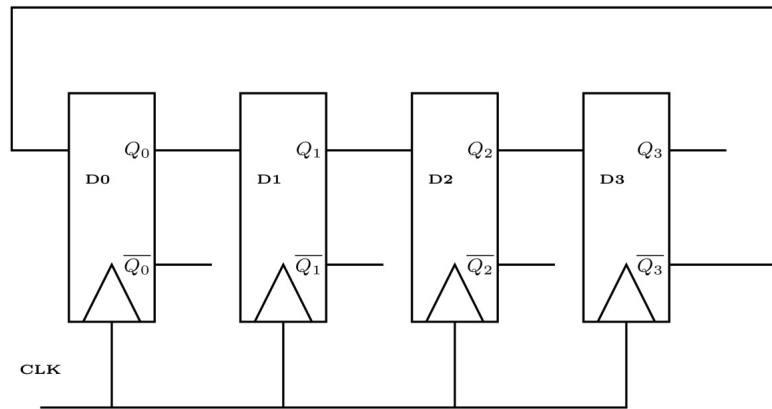
gatecse-2018 digital-logic flip-flop numerical-answers normal one-mark

Answer key

4.17.6 Flip Flop: GATE CSE 2025 | Set 1 | Question: 50



Consider the given sequential circuit designed using D-Flip-flops. The circuit is initialized with some value (initial state). The number of distinct states the circuit will go through before returning back to the initial state is _____. (Answer in integer)



gatecse2025-set1 digital-logic flip-flop sequential-circuit numerical-answers two-marks

Answer key

4.17.7 Flip Flop: GATE IT 2007 | Question: 7



Which of the following input sequences for a cross-coupled $R - S$ flip-flop realized with two $NAND$ gates may lead to an oscillation?

- A. 11,00 B. 01,10 C. 10,01 D. 00,11

gateit-2007 digital-logic normal flip-flop

Answer key

4.18 Floating Point Representation (8)

Practice Test: Test 1 (11Q)

4.18.1 Floating Point Representation: GATE CSE 1987 | Question: 1-vii



The exponent of a floating-point number is represented in excess- N code so that:

- A. The dynamic range is large. B. The precision is high.
C. The smallest number is represented by all zeros. D. Overflow is avoided.

gate1987 digital-logic number-representation floating-point-representation

Answer key

4.18.2 Floating Point Representation: GATE CSE 1989 | Question: 1-vi



Consider an excess -50 representation for floating point numbers with 4 BCD digit mantissa and 2 BCD digit exponent in normalised form. The minimum and maximum positive numbers that can be represented are _____ and _____ respectively.

descriptive gate1989 digital-logic number-representation floating-point-representation

Answer key

4.18.3 Floating Point Representation: GATE CSE 1990 | Question: 1-iv-a



A 32-bit floating-point number is represented by a 7-bit signed exponent, and a 24-bit fractional mantissa. The base of the scale factor is 16, The range of the exponent is _____

gate1990 digital-logic number-representation floating-point-representation fill-in-the-blanks

Answer key

4.18.4 Floating Point Representation: GATE CSE 1990 | Question: 1-iv-b



A 32-bit floating-point number is represented by a 7-bit signed exponent, and a 24-bit fractional mantissa.

The base of the scale factor is 16,
The range of the exponent is _____, if the scale factor is represented in excess-64 format.

gate1990 digital-logic number-representation floating-point-representation fill-in-the-blanks

Answer key

4.18.5 Floating Point Representation: GATE CSE 1997 | Question: 72



Following floating point number format is given

f is a fraction represented by a 6-bit mantissa (includes sign bit) in sign magnitude form, e is a 4-bit exponent (includes sign bit) in sign magnitude form and $n = (f, e) = f.2^e$ is a floating point number. Let $A = 54.75$ in decimal and $B = 9.75$ in decimal

- Represent A and B as floating point numbers in the above format.
- Show the steps involved in floating point addition of A and B .
- What is the percentage error (up to one position beyond decimal point) in the addition operation in (b)?

gate1997 digital-logic floating-point-representation normal descriptive

Answer key

4.18.6 Floating Point Representation: GATE CSE 2003 | Question: 43



The following is a scheme for floating point number representation using 16 bits.

Bit Position	15	14		9	8		0
	s			e			m
	Sign			Exponent			Mantissa

Let s , e , and m be the numbers represented in binary in the sign, exponent, and mantissa fields respectively. Then the floating point number represented is:

$$\begin{cases} (-1)^s (1 + m \times 2^{-9}) 2^{e-31}, & \text{if the exponent} \neq 11111 \\ 0, & \text{otherwise} \end{cases}$$

What is the maximum difference between two successive real numbers representable in this system?

- A. 2^{-40} B. 2^{-9} C. 2^{22} D. 2^{31}

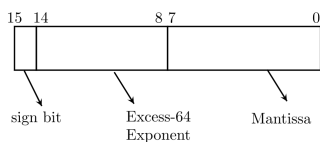
gatecse-2003 digital-logic number-representation floating-point-representation normal

Answer key

4.18.7 Floating Point Representation: GATE CSE 2005 | Question: 85-a



Consider the following floating-point format.



Mantissa is a pure fraction in sign-magnitude form.

The decimal number 0.239×2^{13} has the following hexadecimal representation (without normalization and rounding off):

- A. 0D 24 B. 0D 4D C. 4D 0D D. 4D 3D

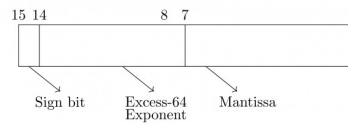
gatecse-2005 digital-logic number-representation floating-point-representation normal

Answer key

4.18.8 Floating Point Representation: GATE CSE 2005 | Question: 85-b



Consider the following floating-point format.



Mantissa is a pure fraction in sign-magnitude form.

The normalized representation for the above format is specified as follows. The mantissa has an implicit 1 preceding the binary (radix) point. Assume that only 0's are padded in while shifting a field.

The normalized representation of the above number (0.239×2^{13}) is:

- A. 0A 20 B. 11 34 C. 49 D0 D. 4A E8

gatecse-2005 digital-logic number-representation floating-point-representation normal

Answer key

4.19 Functional Completeness (7)

Practice Test: Test 1 (5Q)

4.19.1 Functional Completeness: GATE CSE 1989 | Question: 4-iii



Show that {NOR} is a functionally complete set of Boolean operations.

gate1989 descriptive digital-logic functional-completeness

Answer key

4.19.2 Functional Completeness: GATE CSE 1992 | Question: 02-ii



All digital circuits can be realized using only

- A. Ex-OR gates B. Multiplexers C. Half adders D. OR gates

gate1992 normal digital-logic digital-circuits multiple-selects functional-completeness combinational-circuit

Answer key

4.19.3 Functional Completeness: GATE CSE 1993 | Question: 9



Assume that only half adders are available in your laboratory. Show that any binary function can be implemented using half adders only.

gate1993 digital-logic combinational-circuit adder descriptive functional-completeness

Answer key

4.19.4 Functional Completeness: GATE CSE 1998 | Question: 5



The implication gate, shown below has two inputs (x and y); the output is 1 except when $x = 1$ and $y = 0$, realize $f = \bar{x}y + x\bar{y}$ using only four implication gates.



Show that the implication gate is functionally complete.

gate1998 digital-logic functional-completeness descriptive

Answer key

4.19.5 Functional Completeness: GATE CSE 1999 | Question: 2.9



Which of the following sets of component(s) is/are sufficient to implement any arbitrary Boolean function?

- A. XOR gates, NOT gates
- B. 2 to 1 multiplexers
- C. AND gates, XOR gates
- D. Three-input gates that output $(A \cdot B) + C$ for the inputs A, B and C .

gate1999 digital-logic normal functional-completeness multiple-selects

Answer key

4.19.6 Functional Completeness: GATE CSE 2015 | Set 1 | Question: 39

Consider the operations

$$f(X, Y, Z) = X'YZ + XY' + Y'Z' \text{ and } g(X, Y, Z) = X'YZ + X'YZ' + XY$$

Which one of the following is correct?

- A. Both $\{f\}$ and $\{g\}$ are functionally complete
- B. Only $\{f\}$ is functionally complete
- C. Only $\{g\}$ is functionally complete
- D. Neither $\{f\}$ nor $\{g\}$ is functionally complete

gatecse-2015-set1 boolean-algebra difficult functional-completeness

Answer key

4.19.7 Functional Completeness: GATE IT 2008 | Question: 1

A set of Boolean connectives is functionally complete if all Boolean functions can be synthesized using those. Which of the following sets of connectives is NOT functionally complete?

- A. EX-NOR
- B. implication, negation
- C. OR, negation
- D. NAND

gateit-2008 digital-logic easy functional-completeness

Answer key

4.20

IEEE Representation (14)

 **Practice Tests:** [Test 1 \(15Q\)](#) [Test 2 \(6Q\)](#)

4.20.1 IEEE Representation: GATE CSE 2008 | Question: 4

In the IEEE floating point representation the hexadecimal value 0x00000000 corresponds to

- A. The normalized value 2^{-127}
- B. The normalized value 2^{-126}
- C. The normalized value $+0$
- D. The special value $+0$

gatecse-2008 digital-logic floating-point-representation ieee-representation easy

Answer key

4.20.2 IEEE Representation: GATE CSE 2012 | Question: 7

The decimal value 0.5 in IEEE single precision floating point representation has

- A. fraction bits of 000...000 and exponent value of 0
- B. fraction bits of 000...000 and exponent value of -1
- C. fraction bits of 100...000 and exponent value of 0
- D. no exact representation

gatecse-2012 digital-logic normal number-representation ieee-representation

Answer key

4.20.3 IEEE Representation: GATE CSE 2014 | Set 2 | Question: 45



The value of a float type variable is represented using the single-precision 32-bit floating point format of IEEE-754 standard that uses 1 bit for sign, 8 bits for biased exponent and 23 bits for the mantissa. A float type variable X is assigned the decimal value of -14.25 . The representation of X in hexadecimal notation is

- A. C1640000H B. 416C0000H C. 41640000H D. C16C0000H

gatecse-2014-set2 digital-logic number-representation normal ieee-representation

Answer key

4.20.4 IEEE Representation: GATE CSE 2017 | Set 2 | Question: 12



Given the following binary number in 32-bit (single precision) IEEE-754 format :

00111110011011010000000000000000

The decimal value closest to this floating-point number is :

- A. $1.45 * 10^1$ B. $1.45 * 10^{-1}$ C. $2.27 * 10^{-1}$ D. $2.27 * 10^1$

gatecse-2017-set2 digital-logic ieee-representation number-representation floating-point-representation

Answer key

4.20.5 IEEE Representation: GATE CSE 2020 | Question: 29



Consider three registers $R1$, $R2$, and $R3$ that store numbers in IEEE-754 single precision floating point format. Assume that $R1$ and $R2$ contain the values (in hexadecimal notation) $0x42200000$ and $0xC1200000$, respectively.

If $R3 = \frac{R1}{R2}$, what is the value stored in $R3$?

- A. 0x40800000 B. 0xC0800000 C. 0x83400000 D. 0xC8500000

gatecse-2020 digital-logic ieee-representation floating-point-representation two-marks

Answer key

4.20.6 IEEE Representation: GATE CSE 2021 | Set 1 | Question: 24



Consider the following representation of a number in IEEE 754 single-precision floating point format with a bias of 127.

$S : 1$ $E : 10000001$ $F : 111100000000000000000000$

Here S , E and F denote the sign, exponent, and fraction components of the floating point representation.

The decimal value corresponding to the above representation (rounded to 2 decimal places) is _____.

gatecse-2021-set1 digital-logic number-representation ieee-representation numerical-answers one-mark floating-point-representation

Answer key

4.20.7 IEEE Representation: GATE CSE 2021 | Set 2 | Question: 4



The format of the single-precision floating point representation of a real number as per the IEEE 754 standard is as follows:

sign	exponent	mantissa
------	----------	----------

Which one of the following choices is correct with respect to the *smallest* normalized positive number represented using the standard?

- A. exponent = 00000000 and mantissa = 00000000000000000000000000000000
 B. exponent = 00000000 and mantissa = 00000000000000000000000000000001
 C. exponent = 00000001 and mantissa = 00000000000000000000000000000000

D. exponent = 00000001 and mantissa = 00000000000000000000000000000001

gatecse-2021-set2 digital-logic number-representation ieee-representation one-mark floating-point-representation

Answer key

4.20.8 IEEE Representation: GATE CSE 2022 | Question: 31



Consider three floating point numbers A , B and C stored in registers R_A , R_B and R_C , respectively as per IEEE-754 single precision floating point format. The 32-bit content stored in these registers (in hexadecimal form) are as follows.

$$R_A = 0xC1400000 \quad R_B = 0x42100000 \quad R_C = 0x41400000$$

Which one of the following is FALSE?

- A. $A + C = 0$ B. $C = A + B$ C. $B = 3C$ D. $(B - C) > 0$

gatecse-2022 digital-logic ieee-representation number-representation two-marks floating-point-representation

Answer key

4.20.9 IEEE Representation: GATE CSE 2023 | Question: 35



Consider the IEEE-754 single precision floating point numbers $P = 0xC1800000$ and $Q = 0x3F5C2EF4$. Which one of the following corresponds to the product of these numbers (i.e., $P \times Q$), represented in the IEEE-754 single precision format?

- A. $0x404C2EF4$ B. $0x405C2EF4$ C. $0xC15C2EF4$ D. $0xC14C2EF4$

gatecse-2023 digital-logic number-representation ieee-representation two-marks floating-point-representation

Answer key

4.20.10 IEEE Representation: GATE CSE 2024 | Set 2 | Question: 4



The format of a single-precision floating-point number as per the IEEE 754 standard is:

Sign	Exponent	Mantissa
(1 bit)	(8 bits)	(23 bits)

Choose the largest floating-point number among the following options.

- A.

Sign	Exponent	Mantissa
0	01111111	111111111111111111111111
- B.

Sign	Exponent	Mantissa
0	11111110	111111111111111111111111
- C.

Sign	Exponent	Mantissa
0	11111111	111111111111111111111111
- D.

Sign	Exponent	Mantissa
0	01111111	000000000000000000000000

gatecse-2024-set2 digital-logic number-representation ieee-representation floating-point-representation one-mark

Answer key

4.20.11 IEEE Representation: GATE CSE 2025 | Set 2 | Question: 39



Three floating point numbers X , Y , and Z are stored in three registers R_X , R_Y , and R_Z , respectively in IEEE 754 single precision format as given below in hexadecimal:

$$R_X = 0xC1100000, R_Y = 0x40C00000, \text{ and } R_Z = 0x41400000$$

Which of the following option(s) is/are CORRECT?

- A. $4(X + Y) + Z = 0$ B. $2Y - Z = 0$ C. $4X + 3Z = 0$ D. $X + Y + Z = 0$

gatecse2025-set2 digital-logic number-representation ieee-representation multiple-selects two-marks

Answer key

4.20.12 IEEE Representation: GATE CSE 2026 | Set 1 | Question: 26



Consider the real valued variables X, Y and Z represented using the IEEE 754 singleprecision floating-point format. The binary representations of X and Y in hexadecimal notation are as follows:

$$X : 35C00000 \quad Y : 34A00000$$

Let $Z = X + Y$.

Which one of the following is the binary representation of Z , in hexadecimal notation?

- A. 35C80000 B. 35CC0000 C. 35E80000 D. 35EC0000

gatecse-2026-set1 digital-logic ieee-representation floating-point-representation two-marks

Answer key

4.20.13 IEEE Representation: GATE CSE 2026 | Set 2 | Question: 24



The 32-bit IEEE 754 single precision representation of a number is $0xC2710000$. The number in decimal representation is _____. (rounded off to two decimal places)

gatecse-2026-set2 digital-logic ieee-representation numerical-answers one-mark

Answer key

4.20.14 IEEE Representation: GATE IT 2008 | Question: 7



The following bit pattern represents a floating point number in IEEE 754 single precision format

1 10000011 101000000000000000000000

The value of the number in decimal form is

- A. -10 B. -13 C. -26 D. None of the above

gateit-2008 digital-logic ieee-representation number-representation floating-point-representation normal

Answer key

4.21 K Map (17)

Practice Tests: [Test 1 \(15Q\)](#) [Test 2 \(5Q\)](#) [Weekly Quiz 1 \(15Q\)](#) [Weekly Quiz 2 \(15Q\)](#)

4.21.1 K Map: GATE CSE 1987 | Question: 16-a

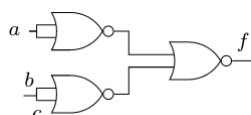


A Boolean function f is to be realized only by NOR gates. Its K -map is given below:

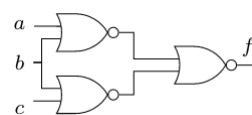
		ab			
		00	01	11	10
c	0	0	0	1	1
	1	0	1	1	1

The realization is

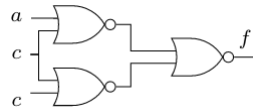
A.



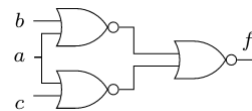
B.



C.



D.



gate1987 digital-logic k-map

Answer key

4.21.2 K Map: GATE CSE 1988 | Question: 3a-b



	BC			
	00	01	11	10
A				
0	1		1	1
1		1	1	

	B	
	0	1
A		
0	$\overline{C}$	1
1	C	C

The Karnaugh map of a function of (A, B, C) is shown on the left hand side of the above figure.

The reduced form of the same map is shown on the right hand side, in which the variable C is entered in the map itself. Discuss,

- The methodology by which the reduced map has been derived and
- the rules (or steps) by which the boolean function can be derived from the entries in the reduced map.

gate1988 descriptive digital-logic k-map

Answer key

4.21.3 K Map: GATE CSE 1992 | Question: 01-i



The Boolean function in sum of products form where K-map is given below (figure) is _____

	C	
	0	1
B		
0	1	0
1	$\overline{A}$	A

gate1992 digital-logic k-map normal fill-in-the-blanks

Answer key

4.21.4 K Map: GATE CSE 1995 | Question: 15-b



What is the equivalent minimal Boolean expression (in sum of products form) for the Karnaugh map given below?

	AB			
	00	01	11	10
CD				
00	1			1
01		1	1	
11		1	1	
10	1			1

gate1995 digital-logic boolean-algebra k-map normal descriptive

Answer key

4.21.5 K Map: GATE CSE 1996 | Question: 2.24



What is the equivalent Boolean expression in product-of-sums form for the Karnaugh map given in Fig

CD \ AB	00	01	11	10
	00	01	11	10
00		1	1	
01	1			1
11	1			1
10		1	1	

A. $B\bar{D} + \bar{B}D$

B. $(B + \bar{C} + D)(\bar{B} + C + \bar{D})$

C. $(B + D)(\bar{B} + \bar{D})$

D. $(B + \bar{D})(\bar{B} + D)$

gate1996 digital-logic k-map easy

Answer key

4.21.6 K Map: GATE CSE 1998 | Question: 2.7



The function represented by the Karnaugh map given below is

A \ BC	00	01	10	11
	00	01	10	11
0	1	0	0	1
1	1	0	0	1

A. $A.B$

B. $AB + BC + CA$

C. $\overline{B \oplus C}$

D. $A.BC$

gate1998 digital-logic k-map normal

Answer key

4.21.7 K Map: GATE CSE 1999 | Question: 1.8



Which of the following functions implements the Karnaugh map shown below?

AB \ CD	00	01	11	10
	00	01	11	10
00	0	0	1	0
01	X	X	1	X
11	0	1	1	0
10	0	1	1	0

A. $\bar{A}B + CD$

B. $D(C + A)$

C. $AD + \bar{A}B$

D. $(C + D)(\bar{C} + D) + (A + B)$

gate1999 digital-logic k-map easy

Answer key

4.21.8 K Map: GATE CSE 2000 | Question: 2.11



Which functions does NOT implement the Karnaugh map given below?

xy \ wz	00	01	11	10
	00	01	11	10
00	0	X	0	0
01	0	X	1	1
11	1	1	1	1
10	0	X	0	0

A. $(w + x)y$

B. $xy + yw$

C. $(w+x)(\bar{w}+y)(\bar{x}+y)$

D. None of the above

gatecse-2000 digital-logic k-map normal

Answer key

4.21.9 K Map: GATE CSE 2001 | Question: 1.11



Given the following karnaugh map, which one of the following represents the minimal Sum-Of-Products of the map?

$wz \backslash wx$	00	01	11	10
00	0	X	0	X
01	X	1	X	1
11	0	X	1	0
10	0	1	X	0

A. $XY + Y'Z$

B. $WX'Y' + XY + XZ$

C. $W'X + Y'Z + XY$

D. $XZ + Y$

gatecse-2001 k-map digital-logic normal

Answer key

4.21.10 K Map: GATE CSE 2002 | Question: 1.12



Minimum sum of product expression for $f(w, x, y, z)$ shown in Karnaugh-map below

$wz \backslash wx$	00	01	11	10
00	0	1	1	0
01	X	0	0	1
11	X	0	0	1
10	0	1	1	X

A. $xz + y'z$

B. $xz' + zx'$

C. $x'y + zx'$

D. None of the above

gatecse-2002 digital-logic k-map normal

Answer key

4.21.11 K Map: GATE CSE 2003 | Question: 45



The literal count of a Boolean expression is the sum of the number of times each literal appears in the expression. For example, the literal count of $(xy + xz')$ is 4. What are the minimum possible literal counts of the product-of-sum and sum-of-product representations respectively of the function given by the following Karnaugh map? Here, X denotes "don't care"

$xy \backslash zw$	00	01	11	10
00	X	1	0	1
01	0	1	X	0
11	1	X	X	0
10	X	0	0	X

A. (11,9)

B. (9,13)

C. (9,10)

D. (11,11)

gatecse-2003 digital-logic k-map normal

Answer key

4.21.12 K Map: GATE CSE 2008 | Question: 5



In the Karnaugh map shown below, X denotes a don't care term. What is the minimal form of the function represented by the Karnaugh map?

ab \ cd	00	01	11	10
00	1	1		1
01	X			
11	X			
10	1	1		X

- A. $\bar{b}.d + \bar{a}.d$
C. $\bar{b}.d + \bar{a}.b.d$

- B. $\bar{a}.\bar{b} + \bar{b}.d + \bar{a}.b.d$
D. $\bar{a}.\bar{b} + \bar{b}.d + \bar{a}.d$

gatecse-2008 digital-logic k-map easy

Answer key

4.21.13 K Map: GATE CSE 2012 | Question: 30



What is the minimal form of the Karnaugh map shown below? Assume that X denotes a don't care term

ab \ cd	00	01	11	10
00	1	X	X	1
01	X			1
11				
10	1			X

- A. $\bar{b}d$
C. $\bar{b}d + ab\bar{c}d$

- B. $\bar{b}d + \bar{b}\bar{c}$
D. $\bar{b}d + \bar{b}\bar{c} + \bar{c}d$

gatecse-2012 digital-logic k-map easy

Answer key

4.21.14 K Map: GATE CSE 2025 | Set 2 | Question: 33



Given the following Karnaugh Map for a Boolean function $F(w, x, y, z)$:

yz \ wx	00	01	11	10
00	1	0	0	1
01	0	1	1	0
11	0	1	1	0
10	1	0	0	1

Which one or more of the following Boolean expression(s) represent(s) F ?

- A. $\bar{w}\bar{x}\bar{y}\bar{z} + w\bar{x}\bar{y}\bar{z} + \bar{w}\bar{x}y\bar{z} + w\bar{x}y\bar{z} + xz$
B. $\bar{w}\bar{x}\bar{y}\bar{z} + \bar{w}\bar{x}y\bar{z} + w\bar{x}y\bar{z} + xz$
C. $\bar{w}\bar{x}\bar{y}\bar{z} + w\bar{x}\bar{y}\bar{z} + w\bar{x}y\bar{z} + xz$
D. $\bar{x}\bar{z} + xz$

gatecse2025-set2 digital-logic k-map multiple-selects two-marks

4.21.15 K Map: GATE IT 2006 | Question: 35



The boolean function for a combinational circuit with four inputs is represented by the following Karnaugh map.

RS \ PQ	00	01	11	10
	00	01	11	10
00	1	0	0	1
01	0	0	1	1
11	1	1	1	0
10	1	0	0	1

Which of the product terms given below is an essential prime implicant of the function?

- A. QRS B. PQS C. $PQ'S'$ D. $Q'S'$

gateit-2006 digital-logic k-map normal

4.21.16 K Map: GATE IT 2007 | Question: 78



Consider the following expression

$$ad\bar{b} + \bar{a}c + b\bar{c}d$$

Which of the following Karnaugh Maps correctly represents the expression?

A.

cd \ ab	00	01	11	10
	00	01	11	10
00	X	X		
01	X	X		
11	X	X		X
10	X			X

B.

cd \ ab	00	01	11	10
	00	01	11	10
00	X	X		
01	X			
11	X	X		X
10	X	X		X

C.

cd \ ab	00	01	11	10
	00	01	11	10
00	X	X		
01	X	X		X
11	X	X		X
10	X			X

D.

cd \ ab	00	01	11	10
	00	01	11	10
00	X	X		
01	X	X		X
11	X	X		X
10	X		X	X

gateit-2007 digital-logic k-map normal

4.21.17 K Map: GATE IT 2007 | Question: 79



Consider the following expression

$$ad\bar{b} + \bar{a}c + b\bar{c}d$$

Which of the following expressions does not correspond to the Karnaugh Map obtained for the given expression?

- A. $\bar{c}\bar{d} + ad\bar{b} + ab\bar{c} + \bar{a}\bar{c}d$
 B. $\bar{a}\bar{c} + \bar{c}\bar{d} + ad\bar{b} + ab\bar{c}d$
 C. $\bar{a}\bar{c} + ad\bar{b} + ab\bar{c} + \bar{c}d$
 D. $\bar{b}\bar{c}\bar{d} + acd\bar{b} + \bar{a}\bar{c} + ab\bar{c}$

gateit-2007 digital-logic k-map normal

4.22

Little Endian Big Endian (1)

4.22.1 Little Endian Big Endian: GATE CSE 2021 | Set 2 | Question: 44



If the numerical value of a 2-byte unsigned integer on a little endian computer is 255 more than that on a big endian computer, which of the following choices represent(s) the unsigned integer on a little endian computer?

- A. 0x6665 B. 0x0001 C. 0x4243 D. 0x0100

Answer key

4.23

Memory Interfacing (5)

Practice Test: Test 1 (5Q)

4.23.1 Memory Interfacing: GATE CSE 1995 | Question: 2.2



The capacity of a memory unit is defined by the number of words multiplied by the number of bits/word. How many separate address and data lines are needed for a memory of $4K \times 16$?

- A. 10 address, 16 data lines
 B. 11 address, 8 data lines
 C. 12 address, 16 data lines
 D. 12 address, 12 data lines

gate1995 digital-logic memory-interfacing normal

Answer key

4.23.2 Memory Interfacing: GATE CSE 2009 | Question: 7, ISRO2015-3



How many $32K \times 1$ RAM chips are needed to provide a memory capacity of $256K$ bytes?

- A. 8
 B. 32
 C. 64
 D. 128

gatecse-2009 digital-logic memory-interfacing easy isro2015

Answer key

4.23.3 Memory Interfacing: GATE CSE 2010 | Question: 7



The main memory unit with a capacity of 4 megabytes is built using $1M \times 1$ -bit DRAM chips. Each DRAM chip has $1K$ rows of cells with $1K$ cells in each row. The time taken for a single refresh operation is 100 nanoseconds. The time required to perform one refresh operation on all the cells in the memory unit is

- A. 100 nanoseconds
 B. 100×2^{10} nanoseconds
 C. 100×2^{20} nanoseconds
 D. 3200×2^{20} nanoseconds

gatecse-2010 digital-logic memory-interfacing normal

Answer key

4.23.4 Memory Interfacing: GATE CSE 2013 | Question: 46



A RAM chip has a capacity of 1024 words of 8 bits each ($1K \times 8$). The number of 2×4 decoders with enable line needed to construct a $16K \times 16$ RAM from $1K \times 8$ RAM is

- (A) 4 (B) 5 (C) 6 (D) 7

gatecse-2013 digital-logic normal memory-interfacing

Answer key

4.23.5 Memory Interfacing: GATE IT 2005 | Question: 9



A dynamic RAM has a memory cycle time of 64 nsec. It has to be refreshed 100 times per msec and each refresh takes 100 nsec. What percentage of the memory cycle time is used for refreshing?

- A. 10
 B. 6.4
 C. 1
 D. 0.64

gateit-2005 digital-logic memory-interfacing normal

Answer key

4.24

Min No Gates (6)

Practice Tests: Test 1 (15Q) Test 2 (7Q)

4.24.1 Min No Gates: GATE CSE 1995 | Question: 15-a



Implement a circuit having the following output expression using an inverter and a nand gate

$$Z = \overline{A} + \overline{B} + C$$

gate1995 digital-logic normal descriptive min-no-gates digital-circuits boolean-algebra

Answer key

4.24.2 Min No Gates: GATE CSE 2000 | Question: 9



Design a logic circuit to convert a single digit BCD number to the number modulo six as follows (Do not detect illegal input):

- Write the truth table for all bits. Label the input bits $I_1, I_2, \dots$ with I_1 as the least significant bit. Label the output bits $R_1, R_2, \dots$ with R_1 as the least significant bit. Use 1 to signify truth.
- Draw one circuit for each output bit using, **altogether**, two two-input AND gates, one two-input OR gate and two NOT gates.

gatecse-2000 digital-logic min-no-gates descriptive

Answer key

4.24.3 Min No Gates: GATE CSE 2004 | Question: 58



A circuit outputs a digit in the form of 4 bits. 0 is represented by 0000, 1 by 0001, ..., 9 by 1001. A combinational circuit is to be designed which takes these 4 bits as input and outputs 1 if the digit ≥ 5 , and 0 otherwise. If only AND, OR and NOT gates may be used, what is the minimum number of gates required?

- 2
- 3
- 4
- 5

gatecse-2004 digital-logic normal min-no-gates

Answer key

4.24.4 Min No Gates: GATE CSE 2009 | Question: 6



What is the minimum number of gates required to implement the Boolean function $(AB+C)$ if we have to use only 2-input NOR gates?

- 2
- 3
- 4
- 5

gatecse-2009 digital-logic min-no-gates normal

Answer key

4.24.5 Min No Gates: GATE CSE 2017 | Set 1 | Question: 21



Consider the Karnaugh map given below, where X represents "don't care" and blank represents 0.

ba \ dc	00	01	11	10
00		X	X	
01	1			X
11	1			1
10		X	X	

Assume for all inputs (a, b, c, d) , the respective complements $(\bar{a}, \bar{b}, \bar{c}, \bar{d})$ are also available. The above logic is implemented using 2-input NOR gates only. The minimum number of gates required is _____.

gatecse-2017-set1 digital-logic k-map numerical-answers normal min-no-gates

Answer key

4.24.6 Min No Gates: GATE IT 2004 | Question: 8



What is the minimum number of NAND gates required to implement a 2-input EXCLUSIVE-OR function without using any other logic gate?

- A. 2 B. 4 C. 5 D. 6

gateit-2004 digital-logic min-no-gates normal

Answer key

4.25

Min Products of Sum Form (2)

4.25.1 Min Products of Sum Form: GATE CSE 1990 | Question: 5-a



Find the minimum product of sums of the following expression

$$f = ABC + \bar{A} \bar{B} \bar{C}$$

gate1990 digital-logic boolean-algebra min-products-of-sum-form canonical-normal-form descriptive

Answer key

4.25.2 Min Products of Sum Form: GATE CSE 2017 | Set 2 | Question: 28



Given $f(w, x, y, z) = \sum_m(0, 1, 2, 3, 7, 8, 10) + \sum_d(5, 6, 11, 15)$; where d represents the 'don't-care' condition in Karnaugh maps. Which of the following is a minimum product-of-sums (POS) form of $f(w, x, y, z)$?

- A. $f = (\bar{w} + \bar{z})(\bar{x} + z)$ B. $f = (\bar{w} + z)(x + z)$
C. $f = (w + z)(\bar{x} + z)$ D. $f = (w + \bar{z})(\bar{x} + z)$

gatecse-2017-set2 digital-logic min-products-of-sum-form

Answer key

4.26

Min Sum of Products Form (16)

Practice Tests: [Test 1 \(15Q\)](#) [Test 2 \(1Q\)](#)

4.26.1 Min Sum of Products Form: GATE CSE 1991 | Question: 5-b



Find the minimum sum of products form of the logic function $f(A, B, C, D) = \sum_m(0, 2, 8, 10, 15) + \sum_d(3, 11, 12, 14)$ where m and d represent minterm and don't care term respectively.

gate1991 digital-logic boolean-algebra min-sum-of-products-form descriptive

Answer key

4.26.2 Min Sum of Products Form: GATE CSE 1997 | Question: 71



Let $f = (\bar{w} + y)(\bar{x} + y)(w + \bar{x} + z)(\bar{w} + z)(\bar{x} + z)$

- A. Express f as the minimal sum of products. Write only the answer.
B. If the output line is stuck at 0, for how many input combinations will the value of f be correct?

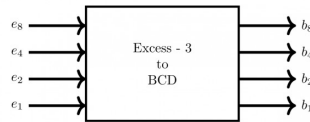
gate1997 digital-logic min-sum-of-products-form numerical-answers

Answer key

4.26.3 Min Sum of Products Form: GATE CSE 2001 | Question: 10



- a. Is the 3-variable function $f = \sum(0, 1, 2, 4)$ its self-dual? Justify your answer.
b. Give a minimal product-of-sum form of the b output of the following excess-3 to BCD converter.



gatecse-2001 digital-logic normal descriptive min-sum-of-products-form

Answer key

4.26.4 Min Sum of Products Form: GATE CSE 2005 | Question: 18



The switching expression corresponding to $f(A, B, C, D) = \Sigma(1, 4, 5, 9, 11, 12)$ is:

- A. $BC'D' + A'C'D + AB'D$ B. $ABC' + ACD + B'C'D$
 C. $ACD' + A'BC' + AC'D'$ D. $A'BD + ACD' + BCD'$

gatecse-2005 digital-logic normal min-sum-of-products-form

Answer key

4.26.5 Min Sum of Products Form: GATE CSE 2007 | Question: 9



Consider the following Boolean function of four variables:

$$f(w, x, y, z) = \Sigma(1, 3, 4, 6, 9, 11, 12, 14)$$

The function is

- A. independent of one variables. B. independent of two variables.
 C. independent of three variables. D. dependent on all variables

gatecse-2007 digital-logic normal min-sum-of-products-form k-map

Answer key

4.26.6 Min Sum of Products Form: GATE CSE 2011 | Question: 14



The simplified SOP (Sum of Product) from the Boolean expression

$$(P + \bar{Q} + R). (P + \bar{Q} + R). (P + Q + R)$$

is

- A. $(\bar{P}.Q + \bar{R})$ B. $(P + \bar{Q}. \bar{R})$ C. $(\bar{P}.Q + R)$ D. $(P.Q + R)$

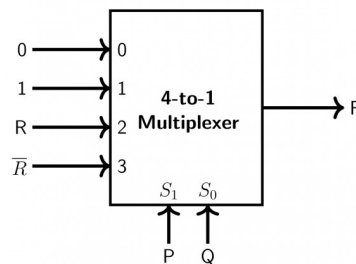
gatecse-2011 digital-logic normal min-sum-of-products-form

Answer key

4.26.7 Min Sum of Products Form: GATE CSE 2014 | Set 1 | Question: 45



Consider the 4-to-1 multiplexer with two select lines S_1 and S_0 given below



The minimal sum-of-products form of the Boolean expression for the output F of the multiplexer is

- A. $\bar{P}Q + Q\bar{R} + P\bar{Q}R$ B. $\bar{P}Q + \bar{P}Q\bar{R} + PQ\bar{R} + P\bar{Q}R$
 C. $\bar{P}QR + \bar{P}Q\bar{R} + Q\bar{R} + P\bar{Q}R$ D. $PQ\bar{R}$

gatecse-2014-set1 digital-logic normal multiplexer min-sum-of-products-form

Answer key

4.26.8 Min Sum of Products Form: GATE CSE 2014 | Set 1 | Question: 7



Consider the following Boolean expression for F :

$$F(P, Q, R, S) = PQ + \bar{P}QR + \bar{P}QR\bar{S}$$

The minimal sum-of-products form of F is

A. $PQ + QR + QS$

B. $P + Q + R + S$

C. $\bar{P} + \bar{Q} + \bar{R} + \bar{S}$

D. $\bar{P}R + \bar{R}\bar{P}S + P$

gatecse-2014-set1 digital-logic normal min-sum-of-products-form

Answer key

4.26.9 Min Sum of Products Form: GATE CSE 2014 | Set 3 | Question: 7



Consider the following minterm expression for F :

$$F(P, Q, R, S) = \sum 0, 2, 5, 7, 8, 10, 13, 15$$

The minterms 2, 7, 8 and 13 are 'do not care' terms. The minimal sum-of-products form for F is

A. $Q\bar{S} + \bar{Q}S$

B. $\bar{Q}\bar{S} + QS$

C. $\bar{Q}\bar{R}\bar{S} + \bar{Q}R\bar{S} + Q\bar{R}S + QRS$

D. $\bar{P}\bar{Q}\bar{S} + \bar{P}QS + PQS + P\bar{Q}\bar{S}$

gatecse-2014-set3 digital-logic min-sum-of-products-form normal

Answer key

4.26.10 Min Sum of Products Form: GATE CSE 2018 | Question: 49



Consider the minterm list form of a Boolean function F given below.

$$F(P, Q, R, S) = \sum m(0, 2, 5, 7, 9, 11) + d(3, 8, 10, 12, 14)$$

Here, m denotes a minterm and d denotes a don't care term. The number of essential prime implicants of the function F is ____

gatecse-2018 digital-logic min-sum-of-products-form k-map numerical-answers two-marks

Answer key

4.26.11 Min Sum of Products Form: GATE CSE 2021 | Set 2 | Question: 52



Consider a Boolean function $f(w, x, y, z)$ such that

$$f(w, 0, 0, z) = 1$$

$$f(1, x, 1, z) = x + z$$

$$f(w, 1, y, z) = wz + y$$

The number of literals in the minimal sum-of-products expression of f is _____

gatecse-2021-set2 digital-logic boolean-algebra min-sum-of-products-form numerical-answers two-marks

Answer key

4.26.12 Min Sum of Products Form: GATE CSE 2024 | Set 1 | Question: 37



Consider a Boolean expression given by $F(X, Y, Z) = \sum(3, 5, 6, 7)$.

Which of the following statements is/are CORRECT?

A. $F(X, Y, Z) = \Pi(0, 1, 2, 4)$

B. $F(X, Y, Z) = XY + YZ + XZ$

C. $F(X, Y, Z)$ is independent of input Y

D. $F(X, Y, Z)$ is independent of input X

gatecse-2024-set1 multiple-selects digital-logic min-sum-of-products-form two-marks

4.26.13 Min Sum of Products Form: GATE CSE 2025 | Set 1 | Question: 32



Consider the following four variable Boolean function in sum-of-product form

$$F(b_3, b_2, b_1, b_0) = \sum(0, 2, 4, 8, 10, 11, 12)$$

where the value of the function is computed by considering $b_3b_2b_1b_0$ as a 4-bit binary number, where b_3 denotes the most significant bit and b_0 denotes the least significant bit. Note that there are no don't care terms. Which ONE of the following options is the CORRECT minimized Boolean expression for F ?

- A. $\bar{b}_1\bar{b}_0 + \bar{b}_2\bar{b}_0 + b_1\bar{b}_2b_3$ B. $\bar{b}_1\bar{b}_0 + \bar{b}_2\bar{b}_0$
 C. $\bar{b}_2\bar{b}_0 + b_1b_2b_3$ D. $\bar{b}_0\bar{b}_2 + \bar{b}_3$

gatecse2025-set1 digital-logic min-sum-of-products-form easy two-marks

4.26.14 Min Sum of Products Form: GATE CSE 2026 | Set 1 | Question: 38



Consider a Boolean function F with the following minterm expression:

$$F(P, Q, R, S) = \sum m(1, 2, 3, 4, 5, 7, 10, 12, 13, 14)$$

Which of the following options is/are the minimal sum-of-products expression(s) of F ?

- A. $\bar{P}S + QR + \bar{P}\bar{Q}R + \bar{Q}R\bar{S}$
 B. $\bar{P}S + QR + \bar{P}\bar{Q}R + PR\bar{S}$
 C. $\bar{P}S + QR + PQ\bar{S} + PR\bar{S}$
 D. $\bar{P}S + QR + PQ\bar{S} + \bar{Q}R\bar{S}$

gatecse-2026-set1 two-marks digital-logic min-sum-of-products-form k-map multiple-selects

4.26.15 Min Sum of Products Form: GATE CSE 2026 | Set 2 | Question: 30



Consider the following 4-variable Boolean function

$$F(A, B, C, D) = \sum m(0, 1, 2, 3, 8, 9, 10, 11)$$

Consider A as MSB, D as LSB. Which one of the following options represents the minimal sum of products form for the above function?

Note: $+$ is OR operation, $\cdot$ is AND operation, $'$ is NOT operation

- A. $A' + B' + C' + D'$ B. B'
 C. $A' \cdot B' + A \cdot B$ D. A'

gatecse-2026-set2 digital-logic min-sum-of-products-form k-map two-marks

4.26.16 Min Sum of Products Form: GATE IT 2008 | Question: 8



Consider the following Boolean function of four variables

$$f(A, B, C, D) = \sum(2, 3, 6, 7, 8, 9, 10, 11, 12, 13)$$

The function is

- A. independent of one variable B. independent of two variables
 C. independent of three variable D. dependent on all the variables

gateit-2008 digital-logic normal min-sum-of-products-form

 **Practice Tests:** [Test 1 \(15Q\)](#) [Test 2 \(13Q\)](#) [Weekly Quiz 4 \(15Q\)](#)

4.27.1 Multiplexer: GATE CSE 1990 | Question: 5-b



Show with the help of a block diagram how the Boolean function :

$$f = AB + BC + CA$$

can be realised using only a 4 : 1 multiplexer.

gate1990 descriptive digital-logic combinational-circuit multiplexer

Answer key 

4.27.2 Multiplexer: GATE CSE 1998 | Question: 1.14



A multiplexer with a 4 – *bit* data select input is a

- A. 4 : 1 multiplexer
- B. 2 : 1 multiplexer
- C. 16 : 1 multiplexer
- D. 8 : 1 multiplexer

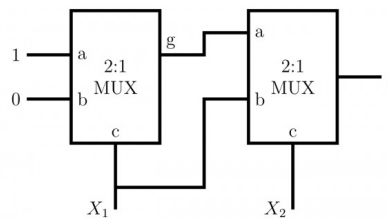
gate1998 digital-logic multiplexer easy

Answer key 

4.27.3 Multiplexer: GATE CSE 2001 | Question: 2.11



Consider the circuit shown below. The output of a 2 : 1 MUX is given by the function $(ac' + bc)$.



Which of the following is true?

- A. $f = X_1' + X_2$
- B. $f = X_1' X_2 + X_1 X_2'$
- C. $f = X_1 X_2 + X_1' X_2'$
- D. $f = X_1 + X_2'$

gatecse-2001 digital-logic normal multiplexer

Answer key 

4.27.4 Multiplexer: GATE CSE 2004 | Question: 60



Consider a multiplexer with X and Y as data inputs and Z as the control input. $Z = 0$ selects input X , and $Z = 1$ selects input Y . What are the connections required to realize the 2-variable Boolean function $f = T + R$, without using any additional hardware?

- A. R to X, 1 to Y, T to Z
- B. T to X, R to Y, T to Z
- C. T to X, R to Y, 0 to Z
- D. R to X, 0 to Y, T to Z

gatecse-2004 digital-logic normal multiplexer

Answer key 

4.27.5 Multiplexer: GATE CSE 2007 | Question: 34



Suppose only one multiplexer and one inverter are allowed to be used to implement any Boolean function of n variables. What is the minimum size of the multiplexer needed?

- A. 2^n line to 1 line
- B. 2^{n+1} line to 1 line
- C. 2^{n-1} line to 1 line
- D. 2^{n-2} line to 1 line

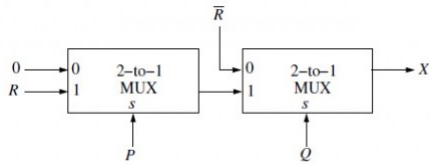
gatecse-2007 digital-logic normal multiplexer

Answer key

4.27.6 Multiplexer: GATE CSE 2016 | Set 1 | Question: 30



Consider the two cascade 2 to 1 multiplexers as shown in the figure .



The minimal sum of products form of the output X is

- A. $\bar{P}\bar{Q} + PQR$ B. $\bar{P}Q + QR$
C. $PQ + \bar{P}\bar{Q}R$ D. $\bar{Q}\bar{R} + PQR$

gatescse-2016-set1 digital-logic multiplexer normal

Answer key

4.27.7 Multiplexer: GATE CSE 2020 | Question: 19



A multiplexer is placed between a group of 32 registers and an accumulator to regulate data movement such that at any given point in time the content of only one register will move to the accumulator. The number of select lines needed for the multiplexer is _____.

gatescse-2020 numerical-answers digital-logic multiplexer one-mark

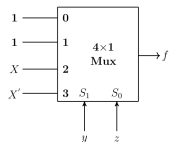
Answer key

4.27.8 Multiplexer: GATE CSE 2021 | Set 2 | Question: 5

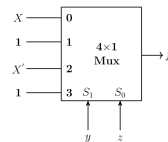


Which one of the following circuits implements the Boolean function given below?

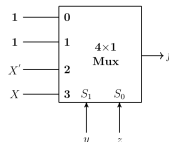
$f(x, y, z) = m_0 + m_1 + m_3 + m_4 + m_5 + m_6$, where m_i is the i^{th} minterm.



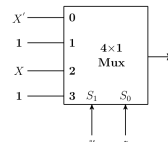
A.



B.



C.



D.

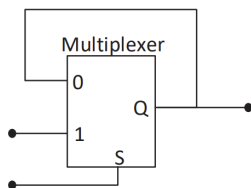
gatescse-2021-set2 digital-logic combinational-circuit multiplexer one-mark

Answer key

4.27.9 Multiplexer: GATE CSE 2023 | Question: 11



The output of a 2-input multiplexer is connected back to one of its inputs as shown in the figure.



Match the functional equivalence of this circuit to one of the following options.

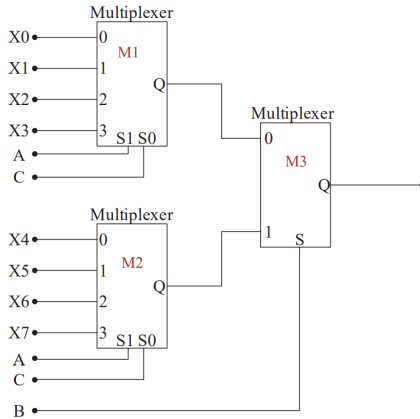
- A. D Flip-flop B. D Latch C. Half-adder D. Demultiplexer

gatescse-2023 digital-logic combinational-circuit multiplexer one-mark

4.27.10 Multiplexer: GATE CSE 2023 | Question: 34



A Boolean digital circuit is composed using two 4-input multiplexers (M1 and M2) and one 2-input multiplexer (M3) as shown in the figure. X0-X7 are the inputs of the multiplexers M1 and M2 and could be connected to either 0 or 1. The select lines of the multiplexers are connected to Boolean variables A, B and C as shown.



Which one of the following set of values of (X0, X1, X2, X3, X4, X5, X6, X7) will realise the Boolean function $\overline{A} + \overline{A} \cdot \overline{C} + A \cdot \overline{B} \cdot C$?

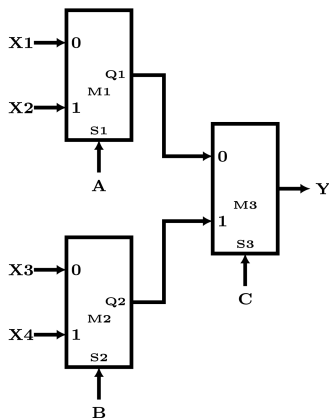
- A. (1,1,0,0,1,1,1,0)
 B. (1,1,0,0,1,1,0,1)
 C. (1,1,0,1,1,1,0,0)
 D. (0,0,1,1,0,1,1,1)

gatecse-2023 digital-logic combinational-circuit multiplexer two-marks

4.27.11 Multiplexer: GATE CSE 2024 | Set 1 | Question: 54



Consider a digital logic circuit consisting of three 2-to-1 multiplexers M1, M2, and M3 as shown below. X1 and X2 are inputs of M1. X3 and X4 are inputs of M2. A, B, and C are select lines of M1, M2, and M3, respectively.



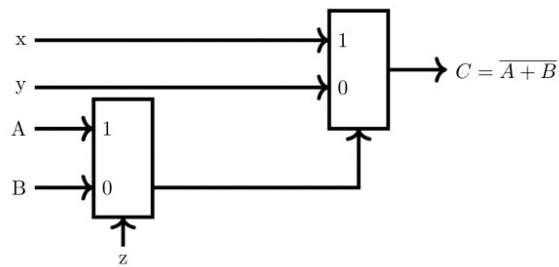
For an instance of inputs $X1 = 1, X2 = 1, X3 = 0$, and $X4 = 0$, the number of combinations of A, B, C that give the output $Y = 1$ is _____.

gatecse-2024-set1 numerical-answers digital-logic multiplexer two-marks

4.27.12 Multiplexer: GATE IT 2005 | Question: 48



The circuit shown below implements a 2-input NOR gate using two 2 – 4 MUX (control signal 1 selects the upper input). What are the values of signals x, y and z?



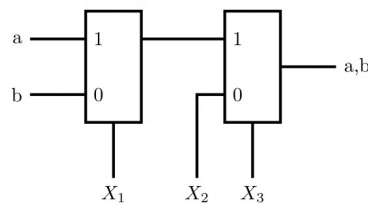
- A. $1, 0, B$ B. $1, 0, A$ C. $0, 1, B$ D. $0, 1, A$

gateit-2005 digital-logic normal multiplexer

Answer key

4.27.13 Multiplexer: GATE IT 2007 | Question: 8

The following circuit implements a two-input AND gate using two 2 – 1 multiplexers.



What are the values of X_1, X_2, X_3 ?

- A. $X_1 = b, X_2 = 0, X_3 = a$ B. $X_1 = b, X_2 = 1, X_3 = b$
 C. $X_1 = a, X_2 = b, X_3 = 1$ D. $X_1 = a, X_2 = 0, X_3 = b$

gateit-2007 digital-logic normal multiplexer

Answer key

4.27.14 Multiplexer: GATE1992-04-b

A priority encoder accepts three input signals (A, B and C) and produces a two-bit output (X_1, X_0) corresponding to the highest priority active input signal. Assume A has the highest priority followed by B and C has the lowest priority. If none of the inputs are active the output should be 00 , design the priority encoder using 4 : 1 multiplexers as the main components.

gate1992 digital-logic combinational-circuit multiplexer descriptive

Answer key

4.28 Number Representation (57)

Practice Tests: [Test 1 \(15Q\)](#) [Test 2 \(15Q\)](#) [Test 3 \(15Q\)](#) [Test 4 \(15Q\)](#) [Test 5 \(9Q\)](#)

4.28.1 Number Representation: GATE CSE 1990 | Question: 1-viii

The condition for overflow in the addition of two 2^n 's complement numbers in terms of the carry generated by the two most significant bits is _____.

gate1990 digital-logic number-representation fill-in-the-blanks

Answer key

4.28.2 Number Representation: GATE CSE 1991 | Question: 01-iii

Consider the number given by the decimal expression:

$$16^3 * 9 + 16^2 * 7 + 16 * 5 + 3$$

The number of 1 's in the unsigned binary representation of the number is _____

Answer key

4.28.3 Number Representation: GATE CSE 1991 | Question: 01-v



When two 4-bit numbers $A = a_3a_2a_1a_0$ and $B = b_3b_2b_1b_0$ are multiplied, the bit c_1 of the product C is given by _____

Answer key

4.28.4 Number Representation: GATE CSE 1992 | Question: 4-a



Consider addition in two's complement arithmetic. A carry from the most significant bit does not always correspond to an overflow. Explain what is the condition for overflow in two's complement arithmetic.

Answer key

4.28.5 Number Representation: GATE CSE 1993 | Question: 6.5



Convert the following numbers in the given bases into their equivalents in the desired bases:

- A. $(110.101)_2 = (x)_{10}$
 B. $(1118)_{10} = (y)_H$

Answer key

4.28.6 Number Representation: GATE CSE 1994 | Question: 2.7



Consider n -bit (including sign bit) 2 's complement representation of integer numbers. The range of integer values, N , that can be represented is _____ $\leq N \leq$ _____.

Answer key

4.28.7 Number Representation: GATE CSE 1995 | Question: 18



The following is an incomplete Pascal function to convert a given decimal integer (in the range -8 to $+7$) into a binary integer in 2 's complement representation. Determine the expressions A , B , C that complete the program.

```
function TWOSCOMP (N:integer):integer;
var
  REM, EXPONENT:integer;
  BINARY :integer;
begin
  if (N>=-8) and (N<=+7) then
  begin
    if N<0 then
      N:=A;
    BINARY:=0;
    EXPONENT:=1;
    while N<>0 do
    begin
      REM:=N mod 2;
      BINARY:=BINARY + B*EXPONENT;
      EXPONENT:=EXPONENT*10;
      N:=C
    end
    TWOSCOMP:=BINARY
  end
end;
```

Answer key

4.28.8 Number Representation: GATE CSE 1995 | Question: 2.12, ISRO2015-9



The number of 1's in the binary representation of $(3 * 4096 + 15 * 256 + 5 * 16 + 3)$ are:

- A. 8 B. 9 C. 10 D. 12

gate1995 digital-logic number-representation normal isro2015

Answer key

4.28.9 Number Representation: GATE CSE 1996 | Question: 1.25



Consider the following floating-point number representation.

31	24	23	0
Exponent		Mantissa	

The exponent is in $2'$ s complement representation and the mantissa is in the sign-magnitude representation. The range of the magnitude of the normalized numbers in this representation is

- A. 0 to 1 B. 0.5 to 1 C. 2^{-23} to 0.5 D. 0.5 to $(1 - 2^{-23})$

gate1996 digital-logic number-representation normal

Answer key

4.28.10 Number Representation: GATE CSE 1997 | Question: 5.4



Given $\sqrt{(224)_r} = (13)_r$.

The value of the radix r is:

- A. 10 B. 8 C. 5 D. 6

gate1997 digital-logic number-representation normal

Answer key

4.28.11 Number Representation: GATE CSE 1998 | Question: 1.17



The octal representation of an integer is $(342)_8$. If this were to be treated as an eight-bit integer in an 8085 based computer, its decimal equivalent is

- A. 226 B. -98 C. 76 D. -30

gate1998 digital-logic number-representation normal

Answer key

4.28.12 Number Representation: GATE CSE 1998 | Question: 2.20



Suppose the domain set of an attribute consists of signed four digit numbers. What is the percentage of reduction in storage space of this attribute if it is stored as an integer rather than in character form?

- A. 80\% B. 20\% C. 60\% D. 40\%

gate1998 digital-logic number-representation normal

Answer key

4.28.13 Number Representation: GATE CSE 1999 | Question: 2.17



Zero has two representations in

- A. Sign-magnitude B. $2'$ s complement
C. $1'$ s complement D. None of the above

gate1999 digital-logic number-representation easy multiple-selects

Answer key

4.28.14 Number Representation: GATE CSE 2000 | Question: 1.6



The number 43 in $2's$ complement representation is

- A. 01010101 B. 11010101 C. 00101011 D. 10101011

gatecse-2000 digital-logic number-representation easy

Answer key

4.28.15 Number Representation: GATE CSE 2000 | Question: 2.14



Consider the values of $A = 2.0 \times 10^{30}$, $B = -2.0 \times 10^{30}$, $C = 1.0$, and the sequence

$X := A + B$ $Y := A + C$
 $X := X + C$ $Y := Y + B$

executed on a computer where floating point numbers are represented with 32 bits. The values for X and Y will be

- A. $X = 1.0, Y = 1.0$ B. $X = 1.0, Y = 0.0$
 C. $X = 0.0, Y = 1.0$ D. $X = 0.0, Y = 0.0$

gatecse-2000 digital-logic number-representation normal

Answer key

4.28.16 Number Representation: GATE CSE 2001 | Question: 2.10



The $2's$ complement representation of $(-539)_{10}$ in hexadecimal is

- A. ABE B. DBC C. $DE5$ D. $9E7$

gatecse-2001 digital-logic number-representation easy

Answer key

4.28.17 Number Representation: GATE CSE 2002 | Question: 1.14



The decimal value 0.25

- A. is equivalent to the binary value 0.1
 B. is equivalent to the binary value 0.01
 C. is equivalent to the binary value 0.00111
 D. cannot be represented precisely in binary

gatecse-2002 digital-logic number-representation easy

Answer key

4.28.18 Number Representation: GATE CSE 2002 | Question: 1.15



The $2's$ complement representation of the decimal value -15 is

- A. 1111 B. 11111 C. 111111 D. 10001

gatecse-2002 digital-logic number-representation easy

Answer key

4.28.19 Number Representation: GATE CSE 2002 | Question: 1.16



Sign extension is a step in

- A. floating point multiplication B. signed 16 bit integer addition
 C. arithmetic left shift D. converting a signed integer from one size to another

gatecse-2002 digital-logic easy number-representation

Answer key

4.28.20 Number Representation: GATE CSE 2002 | Question: 1.21



In 2^s complement addition, overflow

- A. is flagged whenever there is carry from sign bit addition
- B. cannot occur when a positive value is added to a negative value
- C. is flagged when the carries from sign bit and previous bit match
- D. None of the above

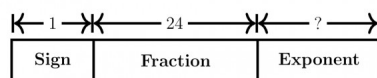
gatecse-2002 digital-logic number-representation normal

Answer key

4.28.21 Number Representation: GATE CSE 2002 | Question: 9



Consider the following 32-bit floating-point representation scheme as shown in the format below. A value is specified by 3 fields, a one bit sign field (with 0 for positive and 1 for negative values), a 24 bit fraction field (with the binary point is at the left end of the fraction bits), and a 7 bit exponent field (in excess-64 signed integer representation, with 16 is the base of exponentiation). The sign bit is the most significant bit.



- A. It is required to represent the decimal value -7.5 as a normalized floating point number in the given format. Derive the values of the various fields. Express your final answer in the hexadecimal.
- B. What is the largest value that can be represented using this format? Express your answer as the nearest power of 10.

gatecse-2002 digital-logic number-representation normal descriptive

Answer key

4.28.22 Number Representation: GATE CSE 2003 | Question: 9



Assuming all numbers are in 2^s complement representation, which of the following numbers is divisible by 11111011?

- A. 11100111
- B. 11100100
- C. 11010111
- D. 11011011

gatecse-2003 digital-logic number-representation normal

Answer key

4.28.23 Number Representation: GATE CSE 2004 | Question: 19



If 73_x (in base- x number system) is equal to 54_y (in base- y number system), the possible values of x and y are

- A. 8,16
- B. 10,12
- C. 9,13
- D. 8,11

gatecse-2004 digital-logic number-representation easy

Answer key

4.28.24 Number Representation: GATE CSE 2004 | Question: 28



What is the result of evaluating the following two expressions using three-digit floating point arithmetic with rounding?

$$(113. + -111.) + 7.51$$

$$113. + (-111. + 7.51)$$

- A. 9.51 and 10.0 respectively
- B. 10.0 and 9.51 respectively
- C. 9.51 and 9.51 respectively
- D. 10.0 and 10.0 respectively

gatecse-2004 digital-logic number-representation normal

Answer key

4.28.25 Number Representation: GATE CSE 2004 | Question: 66



Let $A = 11111010$ and $B = 00001010$ be two 8-bit 2's complement numbers. Their product in 2's complement is

- A. 11000100 B. 10011100 C. 10100101 D. 11010101

gatecse-2004 digital-logic number-representation easy

Answer key

4.28.26 Number Representation: GATE CSE 2005 | Question: 16, ISRO2009-18, ISRO2015-2



The range of integers that can be represented by an n bit 2's complement number system is:

- A. -2^{n-1} to $(2^{n-1} - 1)$ B. $-(2^{n-1} - 1)$ to $(2^{n-1} - 1)$
C. -2^{n-1} to 2^{n-1} D. $-(2^{n-1} + 1)$ to $(2^{n-1} - 1)$

gatecse-2005 digital-logic number-representation easy isro2009 isro2015

Answer key

4.28.27 Number Representation: GATE CSE 2005 | Question: 17



The hexadecimal representation of $(657)_8$ is:

- A. 1AF B. D78 C. D71 D. 32F

gatecse-2005 digital-logic number-representation easy

Answer key

4.28.28 Number Representation: GATE CSE 2006 | Question: 39



We consider the addition of two 2's complement numbers $b_{n-1}b_{n-2}\dots b_0$ and $a_{n-1}a_{n-2}\dots a_0$. A binary adder for adding unsigned binary numbers is used to add the two numbers. The sum is denoted by $c_{n-1}c_{n-2}\dots c_0$ and the carry-out by c_{out} . Which one of the following options correctly identifies the overflow condition?

- A. $c_{out} (\overline{a_{n-1}} \oplus \overline{b_{n-1}})$ B. $a_{n-1}b_{n-1}\overline{c_{n-1}} + \overline{a_{n-1}}\overline{b_{n-1}}c_{n-1}$
C. $c_{out} \oplus c_{n-1}$ D. $a_{n-1} \oplus b_{n-1} \oplus c_{n-1}$

gatecse-2006 digital-logic number-representation normal

Answer key

4.28.29 Number Representation: GATE CSE 2008 | Question: 6



Let r denote number system radix. The only value(s) of r that satisfy the equation $\sqrt{121_r} = 11_r$ is/are

- A. decimal 10 B. decimal 11 C. decimal 10 and 11 D. any value > 2

gatecse-2008 digital-logic number-representation normal

Answer key

4.28.30 Number Representation: GATE CSE 2009 | Question: 5, ISRO2017-57



$(1217)_8$ is equivalent to

- A. $(1217)_{16}$ B. $(028F)_{16}$ C. $(2297)_{10}$ D. $(0B17)_{16}$

gatecse-2009 digital-logic number-representation isro2017

Answer key

4.28.31 Number Representation: GATE CSE 2010 | Question: 8



P is a 16-bit signed integer. The 2's complement representation of P is $(F87B)_{16}$. The 2's complement

representation of $8 \times P$ is

A. $(C3D8)_{16}$

B. $(187B)_{16}$

C. $(F878)_{16}$

D. $(987B)_{16}$

gatecse-2010 digital-logic number-representation normal

Answer key

4.28.32 Number Representation: GATE CSE 2013 | Question: 4



The smallest integer that can be represented by an 8-bit number in 2 's complement form is

A. -256

B. -128

C. -127

D. 0

gatecse-2013 digital-logic number-representation easy

Answer key

4.28.33 Number Representation: GATE CSE 2014 | Set 1 | Question: 8



The base (or radix) of the number system such that the following equation holds is_____.

$$\frac{312}{20} = 13.1$$

gatecse-2014-set1 digital-logic number-representation numerical-answers normal

Answer key

4.28.34 Number Representation: GATE CSE 2014 | Set 2 | Question: 8



Consider the equation $(123)_5 = (x8)_y$ with x and y as unknown. The number of possible solutions is _____.

gatecse-2014-set2 digital-logic number-representation numerical-answers normal

Answer key

4.28.35 Number Representation: GATE CSE 2015 | Set 3 | Question: 35



Consider the equation $(43)_x = (y3)_8$ where x and y are unknown. The number of possible solutions is _____.

gatecse-2015-set3 digital-logic number-representation normal numerical-answers

Answer key

4.28.36 Number Representation: GATE CSE 2016 | Set 1 | Question: 07



The 16-bit 2 's complement representation of an integer is 1111 1111 1111 0101; its decimal representation is _____.

gatecse-2016-set1 digital-logic number-representation normal numerical-answers

Answer key

4.28.37 Number Representation: GATE CSE 2016 | Set 2 | Question: 09



Let X be the number of distinct 16-bit integers in 2 's complement representation. Let Y be the number of distinct 16-bit integers in sign magnitude representation. Then $X - Y$ is_____.

gatecse-2016-set2 digital-logic number-representation normal numerical-answers

Answer key

4.28.38 Number Representation: GATE CSE 2017 | Set 1 | Question: 9



When two 8-bit numbers $A_7 \cdots A_0$ and $B_7 \cdots B_0$ in 2 's complement representation (with A_0 and B_0 as the least significant bits) are added using a **ripple-carry adder**, the sum bits obtained are $S_7 \cdots S_0$ and the carry bits are $C_7 \cdots C_0$. An overflow is said to have occurred if

A. the carry bit C_7 is 1

- B. all the carry bits $(C_7, \dots, C_0)$ are 1
- C. $(A_7 \cdot B_7 \cdot \overline{S_7} + \overline{A_7} \cdot \overline{B_7} \cdot S_7)$ is 1
- D. $(A_0 \cdot B_0 \cdot \overline{S_0} + \overline{A_0} \cdot \overline{B_0} \cdot S_0)$ is 1

gatecse-2017-set1 digital-logic number-representation

Answer key

4.28.39 Number Representation: GATE CSE 2017 | Set 2 | Question: 1



The representation of the value of a 16-bit unsigned integer X in hexadecimal number system is BCA9. The representation of the value of X in octal number system is

- A. 571244 B. 736251 C. 571247 D. 136251

gatecse-2017-set2 digital-logic number-representation

Answer key

4.28.40 Number Representation: GATE CSE 2019 | Question: 22



Two numbers are chosen independently and uniformly at random from the set $\{1, 2, \dots, 13\}$. The probability (rounded off to 3 decimal places) that their 4-bit (unsigned) binary representations have the same most significant bit is _____.

gatecse-2019 numerical-answers digital-logic number-representation probability one-mark

Answer key

4.28.41 Number Representation: GATE CSE 2019 | Question: 4



In 16-bit 2's complement representation, the decimal number -28 is:

- A. 1111 1111 0001 1100 B. 0000 0000 1110 0100
- C. 1111 1111 1110 0100 D. 1000 0000 1110 0100

gatecse-2019 digital-logic number-representation one-mark

Answer key

4.28.42 Number Representation: GATE CSE 2019 | Question: 8



Consider $Z = X - Y$ where X, Y and Z are all in sign-magnitude form. X and Y are each represented in n bits. To avoid overflow, the representation of Z would require a minimum of:

- A. n bits B. $n - 1$ bits C. $n + 1$ bits D. $n + 2$ bits

gatecse-2019 digital-logic number-representation one-mark

Answer key

4.28.43 Number Representation: GATE CSE 2021 | Set 1 | Question: 6



Let the representation of a number in base 3 be 210. What is the hexadecimal representation of the number?

- A. 15 B. 21 C. D2 D. 528

gatecse-2021-set1 digital-logic number-representation normal one-mark

Answer key

4.28.44 Number Representation: GATE CSE 2021 | Set 2 | Question: 18



If x and y are two decimal digits and $(0.1101)_2 = (0.8xy5)_{10}$, the decimal value of $x + y$ is _____

gatecse-2021-set2 numerical-answers digital-logic number-representation easy one-mark

Answer key

4.28.45 Number Representation: GATE CSE 2022 | Question: 8



Let R_1 and R_2 be two 4-bit registers that store numbers in 2's complement form. For the operation $R_1 + R_2$, which one of the following values of R_1 and R_2 gives an arithmetic overflow?

- A. $R_1 = 1011$ and $R_2 = 1110$ B. $R_1 = 1100$ and $R_2 = 1010$
 C. $R_1 = 0011$ and $R_2 = 0100$ D. $R_1 = 1001$ and $R_2 = 1111$

gatecse-2022 digital-logic number-representation one-mark

Answer key

4.28.46 Number Representation: GATE CSE 2023 | Question: 22



A particular number is written as 132 in radix-4 representation. The same number in radix-5 representation is _____.

gatecse-2023 digital-logic number-representation numerical-answers one-mark

Answer key

4.28.47 Number Representation: GATE CSE 2024 | Set 1 | Question: 3



Consider a system that uses 5 bits for representing signed integers in 2's complement format. In this system, two integers A and B are represented as $A=01010$ and $B=11010$. Which one of the following operations will result in either an arithmetic overflow or an arithmetic underflow?

- A. $A + B$ B. $A - B$ C. $B - A$ D. $2 * B$

gatecse-2024-set1 digital-logic number-representation one-mark

Answer key

4.28.48 Number Representation: GATE CSE 2024 | Set 2 | Question: 39



Which of the following is/are EQUAL to 224 in radix - 5 (i.e., base - 5) notation?

- A. 64 in radix -10 B. 100 in radix -8
 C. 50 in radix -16 D. 121 in radix -7

gatecse-2024-set2 digital-logic number-representation easy multiple-selects two-marks

Answer key

4.28.49 Number Representation: GATE CSE 2025 | Set 1 | Question: 15



The number -6 can be represented as 1010 in 4-bit 2's complement representation. Which of the following is/are CORRECT 2's complement representation(s) of -6 ?

- A. $1000\ 1010$ in 8-bits B. $1111\ 1010$ in 8-bits
 C. $1000\ 0000\ 0000\ 1010$ in 16-bits D. $1111\ 1111\ 1111\ 1010$ in 16-bits

gatecse2025-set1 digital-logic number-representation multiple-selects easy one-mark

Answer key

4.28.50 Number Representation: GATE CSE 2026 | Set 1 | Question: 12



Consider the 8-bit signed integers X, Y and Z represented using the sign-magnitude form. The binary representations of X and Y are as follows:

$$X : 10110100 \quad Y : 01001100$$

Which of the following operations to compute Z result(s) in an arithmetic overflow?

- A. $Z = X + Y$ B. $Z = X - Y$ C. $Z = -X + Y$ D. $Z = -X - Y$

gatecse-2026-set1 digital-logic number-representation multiple-selects one-mark

Answer key

4.28.51 Number Representation: GATE CSE 2026 | Set 2 | Question: 18



In a system, numbers are represented using 4-bit two's complement form. Consider four numbers $N_1 = 1011$, $N_2 = 1101$, $N_3 = 1010$ and $N_4 = 1001$ in the system. Which of the following operations will result in arithmetic overflow?

- A. $N_1 + N_2$ B. $N_2 + N_3$ C. $N_3 - N_4$ D. $N_1 + N_4$

gatecse-2026-set2 digital-logic number-representation multiple-selects one-mark

Answer key

4.28.52 Number Representation: GATE IT 2004 | Question: 42



Using a 4 – *bit 2's* complement arithmetic, which of the following additions will result in an overflow?

- i. $1100 + 1100$
ii. $0011 + 0111$
iii. $1111 + 0111$

- A. i only B. ii only C. iii only D. i and iii only

gateit-2004 digital-logic number-representation normal

Answer key

4.28.53 Number Representation: GATE IT 2004 | Question: 43



The number $(123456)_8$ is equivalent to

- A. $(A72E)_{16}$ and $(22130232)_4$ B. $(A72E)_{16}$ and $(22131122)_4$
C. $(A73E)_{16}$ and $(22130232)_4$ D. $(A62E)_{16}$ and $(22120232)_4$

gateit-2004 digital-logic number-representation normal

Answer key

4.28.54 Number Representation: GATE IT 2005 | Question: 47



$(34.4)_8 \times (23.4)_8$ evaluates to

- A. $(1053.6)_8$ B. $(1053.2)_8$ C. $(1024.2)_8$ D. None of these

gateit-2005 digital-logic number-representation normal

Answer key

4.28.55 Number Representation: GATE IT 2006 | Question: 7, ISRO2009-41



The addition of 4 – *bit*, two's complement, binary numbers 1101 and 0100 results in

- A. 0001 and an overflow B. 1001 and no overflow
C. 0001 and no overflow D. 1001 and an overflow

gateit-2006 digital-logic number-representation normal isro2009

Answer key

4.28.56 Number Representation: GATE IT 2007 | Question: 42



$(C012.25)_H - (10111001110.101)_B =$

- A. $(135103.412)_O$ B. $(564411.412)_O$
C. $(564411.205)_O$ D. $(135103.205)_O$

gateit-2007 digital-logic number-representation normal

Answer key

4.28.57 Number Representation: GATE IT 2008 | Question: 15



A processor that has the carry, overflow and sign flag bits as part of its program status word (PSW) performs addition of the following two 2's complement numbers 01001101 and 11101001. After the execution of this

addition operation, the status of the carry, overflow and sign flags, respectively will be:

- A. 1,1,0 B. 1,0,0 C. 0,1,0 D. 1,0,1

gateit-2008 digital-logic number-representation normal co-and-architecture

Answer key

4.29 Number System (1)

 **Practice Tests:** [Weekly Quiz 1 \(15Q\)](#) [Weekly Quiz 2 \(15Q\)](#) [Weekly Quiz 3 \(15Q\)](#) [Weekly Quiz 4 \(15Q\)](#)

4.29.1 Number System: GATE CSE 1988 | Question: 2-vi



Define the value of r in the following: $\sqrt{(41)_r} = (7)_{10}$

gate1988 digital-logic normal number-representation descriptive number-system

Answer key

4.30 Prime Implicants (2)

 **Practice Test:** [Test 1 \(5Q\)](#)

4.30.1 Prime Implicants: GATE CSE 1997 | Question: 5.1



Let $f(x, y, z) = \bar{x} + \bar{y}x + xz$ be a switching function. Which one of the following is valid?

- A. $\bar{y}x$ is a prime implicant of f B. xz is a minterm of f
C. xz is an implicant of f D. y is a prime implicant of f

gate1997 digital-logic normal k-map prime-implicants

Answer key

4.30.2 Prime Implicants: GATE CSE 2004 | Question: 59



Which are the essential prime implicants of the following Boolean function?

$$f(a, b, c) = a'c + ac' + b'c$$

- A. $a'c$ and ac' B. $a'c$ and $b'c$ C. $a'c$ only. D. ac' and bc'

gatecse-2004 digital-logic normal prime-implicants

Answer key

4.31 ROM (4)

4.31.1 ROM: GATE CSE 1993 | Question: 6.6



A ROM is used to store the Truth table for binary multiple units that will multiply two 4-bit numbers. The size of the ROM (number of words $\times$ number of bits) that is required to accommodate the Truth table is M words $\times$ N bits. Write the values of M and N .

gate1993 digital-logic normal rom descriptive

Answer key

4.31.2 ROM: GATE CSE 1996 | Question: 1.21



A ROM is used to store the table for multiplication of two 8-bit unsigned integers. The size of ROM required is

- A. 256×16 B. $64K \times 8$
C. $4K \times 16$ D. $64K \times 16$

gate1996 digital-logic normal rom

Answer key

4.31.3 ROM: GATE CSE 2012 | Question: 19



The amount of ROM needed to implement a 4-bit multiplier is

- A. 64 bits B. 128 bits C. 1 Kbits D. 2 Kbits

gatecse-2012 digital-logic normal rom

Answer key

4.31.4 ROM: GATE IT 2004 | Question: 10



What is the minimum size of ROM required to store the complete truth table of an $8 - bit \times 8 - bit$ multiplier?

- A. $32K \times 16$ bits B. $64K \times 16$ bits
C. $16K \times 32$ bits D. $64K \times 32$ bits

gateit-2004 digital-logic normal rom

Answer key

4.32 Reduction (1)

4.32.1 Reduction: GATE CSE 2026 | Set 1 | Question: 11



Consider the following Boolean expression of a function F :

$$F(P, Q) = (\bar{P} + Q) \oplus (\bar{P}Q)$$

Which of the following expressions is/are equivalent to F ?

- A. $\overline{P \oplus Q}$ B. $P \oplus Q$
C. $\bar{P} \oplus Q$ D. $\bar{P} \oplus \bar{Q}$

gatecse-2026-set1 digital-logic boolean-algebra reduction multiple-selects one-mark

Answer key

4.33 Ripple Counter Operation (1)

4.33.1 Ripple Counter Operation: GATE CSE 2025 | Set 2 | Question: 24



In a 4-bit ripple counter, if the period of the waveform at the last flip-flop is 64 microseconds, then the frequency of the ripple counter in kHz is _____. (Answer in integer)

gatecse2025-set2 digital-logic ripple-counter-operation numerical-answers one-mark

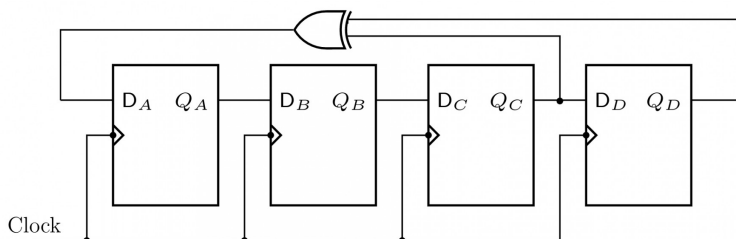
Answer key

4.34 Shift Registers (2)

4.34.1 Shift Registers: GATE CSE 1987 | Question: 13-a



The below figure shows four D-type flip-flops connected as a shift register using a XOR gate. The initial state and three subsequent states for three clock pulses are also given.



State	Q_A	Q_B	Q_C	Q_D
Initial	1	1	1	1
After the first clock	0	1	1	1
After the second clock	0	0	1	1
After the third clock	0	0	0	1

The state $Q_A Q_B Q_C Q_D$ after the fourth clock pulse is

- A. 0000 B. 1111 C. 1001 D. 1000

gate1987 digital-logic circuit-output sequential-circuit digital-counter shift-registers

Answer key

4.34.2 Shift Registers: GATE CSE 1991 | Question: 06,a



Using D flip-flop gates, design a parallel-in/serial-out shift register that shifts data from left to right with the following input lines:

- Clock CLK
- Three parallel data inputs A, B, C
- Serial input S
- Control input $\text{LOAD}/\overline{\text{SHIFT}}$.

gate1991 digital-logic difficult sequential-circuit flip-flop shift-registers descriptive

Answer key

4.35 Static Hazard (1)

4.35.1 Static Hazard: GATE CSE 2006 | Question: 38



Consider a Boolean function $f(w, x, y, z)$. Suppose that exactly one of its inputs is allowed to change at a time. If the function happens to be true for two input vectors $i_1 = \langle w_1, x_1, y_1, z_1 \rangle$ and $i_2 = \langle w_2, x_2, y_2, z_2 \rangle$, we would like the function to remain true as the input changes from i_1 to i_2 (i_1 and i_2 differ in exactly one bit position) without becoming false momentarily. Let $f(w, x, y, z) = \sum(5, 7, 11, 12, 13, 15)$. Which of the following cube covers of f will ensure that the required property is satisfied?

- $\overline{w}xz, wx\overline{y}, x\overline{y}z, xyz, wyz$
- $wxy, \overline{w}xz, wyz$
- $wx\overline{y}z, xz, w\overline{x}yz$
- $wx\overline{y}, wyz, wxz, \overline{w}xz, x\overline{y}z, xyz$

gatecse-2006 digital-logic min-sum-of-products-form difficult static-hazard

Answer key

4.36 Synchronous Asynchronous Circuits (4)

4.36.1 Synchronous Asynchronous Circuits: GATE CSE 1991 | Question: 03-ii



Advantage of synchronous sequential circuits over asynchronous ones is:

- faster operation
- ease of avoiding problems due to hazards
- lower hardware requirement
- better noise immunity
- none of the above

gate1991 digital-logic normal sequential-circuit synchronous-asynchronous-circuits multiple-selects

Answer key

4.36.2 Synchronous Asynchronous Circuits: GATE CSE 1998 | Question: 16



Design a synchronous counter to go through the following states:

1, 4, 2, 3, 1, 4, 2, 3, 1, 4, ...

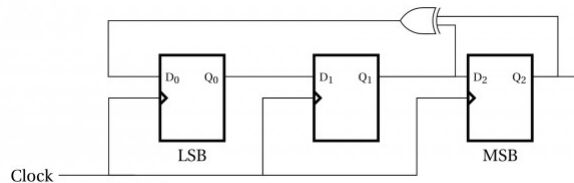
gate1998 digital-logic normal descriptive synchronous-asynchronous-circuits

Answer key

4.36.3 Synchronous Asynchronous Circuits: GATE CSE 2001 | Question: 2.12



Consider the circuit given below with initial state $Q_0 = 1, Q_1 = Q_2 = 0$. The state of the circuit is given by the value $4Q_2 + 2Q_1 + Q_0$



Which one of the following is correct state sequence of the circuit?

- A. 1, 3, 4, 6, 7, 5, 2
B. 1, 2, 5, 3, 7, 6, 4
C. 1, 2, 7, 3, 5, 6, 4
D. 1, 6, 5, 7, 2, 3, 4

gatecse-2001 digital-logic normal synchronous-asynchronous-circuits

Answer key

4.36.4 Synchronous Asynchronous Circuits: GATE CSE 2003 | Question: 44



A 1-input, 2-output synchronous sequential circuit behaves as follows:

Let z_k, n_k denote the number of 0's and 1's respectively in initial k bits of the input ($z_k + n_k = k$). The circuit outputs 00 until one of the following conditions holds.

- $z_k - n_k = 2$. In this case, the output at the k -th and all subsequent clock ticks is 10.
- $n_k - z_k = 2$. In this case, the output at the k -th and all subsequent clock ticks is 01.

What is the minimum number of states required in the state transition graph of the above circuit?

- A. 5
B. 6
C. 7
D. 8

gatecse-2003 digital-logic synchronous-asynchronous-circuits normal

Answer key

Answer Keys

4.1.1	N/A	4.1.2	N/A	4.1.3	B	4.1.4	B	4.1.5	A
4.1.6	B	4.1.7	19.2	4.1.8	B	4.1.9	-1	4.2.1	B
4.2.2	C	4.3.1	C	4.4.1	D	4.4.2	C	4.4.3	N/A
4.4.4	N/A	4.4.5	N/A	4.4.6	D	4.4.7	N/A	4.4.8	A
4.4.9	B	4.4.10	D	4.4.11	C	4.4.12	D	4.4.13	C
4.4.14	C	4.4.15	D	4.4.16	D	4.4.17	D	4.4.18	A
4.4.19	A	4.4.20	D	4.4.21	D	4.4.22	1	4.4.23	A
4.4.24	C	4.4.25	C	4.4.26	D	4.4.27	B	4.4.28	B;C;D
4.4.29	B;C	4.4.30	B;D	4.4.31	B;C;D	4.4.32	B	4.4.33	B
4.4.34	C	4.5.1	N/A	4.5.2	A	4.5.3	B	4.5.4	13:13

4.5.5	A	4.5.6	C	4.5.7	B	4.6.1	N/A	4.6.2	A
4.6.3	C	4.6.4	A	4.6.5	3	4.6.6	A	4.6.7	A
4.6.8	3	4.6.9	B	4.6.10	C;D	4.7.1	A	4.7.2	B
4.8.1	B	4.8.2	N/A	4.8.3	B	4.8.4	N/A	4.8.5	N/A
4.8.6	A;C	4.8.7	B	4.8.8	B	4.8.9	B	4.8.10	011
4.8.11	D	4.8.12	C	4.8.13	N/A	4.8.14	N/A	4.8.15	B
4.8.16	B	4.8.17	A	4.8.18	A	4.8.19	A	4.8.20	D
4.8.21	A	4.8.22	A	4.8.23	D	4.8.24	A	4.8.25	C
4.8.26	C	4.8.27	C	4.8.28	A	4.8.29	A	4.8.30	B
4.8.31	D	4.8.32	B	4.8.33	C	4.8.34	A;B;C	4.8.35	C
4.8.36	X	4.8.37	D	4.8.38	C	4.8.39	A	4.8.40	D
4.9.1	C	4.9.2	A;C	4.10.1	A	4.11.1	C	4.11.2	1034
4.11.3	B	4.12.1	N/A	4.12.2	N/A	4.12.3	D	4.12.4	D
4.12.5	A	4.12.6	C	4.12.7	6:6	4.13.1	C	4.13.2	N/A
4.13.3	N/A	4.13.4	33.33	4.13.5	N/A	4.13.6	N/A	4.13.7	A
4.13.8	C	4.13.9	D	4.13.10	3	4.13.11	3:4	4.13.12	B
4.13.13	A	4.13.14	A	4.13.15	B	4.13.16	D	4.13.17	D
4.13.18	N/A	4.14.1	D	4.15.1	True	4.15.2	B	4.15.3	B
4.15.4	A	4.16.1	D	4.16.2	C	4.17.1	N/A	4.17.2	C
4.17.3	A	4.17.4	B	4.17.5	2	4.17.6	7:8	4.17.7	D
4.18.1	C	4.18.2	N/A	4.18.3	N/A	4.18.4	N/A	4.18.5	N/A
4.18.6	C	4.18.7	D	4.18.8	D	4.19.1	N/A	4.19.2	B;C
4.19.3	N/A	4.19.4	N/A	4.19.5	B;C	4.19.6	B	4.19.7	A
4.20.1	D	4.20.2	B	4.20.3	A	4.20.4	C	4.20.5	B
4.20.6	-7.75 : -7.75	4.20.7	C	4.20.8	B	4.20.9	C	4.20.10	B
4.20.11	A;B;C	4.20.12	C	4.20.13	-60.25:-60.25	4.20.14	C	4.21.1	D
4.21.2	N/A	4.21.3	N/A	4.21.4	N/A	4.21.5	C	4.21.6	C
4.21.7	B	4.21.8	D	4.21.9	A	4.21.10	B	4.21.11	C
4.21.12	A	4.21.13	B	4.21.14	A;D	4.21.15	D	4.21.16	A
4.21.17	C	4.22.1	A;D	4.23.1	C	4.23.2	C	4.23.3	B
4.23.4	B	4.23.5	C	4.24.1	N/A	4.24.2	N/A	4.24.3	B
4.24.4	B	4.24.5	1	4.24.6	B	4.25.1	N/A	4.25.2	A
4.26.1	N/A	4.26.2	9	4.26.3	N/A	4.26.4	A	4.26.5	B
4.26.6	B	4.26.7	A	4.26.8	A	4.26.9	B	4.26.10	3
4.26.11	6 : 6	4.26.12	A;B	4.26.13	A	4.26.14	B;D	4.26.15	B
4.26.16	A	4.27.1	N/A	4.27.2	C	4.27.3	C	4.27.4	A
4.27.5	C	4.27.6	D	4.27.7	5	4.27.8	A	4.27.9	B
4.27.10	C	4.27.11	4	4.27.12	D	4.27.13	A	4.27.14	N/A
4.28.1	N/A	4.28.2	9	4.28.3	N/A	4.28.4	N/A	4.28.5	N/A
4.28.6	N/A	4.28.7	N/A	4.28.8	C	4.28.9	D	4.28.10	C

4.28.11	D
4.28.16	C
4.28.21	N/A
4.28.26	A
4.28.31	A
4.28.36	-11
4.28.41	C
4.28.46	110
4.28.51	B;D
4.28.56	A
4.31.1	N/A
4.33.1	250:250
4.36.2	N/A

4.28.12	C
4.28.17	B
4.28.22	A
4.28.27	A
4.28.32	B
4.28.37	1
4.28.42	C
4.28.47	B
4.28.52	B
4.28.57	B
4.31.2	D
4.34.1	D
4.36.3	B

4.28.13	A;C
4.28.18	D
4.28.23	D
4.28.28	B
4.28.33	5
4.28.38	C
4.28.43	A
4.28.48	A;B;D
4.28.53	A
4.29.1	12
4.31.3	D
4.34.2	N/A
4.36.4	A

4.28.14	C
4.28.19	D
4.28.24	A
4.28.29	D
4.28.34	3
4.28.39	D
4.28.44	3 : 3
4.28.49	B;D
4.28.54	A
4.30.1	C
4.31.4	B
4.35.1	D

4.28.15	B
4.28.20	B
4.28.25	A
4.28.30	B
4.28.35	5
4.28.40	0.502 : 0.504
4.28.45	B
4.28.50	B;C
4.28.55	C
4.30.2	A
4.32.1	A;C
4.36.1	B